FINAL REPORT

Level 04

Investigating and Addressing Information Barriers to Regulatory Compliance Among Sri Lankan SMEs

Enigmatrix

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Faculty of Information Technology

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DECLARATION

We declare that this thesis is our own work and has not been submitted in any form for another degree or diploma at any university or other institution of tertiary education. Information derived from the published or unpublished work of others has been acknowledged in the text and a list of references is given.

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ABSTRACT

Small and medium enterprises make up more than three-quarters of businesses in Sri Lanka, yet regulatory non-compliance among them remains widespread. This study begins from the observation that non-compliance is more often a failure of information than of intent. Rules published in the Government Gazette reach business owners late; guidance received informally is frequently inaccurate; owners cannot see their own compliance risk until a penalty notice arrives; and incorrect regulatory claims circulate widely in Sinhala and Tamil. These four barriers have not previously been measured for Sri Lanka, and no existing system addresses them together.

This research measures each barrier empirically and then builds an integrated platform that is validated against those same measurements. Four modules were developed: a gazette monitor that tracks how long a regulatory change takes to reach an SME and classifies it by category and affected sector; a retrieval-augmented compliance assistant built on an expert-verified knowledge base covering twenty regulatory domains; an explainable risk model that predicts compliance failure; and a claim-verification service checked against the same verified record. All four share one database, one business profile and one regulatory vocabulary, and operate in English, Sinhala and Tamil.

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Chapter 1 - Introduction

1.1 Introduction

Small and medium enterprises are the operating majority of the Sri Lankan economy. They are also the segment least equipped to absorb regulatory change. Compliance is not optional for them: it governs market access, financing, employment obligations, taxation, product standards, import and export permissions and, ultimately, the right to continue trading. Yet the mechanism by which a Sri Lankan SME learns that an obligation has changed is almost entirely informal.

Regulatory change in Sri Lanka is published primarily through the Official Gazette, supplemented by departmental notices, tax circulars, labour regulations, import and export controls and sector-specific standards. These instruments are legally authoritative but were never designed as a communication channel. They are published as unstructured PDF documents, in three languages, with no subscription mechanism, no push notification and no machine-readable metadata describing what changed, who it affects or when it takes effect. A gazette becomes binding on publication regardless of whether any affected business has read it.

Large organisations absorb this cost through dedicated legal, tax, risk and compliance functions. SMEs do not have that capacity. Owners and managers rely on accountants, trade associations, peer networks, messaging groups and social media — channels that are fast but unverified, or reliable but slow and expensive. The result is a lag between legal publication and practical awareness, during which the enterprise is non-compliant without knowing it. Penalties arising in this window are not the product of wilful evasion; they are the product of an information failure.

Enigmatrix is designed around that failure. It is a regulatory intelligence platform that treats the path from official publication to SME action as a measurable pipeline, and attempts to shorten it. The platform detects regulatory changes, structures and classifies them, determines which SME sectors they affect, explains them in plain language in the user's preferred language, assesses the resulting compliance risk, and verifies the accuracy of compliance claims circulating informally. The project is organised into four research modules so that each contribution is independently measurable while sharing a single verified regulatory data layer.

1.2 Background and Motivation

The motivation for this project is the mismatch between how regulation is published and how it is consumed.

Official regulatory documents are optimised for legal validity, not comprehension. A single gazette issue may contain several unrelated instruments, cross-reference earlier amendments by number, use statutory drafting conventions, and appear as a scanned image rather than selectable text. Sinhala and Tamil issues frequently use legacy font encodings - notably Wijesekara - that render correctly on screen but produce meaningless byte sequences when extracted programmatically, and do so silently. Even once a document is located, the SME owner must still determine four things: whether it applies to their sector, what category of change it represents, what duty or deadline it creates, and what action to take.

Existing approaches address parts of this but not the whole:

- **Government portals** publish authoritative content but assume the user already knows what to search for and can interpret what they find. Discovery remains a pull operation.
- **Professional advisory services** - accountants, tax consultants, legal advisers - provide interpretation, but at a cost and cadence many SMEs cannot sustain. Advice typically arrives at filing time, not at publication time.
- **Informal channels** - peer groups, messaging applications, social media - are fast and free, and are consequently the dominant channel. They are also unverified, and are the primary vector for regulatory misinformation.
- **Enterprise GRC and RegTech systems** automate regulatory monitoring effectively, but are built for large regulated institutions, assume English-first structured regulatory feeds, and presume in-house compliance staff to act on their output.

The research gap is therefore not a missing piece of software; it is a missing measurement. Compliance research establishes that knowledge and tax literacy predict voluntary compliance [22], [23], [24], and the slippery-slope framework explains compliance as a function of trust and enforcement power [25]. But this literature treats knowledge as a static attribute of the owner. It rarely asks how regulatory information physically travelled from the state to the enterprise, how long it took, through which channel, or whether it was distorted in transit. Akerlof's account of information asymmetry [26] describes exactly this structure: one party holds information the other needs, and the outcome degrades when the transfer fails.

Enigmatrix responds by building the transfer mechanism *and* instrumenting it. Every stage of the pipeline records a timestamp, every downstream appearance of a regulation on a secondary channel is captured as a propagation event, and the resulting dataset makes the diffusion lag an empirical quantity.

1.3 Problem in Brief

Sri Lankan SMEs lack a timely, reliable, multilingual and sector-aware mechanism for discovering regulatory changes and converting them into practical compliance action.

The problem decomposes into four connected gaps, which map one-to-one onto the four modules of the platform:

1. **Regulatory change awareness gap (Module 1).** SMEs do not reliably learn that a relevant tax, labour, import/export, product-standard or sector rule has changed, and there is no measurement of how long that discovery actually takes.
2. **Compliance knowledge accuracy gap (Module 2).** Even when a regulation is found, its meaning and the required action are not clear to a non-specialist reader.
3. **Compliance risk invisibility gap (Module 3).** SMEs cannot see which of their compliance weaknesses are most urgent, so effort is allocated by anxiety rather than by exposure.
4. **Regulatory misinformation gap (Module 4).** Informal channels circulate inaccurate compliance claims that are indistinguishable, to the recipient, from correct ones.

A system that only stores and displays documents does not close any of these gaps. The platform must convert fragmented regulatory publications into verified, searchable, explainable and enterprise-specific intelligence.

1.4 Aim & Objectives

1.4.1 Aim

To design, implement and evaluate a modular, trilingual regulatory intelligence platform that enables Sri Lankan SMEs to identify, understand, assess and respond to relevant regulatory changes **in a timely and trustworthy manner, and to measure the reduction in regulatory information lag that the platform achieves.**

1.4.2 Objectives

1. To investigate the regulatory information barriers faced by Sri Lankan SMEs - awareness delay, interpretation difficulty, risk prioritisation and misinformation exposure - and to formalise them as measurable research questions.
2. To design an end-to-end platform architecture connecting regulatory ingestion, SME profiles, multilingual explanation, knowledge retrieval, risk assessment and misinformation verification over a single verified regulatory data layer.
3. To implement a regulatory change awareness module that scrapes, extracts, preprocesses, classifies and structures official regulatory content, including scanned and legacy-encoded Sinhala and Tamil documents.
4. To construct a labelled regulatory dataset through dual annotation and adjudication, and to establish its reliability using Cohen's kappa and field-level reconciliation statistics.
5. To train and evaluate baseline and transformer-based models for regulatory change classification and SME sector relevance detection, using macro-F1 as the primary metric under class imbalance.
6. To support multilingual regulatory understanding through controlled English summarisation and Sinhala/Tamil translation of titles and summaries.
7. To implement propagation watchers, sector-matched alerting and lag analytics so that regulatory information diffusion can be measured empirically.
8. To design and implement a sealed-baseline evaluation framework that quantifies extraction and classification accuracy at field, record and stage granularity.
9. To provide defined integration points for the compliance knowledge, compliance risk and misinformation verification modules.
10. To document the design, implementation, evaluation, limitations and future work in a reproducible dissertation.

Table 1.1 — Research questions addressed by Module 1

ID	Research question
RQ1	Can a single trilingual classifier assign gazette changes to an 8-domain, 3-sector taxonomy at a macro-F1 of at least 0.92, with no language slice more than 8 percentage points below the overall score?
RQ2	Does extraction quality — in particular Sinhala/Tamil OCR and Wijesekara-to-Unicode conversion — support reliable downstream classification?
RQ3	What is the measured regulatory information-diffusion lag across publication channels in Sri Lanka?
RQ4	Do targeted, sector-matched alerts reduce that lag relative to the unassisted baseline?

1.5 Proposed Solution

Enigmatrix is a modular web platform composed of a FastAPI backend, a Next.js frontend, relational and vector storage, an asynchronous task layer and a Python machine-learning package. SMEs register with a business profile — sector, region, scale and language preference. Regulatory information is collected from official sources, extracted into structured records, classified by change category and affected sector, summarised into plain language, translated where required, and delivered through dashboards, alerts, search and advisory workflows. Administrative and expert users verify, correct, annotate and measure the pipeline through a dedicated admin surface.

The design principle is that regulatory information is structured **once**, into a verified regulatory intelligence store, and then reused by every downstream module. Module 1 produces that store; Modules 2, 3 and 4 consume it.

1.5.1 Regulatory Change Awareness Gap

Module 1 is owned by **Mohomed M.R.I (215075J)**. It owns the upstream pipeline and the diffusion-measurement research programme. Its functions are:

1. **Regulation source management.** Administrative CRUD over regulation records and sources, soft deletion via an is_active flag, an expert-verification gate, bulk verification, restore, and audit logging on every mutation.
2. **Scraping and ingestion.** Four Scrapy spiders (gazette, weekly gazette, acts, bills) targeting gazette.lk and documents.gov.lk, with date scoping, English→Sinhala→Tamil fallback, and completeness verification with re-fetch.
3. **Extraction.** A per-document and per-page routing chain that classifies each PDF as text, hybrid or scanned and dispatches to PyMuPDF, pdfplumber, pypdfium2 or Tesseract 5, with a Surya OCR fallback profile and font-aware Wijesekara-to-Unicode conversion.
4. **Preprocessing.** Cleaning, fastText language identification, metadata extraction (gazette number, dates, penalties including the multi-penalty case), chunking and sub-document splitting into a classification_chunk.
5. **Annotation.** A Label Studio project covering 8 domains, 3 sectors, SME relevance, annotator confidence and free-text rationale; a 20-document trilingual calibration set with expert labels; stratified, k-means and hybrid active-learning samplers with minority-domain targeting.
6. **Gold dataset construction.** Dual annotation of batches 02–05, automated reduction through resolve_iaa.py, and manual adjudication of disagreements into a frozen 800-row gold dataset with zero lead-annotator fallback rows.
7. **Classification.** Eight regulatory change categories and three SME study sectors, modelled by TF-IDF baselines and an XLM-RoBERTa encoder with LoRA adapters and a dual head, exported to ONNX for CPU inference.
8. **Summarisation and translation.** Controlled English summary generation from the classified record, and NLLB-200 translation of titles and summaries into Sinhala and Tamil, with an administrative translation review queue.
9. **Watchers, alerts and analytics.** Portal and RSS watchers feeding a propagation-event table via a two-step matcher, an idempotent sector-matched alert dispatcher across in-app, email and SMS channels, and materialised lag-analytics views refreshed nightly.
10. **Measurement and retraining.** A dataset registry with immutable sealed versions, an extraction-profile registry, a measurement engine producing per-field metrics, a

downloadable accuracy report, a Kullback–Leibler drift monitor and a canary promotion decision function.

The research contribution of Module 1 is not the classifier alone. It is the construction of a measurable awareness pipeline together with the first instrumented dataset of Sri Lankan regulatory information diffusion.

1.5.2 Compliance Guidance Platform

The Compliance Guidance Platform is a retrieval-augmented, multilingual question-answering system that tells a Sri Lankan small or medium enterprise (SME) not merely which regulations exist, but exactly how to comply with them — which form to file, at which office, by which deadline, and with which supporting documents. It is built on the observation that SMEs rarely fail compliance because they are unaware that a rule exists; they fail because they do not know the procedure for acting on it. The module is organised around the distinction between declarative knowledge (knowing that a rule exists) and procedural knowledge (knowing how to satisfy it). To operationalise this, every answer is required to be procedurally complete: a numbered step-by-step procedure, the exact form and office for each step, a plain-language explanation of every named form, and a worked example. Every figure quoted in an answer — rates, thresholds, deadlines, form codes — is retrieved from published government text at query time and is traceable to its source, so the system reports what regulations say rather than offering advice. The platform additionally embeds an awareness-measurement instrument. A structured survey scores an SME owner's declarative and procedural knowledge separately across the regulatory domains that apply to their sector, allowing the gap between the two to be measured empirically. The system supports English, Sinhala and Tamil and exposes a fact-checking endpoint consumed by the Regulatory Misinformation module. Its scope spans twenty regulatory domains grouped into eight categories across three retail sectors.

1.5.3 SME Compliance Risk Prediction

Small and medium enterprises in Sri Lanka rarely discover that they are heading towards a compliance failure until a penalty notice arrives. There is no early warning mechanism available to them, and no published evidence identifying which kinds of businesses are most exposed. The risk is, in effect, invisible until it has already materialised as a cost. The SME Compliance Risk Prediction module addresses this invisibility by measuring which characteristics of a business actually precede compliance failure, and by converting that measurement into an early warning that reaches the owner before a deadline is missed.

The module is built around a question that the wider literature has not answered for Sri Lanka: whether compliance failure is better explained by what a business is, or by what its owner knows. Conventional risk profiling relies on demographic attributes such as sector, size, age and location. This module tests those attributes against a second and less studied group of variables describing the owner's information environment, namely how and when they learn that a regulation has changed, and whether that information reaches them in a language they work in. The comparison is treated as a formal hypothesis with thresholds fixed before the analysis was run, so that the result could fail.

The evidence collected supports the informational explanation. On a survey of 300 SMEs in the grocery retail, food service and general retail sectors, adding information-access variables to a demographic model raised discrimination from an AUC of 0.637 to 0.720, an improvement of 0.083 that is statistically significant and stable across repeated cross-validation. Three of the five most influential variables were informational, and business sector, which is the attribute an analyst would most likely tabulate first, was not statistically associated with failure at all. Compliance risk in this population is therefore substantially a problem of information flow rather than of business structure.

The solution built from that finding is a web-based compliance risk platform. An owner registers a short business profile and immediately receives a risk score, an explanation of the specific factors driving that score, and guidance written in English, Sinhala or Tamil. Because the finding identifies late awareness as a principal driver, the platform does not stop at a single assessment: it maintains the registered profile and, whenever a regulatory change is published, determines which registered businesses that change actually affects and delivers a targeted alert in the owner's own language. Matching is precise rather than broadcast, so a sole trader with no payroll is not alerted about an employees' provident fund change, and a general retailer is not alerted about a food labelling rule.

For an SME owner the value of the module is that it makes an otherwise invisible exposure visible early enough to act on, and expresses it in terms the owner can use. For the wider platform it supplies the risk layer: it consumes regulatory change events detected by the Regulatory Change Awareness module and converts them into individually targeted, explainable warnings.

1.5.4 Regulatory Misinformation Spread

The Regulatory Misinformation Spread module aims to combat the dissemination of misinformation on regulatory matters in Sri Lankan SME social media markets, where users are likely to receive informal advice regarding taxation, import restrictions, labor obligations, product standards, and product registration requirements. These assertions are problematic as they can seem credible but only partly so, or they can be out of date or contextually confusing and can therefore shape compliance actions in ways that can cost small businesses.

The proposed solution is a multi-lingual misinformation detection and verification component, which will be based on a mix of supervised classification and retrieval-based grounding. The module is not just a single undifferentiated category of misleading content but aims to classify posts at the annotation stage as accurate, partially accurate, misleading, and false, then, via a smaller, 3-way model space, to help the classifiers be developed, in which misleading and false are grouped together as harmful for model comparison. This makes the module appropriate for the regulatory domain where it is often important that the information is factually correct and that the wording is exactly the correct one in the law, numeric thresholds and date of posting a claim are important.

The aim of the module is to offer a reliable tool to detect misinformation which can influence decisions made by SME. The second is to provide verification in a retrieval-augmented workflow allowing for support of predictions based on regulatory evidence, not just textual style or simple patterns. The module will therefore aim to enhance the predictability of the detection process as well as the interpretability and usefulness of the prediction process for the SME.

The high-level process starts with gathering public social media posts and fact-checking content, then preprocessing and multilingual annotating them, and finally training and comparing three approaches: fine-tuned XLM-RoBERTa, retrieval-augmented generation with Module 2, and direct prompting with Gemini without retrieval. Data collection is the raw material stage, where preprocessing is used to standardise it, retrieval is used to obtain the evidence of regulation, and prediction is used to obtain the veracity judgement of the data.

This module is crucial for SME users as regulatory misinformation is not simply a content quality problem, it can result in incorrect pricing, delayed registration, wrong import decisions, or non compliance of labour. A module that can detect and categorize such claims will help in quicker and better business decisions. Its usefulness is that it allows SMEs to differentiate between authoritative information and that which needs to be taken with a grain of salt.

1.5.5 Flow of the Overall System

The platform begins with two inputs: official regulatory publications and SME business profiles. Module 1 converts the former into structured, classified, verified records held in PostgreSQL and mirrored into ChromaDB for retrieval. A personalisation layer joins those records to SME profiles by sector, region and language, and drives alerting. Modules 2, 3 and 4 consume the same verified store for grounded question answering, risk assessment and claim verification respectively. All four surface through a single trilingual frontend, while administrators and domain experts feed corrections, verifications and annotations back into the store. In parallel, propagation watchers observe secondary channels and record when each regulation appears on them, producing the diffusion dataset that answers RQ3 and RQ4.

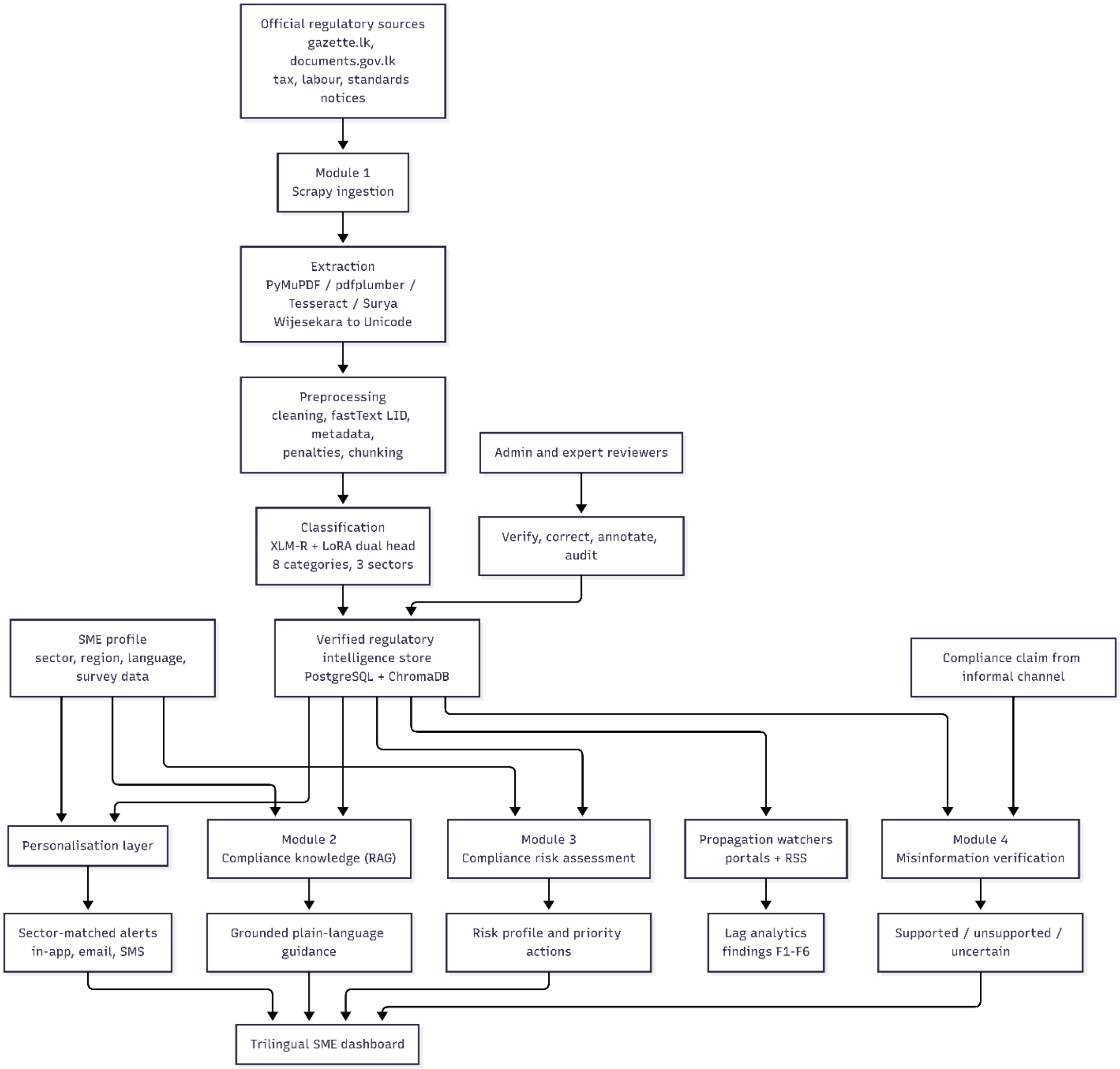


Figure 1 Overall flow of the Enigmatrix regulatory intelligence platform

Chapter 2 - Related Work

2.1 Introduction

This chapter reviews the literature relevant to Enigmatrix and to its four modules. It begins with the SME regulatory-compliance problem in developing economies, where compliance cost, regulatory complexity and limited access to specialist advice interact. It then reviews regulatory technology, information asymmetry, multilingual natural language processing, retrieval-augmented question answering, explainable compliance risk modelling and misinformation detection.

The purpose is not to enumerate existing systems but to locate a gap. The compliance literature explains *that* knowledge and trust drive compliance; the RegTech literature shows *how* regulatory text can be processed automatically. Neither measures the movement of regulatory information from publication to SME understanding in a multilingual, low-resource setting. That measurement is what this project adds.

2.2 Overall Related Work

2.2.1 SME compliance behaviour and the knowledge constraint

Research on SME compliance consistently identifies knowledge, complexity and cost as the dominant determinants. Musimenta [22] finds that knowledge requirements, tax complexity and compliance cost jointly explain tax compliance among Ugandan SMEs. Agusti and Rahman [23] reach a comparable conclusion for Indonesian enterprises, linking tax attitude to procedural understanding. Mokoena [24] extends the analysis to tax literacy, amnesty, reward and service delivery, again finding literacy to be a first-order determinant of voluntary compliance. Kirchler et al. [25] provide the theoretical frame through the slippery-slope model, in which compliance is produced either by trust in authorities or by perceived enforcement power.

Sri Lanka-specific evidence points the same way. SMEs constitute the bulk of enterprises but operate under a compliance burden calibrated for larger firms [1], [2], [6]. Regulatory and policy barriers are documented as a constraint on SME digital and commercial development in comparable economies [10], and the OECD identifies regulatory quality and accessibility as a direct determinant of the SME business environment [7]. Financial-compliance studies note that navigating the requirements is itself a specialised skill that SMEs typically lack [4]. Concrete Sri Lankan instruments illustrate the volatility: the Social Security Contribution Levy and successive budget changes altered SME obligations at short notice [3], [8], [9]. Access to finance and the wider business environment reinforce the same constraint [21].

The limitation of this body of work, for the present purpose, is that it treats knowledge as a property of the owner measured at a point in time. It asks whether the owner knows the rule. It does not ask when the rule reached them, through which channel, in which language, or whether it survived transmission intact.

2.2.2 Information asymmetry as the organizing theory

Akerlof's analysis of markets under asymmetric information [26] is the theoretical anchor. Where one party holds material information the other lacks, outcomes degrade irrespective of either party's intent. Subsequent work applies the framework to pharmaceuticals [27], to the regulator's own signalling problem [28] and to SME disclosure, where improved information disclosure measurably reduces asymmetry [29].

Applied to regulatory compliance, the asymmetry is between the state and the enterprise. The state discharges its duty by publishing; the enterprise bears the consequence of not receiving. Publication is a necessary but insufficient condition for compliance. Enigmatrix is, in these terms, an intermediary that reduces the asymmetry by making the transfer active, structured and measurable rather than passive and assumed.

2.2.3 Regulatory technology and regulatory NLP

RegTech is the closest technical field. Arner, Barberis and Buckley [11] frame RegTech as the response to escalating compliance scale and complexity following the financial crisis. McKinsey [16] characterises it operationally as software that monitors, manages and reports compliance. Butler and O'Brien [20] argue for semantic technologies and NLP as the means of rendering regulatory content machine-processable.

More recent work operationalises this. Automated compliance-monitoring pipelines have been proposed for PDF ingestion, entity extraction and regulatory-impact assessment [30], [31], and NLP has been applied to uncover latent structure in regulatory corpora [32]. RegNLP [33] investigates retrieval and answer generation over regulatory documents specifically, establishing that regulatory text is amenable to modern information-retrieval and generation methods.

Two limitations recur. First, the target user is an institution — typically a bank — with compliance staff, structured regulatory feeds and English-language source material. Second, evaluation is framed around document-processing accuracy, not around whether the regulated party became aware in time. Neither assumption holds for a Sri Lankan SME reading a scanned Sinhala gazette.

2.2.4 Multilingual and low-resource NLP

Enigmatrix must process English, Sinhala and Tamil. XLM-RoBERTa [19] provides cross-lingual representations trained at scale and is the standard choice for low-resource transfer. For Sinhala specifically, de Silva [5] evaluates pre-trained language models for text classification and finds transformer-based models outperform earlier multilingual baselines. Recent Sinhala and code-mixed work on sentiment, aspect classification and keyword extraction confirms the viability of this family of models on Sri Lankan text [47], [48]. Low-resource fine-tuning strategies for multilingual pre-trained models are documented in the SemEval setting [49], and lifelong-learning approaches address multilingual drift [44]. BERT [43] remains the architectural reference point for the encoder family.

The practical implication is that a single multilingual encoder is preferable to three language-specific pipelines at prototype scale: it shares representation across languages where labelled data is scarce, and it reduces the maintenance surface to one model, one tokenizer and one deployment artefact.

2.2.5 Retrieval-augmented generation for grounded compliance answers

Lewis et al. [17] introduced retrieval-augmented generation as a means of grounding generated text in retrieved evidence. In legal and financial domains, where an unsupported assertion is a liability rather than a stylistic flaw, grounding is essential: CBR-RAG [18] combines case-based reasoning with retrieval for legal question answering, FinSage [46] addresses multi-aspect retrieval over financial filings, and recent systems apply RAG to radio-spectrum regulation [50] and to judicial forensics with explicit trustworthiness constraints [51].

Evaluation methodology has matured alongside. RAGAS [52] and ARES [53] define automated measures of faithfulness, answer relevancy, context precision and context recall; VERA [54] adds validation of retrieval-augmented systems; and a recent survey [55] consolidates trustworthiness criteria.

2.2.6 Explainable risk modelling

Bussmann et al. [12] demonstrate explainable machine learning in credit risk management, and Bonifazi et al. [36] apply interpretable models to SME credit default. SHAP [15] is the dominant attribution method, and its stability in credit-risk settings has itself been studied [39]. Miah et al. [38] extend explainable AI to managerial decision-making in financial organisations. Class imbalance is the recurring methodological obstacle: SMOTE [40] remains the reference oversampling technique, and comparative studies of imbalance handling in fraud and risk detection quantify the trade-offs [37], [41], [42].

The gap is that these models predict *credit* risk from proprietary financial data. Compliance risk in an SME setting must be inferred from survey responses, behavioural signals and regulatory exposure, which are sparser and noisier.

2.2.7 Misinformation detection

Network-oriented analysis of WhatsApp [13] shows how closed messaging networks amplify unverified claims, and studies of misinformation sharing during COVID-19 [14], [35] and public-health surveys [34] characterise the antecedents and consequences of that behaviour. Technically, transformer-based classifiers [43] and multilingual disinformation detection systems such as PolyTruth [45] provide the modelling foundation, and evidence-grounded retrieval [46] supplies the verification mechanism.

The literature concentrates on political, health and general financial misinformation. Routine regulatory misinformation — "the VAT threshold has changed", "registration is no longer required" - is under-studied, despite being the category most likely to cause direct financial harm to an SME.

2.2.8 Synthesis

Related work establishes that automated regulatory monitoring is feasible, that multilingual transformers can handle Sinhala and Tamil, that grounded generation can produce trustworthy compliance answers, that risk can be modelled explainably, and that misinformation can be detected. What no existing system does is connect these into a single pipeline for a low-resource, multilingual SME population *and measure the resulting change in awareness lag*. That is the space Enigmatrix occupies.

2.3 Module-wise Related Work

2.3.1 Module 1 - Regulatory Change Awareness Gap

Module 1's related work falls into four strands.

RegTech monitoring. Arner et al. [11], McKinsey [16] and Butler and O'Brien [20] collectively justify the premise that regulatory monitoring should be a continuous machine process rather than a periodic human search. Module 1 implements exactly that: scheduled spiders, an automatic extraction and classification chain, and event-driven alerting.

Regulatory document NLP. Automated pipelines for regulatory PDF ingestion, entity extraction and impact assessment [30], [31], [32] and retrieval and generation over regulatory corpora [33] establish that regulatory text is tractable. What they do not address is the document-quality problem that dominates the Sri Lankan case: scanned pages, mixed scripts and legacy font encodings. Module 1's per-page routing chain, Surya fallback and font-aware Wijesekara conversion are responses to a problem the existing literature largely assumes away.

Multilingual classification. XLM-R [19] and Sinhala classification results [5], [47], [48], [49] justify a single multilingual encoder over per-language models. LoRA adaptation is adopted so that fine-tuning remains feasible on a single free-tier GPU, which is the realistic constraint for a final-year project. The dual head - single-label category with cross-entropy, multi-label sector with binary cross-entropy - reflects the structure of the task rather than an off-the-shelf configuration.

Annotation reliability. Because the labelled dataset becomes ground truth for both training and evaluation, its reliability must itself be reported. Cohen's kappa is appropriate because it discounts agreement expected by chance. The Module 1 dataset achieves a category kappa of 0.8715, a mean sector kappa of 0.8638 and an SME-relevance kappa of 0.7235 over 800 dual-annotated tasks. The lower relevance figure is informative rather than embarrassing: it quantifies the genuine subjectivity of the SME/non-SME boundary and motivates the resolver rules documented in Chapter 6.

Information diffusion. The compliance literature [22], [23], [24] measures knowledge but not its transit. Module 1's propagation-event table, two-step matcher and lag analytics constitute a measurement instrument for that transit, answering RQ3 and RQ4.

Module 1 research gap. Existing RegTech and regulatory-NLP systems can process regulatory text, but there is no evidence of a Sri Lankan, SME-focused, trilingual pipeline that both reduces and measures the delay between official regulatory publication and SME awareness.

2.3.2 Compliance Guidance Platform

A growing body of work applies conversational AI to make legal and regulatory knowledge accessible to non-experts. Li et al. [31] study question-answering systems for access to justice, examining how large language models can answer laypeople's legal questions reliably. Westermann and Benyekhlef [32] propose JusticeBot, a methodology for building augmented-intelligence tools that guide ordinary users through legal decisions. Closest to this module, Panchal et al. [33] present LawPal, a retrieval-augmented legal chatbot for the Indian context that grounds its answers in official legal documents. These systems share this module's goal of turning dense regulatory text into actionable guidance, but they stop at answering the user's question; none of them first measures what the user already knows in order to target the guidance, and none enforces the procedural completeness - the exact form, office and deadline for each step - that SME compliance demands.

On the assessment side, the accounting and public-administration literature has long studied how to measure a business owner's regulatory knowledge and relate it to compliance risk. Bornman and Ramutumbu [34] propose a conceptual framework that decomposes tax knowledge into general knowledge (awareness that an obligation exists), procedural knowledge (knowing how to meet it) and legal knowledge. This decomposition directly motivates this module's awareness survey, which scores an SME owner's declarative and procedural knowledge separately for each applicable domain and treats a low procedural score as an indicator of compliance risk. Prior instruments of this kind are typically paper questionnaires analysed offline; the contribution here is to couple such a knowledge-risk assessment with an on-demand guidance system, so that the gaps a respondent reveals can be addressed immediately by the same platform.

Retrieval-Augmented Generation (RAG), introduced by Lewis et al. [7], addresses a central weakness of large language models for factual tasks: knowledge stored in model weights cannot be updated without retraining and cannot be cited. By retrieving relevant passages from an external corpus at inference time and conditioning generation on them, RAG keeps facts current and attributable. For a compliance assistant, where regulations change through gazette amendments and every figure carries liability, this property is decisive; it motivates the module's design principle that facts live in retrievable text and never in the model's weights.

The retrieval layer depends on dense sentence embeddings and nearest-neighbour search. Sentence-BERT [8] established the bi-encoder architecture used here, and its multilingual variants allow a query in Sinhala or Tamil to retrieve English regulatory text; MPNet [25] provides the pre-training objective behind the embedding model adopted. Vector databases such as ChromaDB [26] supply the persistence and similarity-search layer.

Parameter-efficient fine-tuning makes it feasible to adapt an eight-billion-parameter instruction model on a single GPU. Low-Rank Adaptation (LoRA) [23] and its quantised form QLoRA [24] freeze the base weights and train a small number of low-rank adapters, reducing the trainable-parameter count by more than two orders of magnitude while preserving quality. In this module fine-tuning is used deliberately not to inject facts but to teach answer structure, completeness and refusal discipline, leaving all factual content to the retrieval layer.

Serving Sinhala and Tamil requires machine translation that preserves numeric and form-code fidelity. The No Language Left Behind (NLLB) project [12] provides open multilingual translation covering both languages; crucially, unlike IndicTrans2, NLLB supports Sinhala, which determined its selection for this module.

2.3.3 SME Compliance Risk Prediction

Research on predicting regulatory and tax non-compliance has developed largely within revenue administrations in high-income economies, where the modelling is supported by confidential taxpayer records, audit outcomes and longitudinal filing histories. That work establishes that non-compliance is partially predictable from administrative signals, but it is not directly transferable to the Sri Lankan SME context, where no equivalent entity-level dataset is publicly available. A separate strand of development-economics literature examines informality and the administrative burden faced by small firms, but treats compliance as an outcome to be explained descriptively rather than predicted at the level of an individual business.

The theoretical basis for treating information as a determinant of compliance comes from work on information asymmetry and on the communication of regulation. That literature argues that an obligation is only actionable once the obligated party is aware of it, understands it, and receives it in time to respond. Studies of regulatory communication in multilingual jurisdictions further note that official guidance issued in a language the recipient does not operate in imposes a comprehension cost that is functionally equivalent to not having received the guidance at all. These arguments motivate the information-barrier variables used in this module, but they have not previously been operationalised as measured predictors of compliance failure in a South Asian SME setting.

A methodological problem specific to this setting is that the outcome label is one-sided. Businesses that disclose a failure are confirmed positives, whereas businesses that report no failure are not confirmed compliant: they may have obligations they are unaware of, or may have chosen not to disclose. Treating such cases as negatives biases a conventional classifier. Elkan and Noto [16] formalise learning from positive and unlabelled data and show that, under the assumption that labelled positives are selected at random from all positives, a classifier trained against the unlabelled pool can be corrected by an estimated label frequency. This provides the appropriate treatment for the present data and is applied in this module as a robustness analysis.

Because a risk score intended for a small-business owner must be actionable, model interpretability is a functional requirement rather than a secondary concern. Lundberg and Lee [17] introduce SHAP, a unified additive attribution framework grounded in cooperative game theory that assigns each feature a contribution to an individual prediction and satisfies local accuracy and consistency. For linear models these attributions have an exact closed form, which makes per-business explanation inexpensive and removes any sampling approximation.

Comparing two models on the same respondents requires a test that accounts for the correlation between their predictions. DeLong, DeLong and Clarke-Pearson [18] derive a non-parametric procedure for comparing correlated receiver operating characteristic curves, and Sun and Xu [19] give a computationally efficient formulation of the same statistic. This is the correct inferential instrument for the hypothesis tested in this module, where a demographic model and an information-augmented model score identical businesses.

Finally, the literature on model stability in small samples constrains the design. Peduzzi et al. [20] demonstrate that logistic regression coefficients become unstable when the number of outcome events per predictor variable falls too low, which for a survey-scale dataset implies a deliberately restricted feature set and a preference for regularised linear models over higher-capacity alternatives. This consideration, rather than any performance argument, governs the modelling choices described in Chapter 4.

The gap this module addresses is therefore specific. Compliance-risk prediction has been demonstrated where administrative data exists; information asymmetry has been argued theoretically; and the statistical machinery for testing an incremental contribution is well established. What has not been done is to measure whether information-access variables predict compliance failure beyond business characteristics for Sri Lankan SMEs, using data that can be collected without privileged access to revenue records.

2.3.4 Regulatory Misinformation Spread

The majority of the studies conducted on misinformation detection have centered on political and health-related and/or crisis-related misinformation, and many of the initial systems have posed the task as a binary classification problem, distinguishing between true and false. Previous research on fake news and misinformation classification tended to focus on the superficial text cues, propagation patterns and stance signals, but later studies demonstrated that misinformation is not always entirely false. Multi-class labelling is suited to applied fact-checking tasks as a result of the various forms of misinformation and related information disorders outlined by Wardle and Derakhshan (2017) such as misleading context, manipulation and false framing. This distinction is particularly important in the context of regulatory misinformation, because a post may contain accurate terminology but present an error in practical terms concerning the tax, labour, import or registration requirements.

The study of social media misinformation has also highlighted the challenge of dealing with such short, noisy and variable user-generated content. Both Castillo et al. (2011) and Shu et al. (2017) noted that the brevity and informality of posts on Twitter influence credibility assessment, and misinformation on social media is disseminated quickly via reposting, engagement, and social endorsement, respectively. Under these circumstances, posts can be code-mixed, shortened, or include dates and numerical references that are critical for understanding. Social media misinformation is especially difficult to navigate when there are regulatory claims because one number out of place or wrong date can render the entire post inaccurate.

Multilingual misinformation detection is becoming a more critical challenge given the fact that misinformation is not present in just one language even in multilingual countries and regions. However, research on multilingual NLP has demonstrated that the ability of transfer across language is challenging if a model is only trained in high resource languages, and that performance can be negatively impacted in low resource languages without multilingual representation learning. Conneau et al. (2020) proposed XLM-R, a multilingual transformer model that is pre-trained on large-scale cross-lingual data, which achieves good cross-lingual transfer for many downstream tasks. This is applicable to the current module as the data set contains posts in English, Sinhala and Tamil, and using a multilingual encoder is more appropriate than a monolingual encoder.

A second topic that is central in the literature is the preparation of datasets. The ability to detect misinformation requires rigorous selection of the data set, filtering of irrelevant data, de-duplication, and retention of metadata that could be useful for understanding it. Potthast et al. (2018) and Zubiaga et al. (2018) both pointed out the importance of robust corpus construction methods for misinformation corpora, since the lack of quality corpus construction may lead to the inclusion of noise that may harm the models that follow. Consequently, in applied domains, the process of preparing datasets is not simply a preprocessing step, it is an integral part of the research design, as the quality of the corpus will not only influence the ability of the model to learn meaningful distinctions, but also influence the kind of artefacts that it will learn. The present module is based on the same approach, by creating a hand-collected corpus of social media and fact-checking sources out of that, and cleaning and structuring it prior to annotation.

When annotators have a higher degree of confidence in one language than another, translation is often incorporated into the multilingual annotation workflow to increase consistency. For multilingual NLP, the use of translation-assisted annotation to address ambiguities and fine-tune consistency of labels across languages, particularly for low-resource languages, is applied. Neural machine translation systems like NLLB-200 (Costa-jussà et al., 2022) can be a practical solution to such workflows, while not losing the original text for subsequent modelling. In multilingual misinformation projects, this helps translate the information to support the original language and not replace it.

In the field of misinformation research, the reliability of the labels is a topic that has received a lot of discussion, with the quality of the annotation playing a significant role in the validity of the model. Cohen's Kappa is one of the most popular measures of agreement between two annotators, as it is easy to understand, appropriate for binary and multi-class settings, and corrects for chance agreement (Cohen, 1960). Within annotation studies, where it is hard to separate out the classes, agreement statistics can be used as evidence that the label scheme is sufficiently constant to be modelled. Particularly in the context of regulatory misinformation, it matters what is partially accurate, misleading, and inaccurate, the context of the post, and the specific language of the statement.

The transformer-based models are the most popular type of text classifiers used in recent misinformation detection studies because they are more adept than previous bag-of-words or shallow neural models in capturing contextual relationships. Devlin et al. (2019) showed that BERT is one of the most significant breakthroughs in the area of contextual language representation and subsequently multilingual variants were developed for addressing cross-lingual tasks. XLM-RoBERTa is particularly interesting for the multilingual misinformation detection task as it is pre-trained on various languages and hence is able to handle Sinhala, Tamil, and English without having to train language-specific models. Previous research has demonstrated that XLM-R is a very competitive base for multilingual classification tasks, especially in the case of transfer across scripts and across language varieties.

When classification is based on knowledge beyond the text, but not just style, then RAG proves to be a promising alternative. Lewis et al. (2020) formalized the RAG framework using retrieval + sequence generation, and demonstrated that external evidence retrieval can enhance knowledge-intensive NLP tasks. In regulatory or legal contexts, the solution may be found in an authoritative source and not in the formulation of the claim. RAG is thus a suitable method to achieve verification when the model needs to tie its output to a retrieved context, in particular when the regulation changes over time or when the post mentions a numeric threshold or legal date.

Given their general language ability, classification, extraction, and reasoning tasks are large language models that have been increasingly applied to Gemini. Recent studies on LLM have demonstrated that they can achieve good performance on zero shot and few shot tasks, however their results are still sensitive to prompt design and to grounding the prompt in the specific domain (Brown et al., 2020; OpenAI, 2023). But earlier studies also indicate that direct prompts may not be effective in cases where the task demands a factual basis for an answer or a more specific domain-specific legal interpretation. Therefore, there has been a recent trend in developing systems that incorporate retrieval alongside prompting in conjunction with an LLM. In the current module, the authors compare RAG with direct prompting with Gemini and with a fine-tuned multilingual transformer.

This module is narrower than previous modules in terms of topic but more specific with regard to design. It does not try to generally detect misinformation on the web, but it works with regulatory misinformation in Sri Lankan SMEs, where the need to identify factual and multilingual information is crucial. The module thus provides an applied comparison of the annotation, multilingual pre-processing, transformer classification, and retrieval-grounded verification in one domain-specific pipeline.

2.4 Summary

The reviewed literature supports four conclusions that shape this project. Compliance failure among SMEs is substantially an information failure, not primarily an intent failure [22]–[26]. Regulatory monitoring can be automated, but existing RegTech assumes institutional users and English structured feeds [11], [16], [20], [30]–[33]. Multilingual transformers are the appropriate modelling family for Sinhala and Tamil regulatory text [5], [19], [47]–[49]. Grounded generation, explainable risk modelling and misinformation detection each have mature methods and evaluation frameworks [17], [18], [46], [52]–[55], [12], [15], [36]–[42], [13], [14], [34], [35], [43], [45].

What is absent is integration and measurement. Enigmatrix combines awareness, knowledge, risk and verification over a shared verified data layer, and Module 1 supplies both that layer and the instrumentation needed to quantify whether the information barrier has actually narrowed.

Chapter 3 - Technology Adapted

3.1 Introduction

This chapter records the technologies selected for the platform and the reasoning behind each choice. Selections were driven by four constraints: the source documents are multilingual and frequently scanned; the project must run on student-accessible hardware and free or academic-tier services; every result must be reproducible for examination; and each module must remain operable by a single developer.

Table 3.1 — Technology stack by layer

Layer	Technology
Backend	FastAPI, SQLAlchemy 2.0 async (asyncpg), Alembic, Pydantic v2, Celery + Beat (Redis), slowapi, passlib/bcrypt, python-jose
Scraping	Scrapy — gazette, weekly-gazette, acts and bills spiders (Module 1); per-authority procedure scrapers (Module 2)
Extraction	PyMuPDF, pdfplumber, pypdfium2, Tesseract 5 (eng+sin+tam), Surya OCR, font-aware Wijesekara→Unicode conversion
Classification (M1)	XLM-RoBERTa base, PEFT (LoRA), PyTorch, Transformers, ONNX Runtime (INT8), fastText language ID
Retrieval + generation (M2)	ChromaDB, LangChain recursive splitter, sentence-transformers paraphrase-multilingual-mpnet-base-v2, Llama-3.1-8B-Instruct + QLoRA adapter, gradio_client, Hugging Face Spaces (ZeroGPU)
Risk modelling (M3)	scikit-learn (logistic regression), joblib, NumPy, pandas, closed-form linear SHAP, DeLong correlated-ROC test, Elkan-Noto PU learning
Verification (M4)	httpx proxy to the Module 2 fact-check contract; verdict parsed from a fixed first line
Translation	NLLB-200 distilled 600M with a figure-masking pipeline (mask → split → translate → restore → verify)
Frontend	Next.js 14 App Router, React 18, Tailwind with shadcn-pattern HSL tokens, next-intl (EN/SI/TA), TanStack Query v5, recharts, Playwright, Vitest
Storage	PostgreSQL 15+, ChromaDB, Redis, local object storage
Annotation	Label Studio

Compute	Google Colab GPU, Kaggle Notebooks GPU, Hugging Face ZeroGPU, local CPU workstation
Deployment	Render / Railway (full backend), Vercel (frontend and a slim API build), managed PostgreSQL, Railway (Module 2 retrieval sidecar)
Quality	pytest, testcontainers, Playwright, ruff, Great-Expectations-style JSON suites

3.2 Platform Architecture and Repository Layout

The system is a monorepo with git submodules:

Table 3.2 — Repository components

Component	Purpose
enigmatrix-backend	FastAPI, SQLAlchemy 2.0 async, Alembic, Pydantic v2, Celery + Beat, Scrapy spiders. Owns the PostgreSQL schema and every migration. Hosts the per-module packages app/m1, app/m2 and app/m3.
enigmatrix-frontend	Next.js 14 App Router, Tailwind with shadcn-pattern HSL tokens, next-intl (EN/SI/TA), TanStack Query, Playwright. The single user interface for all four modules.
enigmatrix-ml	Python packages m1–m4 — extraction, preprocessing, evaluation, model training and the research notebooks that produce the thesis figures.
enigmatrix-docs	MkDocs documentation set
enigmatrix-infrastructure	Infrastructure configuration and local orchestration
research/data	Label Studio configuration, calibration set, annotation batches
scripts	Cross-repository scripts (sampling, thesis artefact regeneration, translation helpers)

Placing extraction and evaluation in enigmatrix-ml rather than in the backend was a deliberate decision: the ML package is pip-installable and backend-independent, so it can be executed in Google Colab or a notebook without importing the web application. The backend app/extraction module is a thin re-export adapter that preserves existing imports and tests.

3.3 Backend and Asynchronous Processing

FastAPI with SQLAlchemy 2.0 async was selected over Django or Flask for three reasons: native asynchronous request handling suits an I/O-bound workload dominated by scraping and database access; Pydantic v2 gives schema validation and OpenAPI documentation without additional code; and the async ORM allows long extraction transactions without blocking the request loop.

Celery with Redis and Beat provides the task layer. Extraction of a single scanned gazette can take minutes; performing that in a request would be untenable. Celery also supplies the scheduled cadence the research programme requires:

Table 3.3 — Scheduled task cadence

Task	Schedule
Scraper	every 6 hours
Portal watcher / RSS watcher	every 2 hours (offset)
Retire old dataset versions	20:30 UTC daily
Refresh lag analytics	21:00 UTC daily
Retraining	quarterly (1 Jan / Apr / Jul / Oct, 03:00)

Alembic maintains a linear migration chain. slowapi enforces rate limits. Audit logging is written on every authentication event and every administrative mutation through a single audit_service.record() entry point that is never bypassed - a requirement for a system whose outputs may be cited as compliance evidence.

3.4 Scraping

Scrapy was chosen over ad-hoc requests scripts for its built-in scheduling, retry, throttling and pipeline abstractions. Four spiders cover the source surface: gazette_spider, weekly_gazette_spider, acts_spider and bills_spider, targeting gazette.lk and documents.gov.lk. Each supports date scoping, closes on scope exhaustion, and falls back English→Sinhala→Tamil when a language edition is unavailable. Completeness verification and re-fetch endpoints reconcile what was expected against what was retrieved.

3.5 Extraction and OCR

No single extraction engine handles the Sri Lankan gazette corpus. Documents fall into three classes, and pages within a single document may differ:

Table 3.4 — Extraction engine routing

Class	Characteristics	Engine route
Text PDF	Born-digital, selectable text, usually English	PyMuPDF with TEXTFLAGS_TEXT, then pdfplumber for layout and tables
Hybrid	Mixed selectable and image pages	Per-page routing between text and OCR engines
Scanned	Image-only, frequently Sinhala or Tamil	Tesseract 5 --oem 1 --psm 6 -l eng+sin+tam at 300 dpi; Surya OCR as fallback profile

A classify_pdf step assigns the route using calibratable thresholds. Extraction profiles are registered and versioned-legacy_v1, page_routing_v1, surya_fallback_v1, wijesekara_routing_v1 - so that any measurement run can be attributed to a specific extraction configuration.

Wijesekara conversion. Legacy Sinhala documents encode text in the Wijesekara keyboard layout with non-Unicode fonts. Extracted bytes are meaningless without conversion, and the failure is silent: text appears to extract successfully but is unusable. A font-aware converter detects the encoding from embedded font metadata and maps to Unicode. Residual (cid:…) glyph spans are a known extraction risk and are logged for re-extraction.

Character Error Rate is computed by a dedicated calculator so extraction quality is measured, not assumed.

3.6 Language Identification and Preprocessing

fastText performs language identification. It was preferred over langdetect for short-text stability and for its handling of Sinhala and Tamil scripts. Preprocessing then performs Unicode normalisation and whitespace collapse, metadata extraction (gazette number, publication and effective dates, penalties including the multi-penalty case), chunking to the model's 512-token window, and sub-document splitting where a single gazette issue contains multiple independent instruments.

Enigmatrix is a modular web platform composed of a FastAPI backend, a Next.js frontend, relational and vector storage, an asynchronous task layer and a Python machine-learning package. SMEs register with a business profile — sector, region, scale and language preference. Regulatory information is collected from official sources, extracted into structured records, classified by change category and affected sector, summarised into plain language, translated where required, and delivered through dashboards, alerts, search and advisory workflows. Administrative and expert users verify, correct, annotate and measure the pipeline through a dedicated admin surface.

The design principle is that regulatory information is structured once, into a verified regulatory intelligence store, and then reused by every downstream module. Module 1 produces that store; Modules 2, 3 and 4 consume it.

3.7 Classification Model

XLM-RoBERTa base with LoRA adapters. XLM-R [19] covers English, Sinhala and Tamil in one encoder, which is decisive when labelled data per language is limited. Full fine-tuning of a 270M-parameter encoder is impractical on the available hardware; LoRA [low-rank adaptation] injects trainable rank-decomposition matrices into the attention query and value projections and freezes the base weights, reducing trainable parameters by orders of magnitude and making training feasible on a single Colab GPU.

The head is dual, matching the label structure:

Table 3.5 — Classification heads, tasks and loss functions

Head	Task	Loss
Category head	Single-label over 8 categories	Cross-entropy
Sector head	Multi-label over 3 sectors	Binary cross-entropy

Total loss is the sum, with a configurable sector_loss_weight. **ONNX Runtime** serves inference on CPU with an optional INT8 quantisation, removing the need for a GPU in production and keeping hosting within free and hobby tiers.

Baselines. TF-IDF with logistic regression and with LinearSVC establish the non-transformer reference point. Reporting them is a methodological requirement: without a baseline, a transformer result has no interpretable scale.

3.8 Annotation Tooling

Label Studio hosts the annotation project. Its XML configuration encodes 8 domains, 3 sectors, an SME-relevance boolean, an annotator confidence rating and a free-text rationale field. A 20-document trilingual calibration set with expert labels and written rationales is used to align annotators before production labelling. Sampling for annotation uses stratified and k-means samplers with minority-domain targeting and a hybrid active-learning mode, so that annotation effort is directed at the documents that most improve the model rather than at whatever arrives first.

3.9 Retrieval-Augmented Guidance

Module 1 answers what changed. Module 2 answers how to comply — which form, which office, which deadline, which fields — for 20 regulatory domains grouped under the same 8 categories Module 1 classifies into. The technology choice that shapes everything else is that facts are retrieved at query time and never stored in model weights.

The reasoning is domain-specific rather than architectural fashion. Regulatory figures change: VAT rates, registration thresholds, gazette amendments. A figure baked into weights goes stale silently and cannot be cited. Compliance guidance also carries real liability, so every figure must be traceable to an official source. Retrieval-augmented generation satisfies both; fine-tuning is used only to teach the model structure, completeness and refusal discipline. That split maps exactly onto the module's research claim, in which the retrieval layer supplies declarative content and the fine-tuned behaviour supplies procedural shape.

Retrieval

Embeddings use sentence-transformers/paraphrase-multilingual-mpnet-base-v2. A multilingual sentence encoder was required rather than preferred: a Sinhala or Tamil question must retrieve rules held in English, which is only possible when questions and documents share one embedding space. Text is chunked with the LangChain recursive splitter at 400 characters with 60 characters of overlap — small chunks because a procedural answer depends on retrieving a specific step rather than a broad passage.

ChromaDB stores one collection per regulatory domain, with each chunk carrying its domain, category and applicable sectors as metadata. Retrieval operates at three scopes: a single domain, a category (pooling across its member domains), or across all domains when the question is unscoped. Chroma was chosen over a managed vector service for cost and reproducibility, and because an embedded store can be built at container-image build time, so a deployed instance never has to re-embed the corpus at boot.

Generation

The generator is Llama-3.1-8B-Instruct with a QLoRA adapter. The base model is quantised to 4-bit NF4 and frozen; low-rank matrices of rank 16 ($\alpha = 32$, dropout 0.05) are injected into all seven projection matrices, leaving 41.9 million of 8.07 billion parameters trainable — 0.52 per cent. Training ran 75 steps over three epochs in approximately sixteen minutes on a single A100 in bf16, with an effective batch size of 16. The published adapter is the epoch-2 checkpoint rather than the final one, selected automatically on validation loss, which rose at epoch 3.

Serving uses a Hugging Face Space on ZeroGPU, reached with gradio_client. ZeroGPU allocates a GPU per decorated call rather than holding one, which is what makes an 8-billion-parameter model affordable for a student project. It also imposes a constraint that shaped the code: there is no GPU at boot, so the adapter must be loaded onto CPU explicitly and moved by the framework on the first GPU call.

The prompt as a research instrument

The prompt is treated as part of the method, not as user-interface text. It fixes the composition order, states a non-advice framing (the system reports what published regulations say and does not advise), permits exactly one refusal sentence and no other, forbids the model from naming internal routing codes in prose, and imposes a six-point procedural contract requiring a complete numbered procedure, the exact form, office and deadline per step, a plain-language note on every named form, a worked example, and figures drawn only from the retrieved context. Because the evaluation harness scores each of those requirements as a separate column, re-wording the prompt changes the instrument, and the merged codebase therefore holds it verbatim under version control.

Refusal is handled at three layers because no single layer was sufficient. The base instruction-tuned model refused legitimate questions as financial advice, which destroyed answer coverage in Sinhala and Tamil in particular. The prompt frames the task; the fine-tune trains refusal discipline; and a post-processing regular expression detects a residual refusal and triggers exactly one bounded retry, keeping the retry only if it no longer refuses. A second post-processing pass rewrites internal identifiers that leaked into prose.

Evaluation design

Three runs are compared, not two. Going directly from the previously deployed model to the fine-tuned one moves two variables at once — different base weights and a new adapter — so a stock Llama-3.1-8B-Instruct control is evaluated under the identical prompt and identical retrieval. Only that comparison makes an improvement attributable to the adapter. Grading is by regular expression across ten columns, which is a disclosed limitation rather than a claim of human-level assessment.

3.10 Risk Modelling

Module 3 estimates the probability that a business will fail a compliance obligation within twelve months and, more importantly, states why. Its technology choices run deliberately against the direction of the rest of the platform, and the justification is worth stating plainly: the deployed model is a logistic regression.

Three constraints force that. The sample is 300 surveyed businesses with ten predictors, where a high-capacity model would fit noise. The research question is whether information barriers add explanatory power over demographics, which is a question about coefficients and therefore needs a model whose coefficients mean something. And the product requirement is an explanation an owner can act on, which is not a post-hoc rationalisation of a black box but the model itself.

That last point yields a concrete technical benefit. For a linear model, Shapley values have a closed form: the contribution of feature i is its coefficient multiplied by the deviation of that feature from its dataset mean, measured in log-odds. The attribution is therefore exact rather than sampled, costs nothing to compute at request time, and is reproducible to the last decimal. The shap package was consequently removed from the dependency set during integration — an approximation library is unnecessary when the quantity is available analytically.

Table 3.6 — The model ladder

Model	Feature set	Purpose
M0	Naïve cross-tabulation on sector	The obvious answer a practitioner would reach for, stated so it can be beaten
M1	Demographics only — sector, location, owner education, accountant use, business age, headcount	What a business is

M2	M1 plus the information block — awareness lag, language match, responsiveness, informal reliance	What the owner knows, and how they came to know it

The deployed artefact is model M2, serialised with joblib. It is a 2.9 KB file carrying the fitted estimator, the trained column order, the feature means used as the SHAP baseline, the mapping from encoded column back to source feature, the block each feature belongs to, and the cross-validated metrics. Storing the baseline and the column order inside the artefact rather than recomputing them at load time is what makes an explanation reproducible across deployments.

Statistical comparison between rungs uses the DeLong test for two correlated ROC curves, computed on cross-validated out-of-fold predictions. Correlated is the operative word: the two models score the same businesses, so an unpaired test would be invalid. Robustness uses Elkan-Noto positive-unlabelled learning, chosen because the outcome label is asymmetric by construction — a positive is a confirmed self-reported failure, but a negative means only that no failure was reported, and an unlabelled case is not a confirmed compliant one. Treating unlabelled data as negative would bias any standard classifier, so the robustness check asks whether the result survives when that assumption is dropped.

Explanations are template-based and multilingual rather than generated. This is the opposite of Module 2's choice and for the same underlying reason: a risk explanation must be deterministic and incapable of inventing a compliance fact, whereas a procedural answer benefits from fluent generation over retrieved and cited text. Because the model is small and CPU-only, Module 3 runs in-process inside the backend with no inference service, no GPU and a scoring latency dominated by the database round-trip.

3.11 Misinformation Verification

Module 4 checks a claim a business owner has encountered — typically forwarded through a messaging application — against the verified corpus. Architecturally it is a consumer rather than a producer: it calls Module 2's fact-check endpoint over HTTP through an httpx client in the backend and maps the response into the platform's own schema.

The interface is deliberately rigid. The verifying model must emit its verdict — Accurate, Outdated, Misleading or Unverifiable — as the first line of its response, followed by reasoning and the source. Module 4 parses that line, which makes it a hard contract rather than a formatting preference: relaxing the format silently breaks the consumer. Integration preserved this contract untouched, and the Module 2 service Module 4 depends on was explicitly excluded from the migration work so that a live dependency was never interrupted.

3.12 Summarization and Translation

Summaries are generated in a controlled form from the classified record — title, regulatory domain, affected sectors, amendment type and cleaned regulatory text — rather than by free generation over the raw document, because an unconstrained generator in a compliance context risks producing plausible but unsupported obligations.

NLLB-200 distilled 600M performs English→Sinhala and English→Tamil translation of titles and summaries. It was chosen over commercial translation APIs for cost, reproducibility and offline operation, and over larger NLLB variants because the distilled model fits comfortably in Colab memory while retaining acceptable quality for short administrative text. All machine translations enter an administrative review queue before being surfaced to SMEs.

3.13 Frontend

Next.js 14 App Router with server components reduces client bundle size on the content-heavy regulation pages. **next-intl** provides English, Sinhala and Tamil localisation with script-appropriate fonts. **Tailwind with shadcn-pattern HSL tokens** gives a trust-oriented blue and amber palette in light and dark modes. **TanStack Query** manages server state and cache invalidation for the admin pipeline views; **WebSocket** and **SSE** channels carry live extraction progress. **Playwright** covers end-to-end flows.

3.14 Storage and Deployment

Table 3.7 — Storage and deployment layers

Layer	Technology	Rationale
Relational	PostgreSQL 15+ (asyncpg)	Referential integrity across the regulation status machine and across module boundaries; materialised views for lag analytics; JSONB where a schema is genuinely open-ended
Vector	ChromaDB	Retrieval for Module 2; embedded deployment avoids a managed vector service and allows vectors to be baked at image build time
Cache / broker	Redis	Celery broker and result backend
Object storage	Local storage volume	Raw PDFs and model artefacts
Backend hosting	Render / Railway (uvicorn + Celery worker + Beat)	Cost-appropriate for the project; documented limitation under load
Serverless API	Vercel (slim build)	Stateless read paths; excludes the scientific and scraping dependency sets to stay inside the function size limit

M2 retrieval sidecar	Railway	Holds ChromaDB, the sentence encoder and the Space client — approximately two gigabytes of dependencies that cannot share the API deployment
Frontend hosting	Vercel	Native Next.js deployment target

Data quality is enforced by Great-Expectations-style JSON suites in enigmatrix-backend/data_quality/, validated automatically after each dataset version is sealed.

3.15 Development and Reproducibility Tooling

uv manages Python environments and dependency groups (serving, training, research), enabling a workspace layout in which the backend and ML package share resolution. pytest covers backend and ML unit tests; pnpm and Playwright cover the frontend. Model training is executed in **Google Colab** with a GPU runtime because no CUDA device is available locally; the local machine is used only for smoke testing, and Chapter 6 reports both configurations explicitly.

3.16 Summary

Each selection follows from a constraint of the problem domain rather than from familiarity. Multi-engine extraction exists because the corpus is heterogeneous. Wijesekara conversion exists because the corpus is partly legacy-encoded. A single multilingual encoder with LoRA exists because labelled data and GPU hours are both scarce. ONNX CPU inference exists because production hosting has no GPU. Sealed dataset versions and an atomic evaluation fact table exist because the results must be defensible under examination.

Chapter 4 - Approach

4.1 Introduction

This chapter states, for each module, what it consumes, what it does and what it produces. The platform is a chain: Module 1 turns unstructured official publications into verified structured records, and Modules 2, 3 and 4 each transform those records, joined with SME context, into a different decision-support output. Describing the modules in input–process–output terms makes the dependency explicit and defines the contract each module must honor.

4.2 Regulatory Change Awareness Gap

4.2.1 Input

Table 4.1 — Module 1 inputs

Input	Source	Form
Gazette issues, acts and bills	gazette.lk, documents.gov.lk	PDF; born-digital English, scanned Sinhala and Tamil, some legacy-font encoded
Publication metadata	Source listing pages	Gazette number, document number, publication date, source URL, language edition
SME business profiles	Platform registration	Sector, region, scale, preferred language, alert channel preferences
Secondary-channel content	IRD, EPF, ETF and eROC portals; five news RSS feeds	HTML and RSS items used only for propagation measurement
Annotation ground truth	Label Studio	Dual annotations of change category, affected sectors, SME relevance, confidence, notes
Sealed evaluation baseline	Curated ground-truth workbook	~800 manually curated field-level records, checksummed

4.2.2 Process

The module executes a five-stage pipeline with a parallel measurement loop.

1. **Ingest.** Date-scoped Scrapy spiders discover and download issues, falling back across language editions, and write a record at status `ingested` with the raw PDF stored on disk. Completeness verification reconciles the expected issue list against what was retrieved and re-fetches gaps.

2. **Extract.** classify_pdf routes each document — and, in hybrid documents, each page — to a text or OCR engine. Font metadata is inspected and legacy Wijesekara text is converted to Unicode. The record advances to extracted with raw_text, extraction_method, sha256, pdf_pages and language.
3. **Preprocess.** Text is cleaned and Unicode-normalised, the language is confirmed by fastText, metadata is extracted (gazette number, publication and effective dates, penalties including the multi-penalty case), sub-documents are split where one issue carries several independent instruments, and the text is chunked to the model's 512-token window as classification_chunk. The record advances to preprocessed.
4. **Classify.** The ONNX-exported XLM-R + LoRA model assigns one of eight change categories and a multi-label set of affected sectors, with a confidence score. Predictions below the 0.55 threshold are routed to expert review; records already marked expert-verified are never overwritten. The record advances to classified.
5. **Explain and alert.** A controlled English summary is generated from the title, domain, affected sectors, amendment type and cleaned text; the title and summary are translated into Sinhala and Tamil by NLLB-200 and queued for administrative review. The alert dispatcher matches the classified record against SME profiles by sector and emits idempotent in-app, email and SMS alerts.

In parallel, portal and RSS watchers poll secondary channels every two hours and attempt to match observed content back to a known regulation, first on exact gazette number and then on fuzzy title similarity, writing a propagation event on first observation. Nightly jobs refresh the lag-analytics materialised views, and a Kullback–Leibler drift monitor over the confidence distribution triggers retraining when it exceeds 0.15.

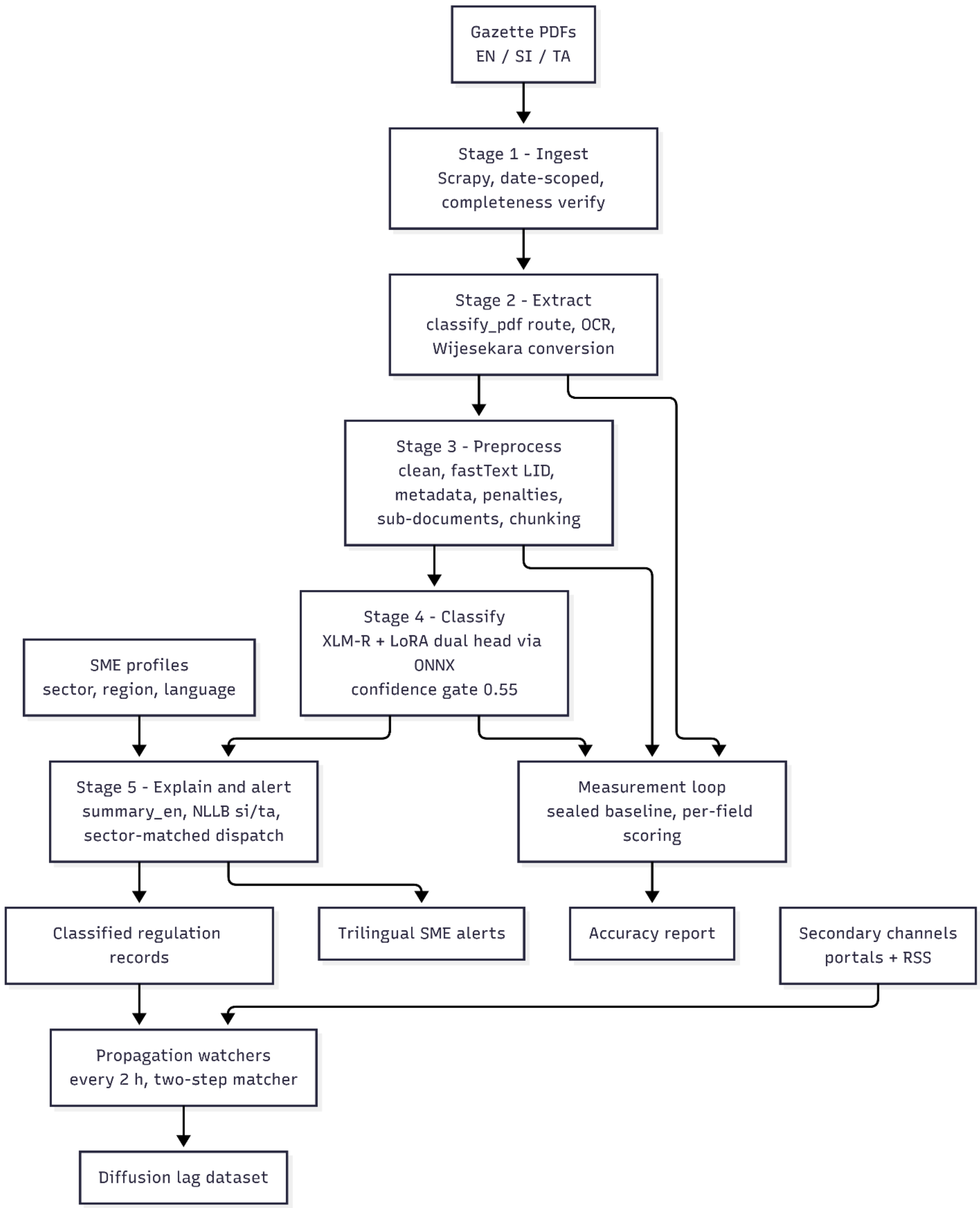


Figure 2 Module 1 input, process and output

4.2.3 Output

Table 4.2 — Module 1 outputs

Output	Consumer	Description
Verified regulation records	Modules 2, 3 and 4; SME dashboard	Structured rows with category, affected sectors, SME relevance, confidence, dates, penalties and full text
Trilingual titles and summaries	SME alerts and dashboard	title_en/si/ta, summary_en/si/ta, reviewed before publication
Sector-matched alerts	SME users	Idempotent in-app, email and SMS notifications keyed on regulation × recipient × channel
Propagation events	Research analysis	First-observation timestamps per regulation per secondary channel
Lag analytics views	Findings notebooks	v_m1_regulation_lag_summary, v_m1_channel_effectiveness
Accuracy reports	Examiners and maintainers	Per-field, per-record and per-stage scores with error taxonomy and worst-N leaderboard
Labelled gold dataset	Model training and evaluation	800 reconciled records with adjudication provenance

4.3 Compliance Guidance Platform

The module transforms a natural-language compliance question into a procedurally complete, source-grounded answer in the user's chosen language. Its pipeline covers language handling, retrieval, prompt construction, generation by a fine-tuned language model, and post-processing to guarantee grounding and to remove internal artefacts.

4.3.1 Input

The primary input is a compliance question posed in English, Sinhala or Tamil, optionally scoped to one of eight regulatory categories. A second input path is the awareness survey, which takes a respondent's business profile (sector, employee band and related attributes) together with their multiple-choice and open-response answers. All regulatory content is supplied to the system as curated text files compiled exclusively from official Sri Lankan government sources.

4.3.2 Process

A non-English question is first translated into English so that retrieval operates in the language of the corpus. The retriever embeds the question and selects the most relevant passages from the appropriate domain or category collection. These passages are assembled into a strict prompt that instructs the model to report only what the retrieved text says, to produce a complete numbered procedure giving the exact form, office and deadline for each step, to explain every named form in plain language, and to refuse with a single fixed sentence when the answer is not present. The fine-tuned Llama-3.1 model generates the answer; a post-processing stage then detects and repairs spurious safety refusals and rewrites any internal routing codes that leak into the text. For Sinhala and Tamil, the English answer is translated back through a figure-protecting pipeline that masks numbers, form codes, URLs and amounts before translation and restores them afterwards, falling back to English for any sentence in which a critical figure would otherwise be lost.

4.3.3 Output

The output is a grounded, procedurally complete answer in the requested language, with the responsible authority named and no figure present that does not appear in the retrieved source. For the survey the output is a scored report card that reports declarative and procedural knowledge separately for each applicable domain, and for the fact-check endpoint it is a verdict — Accurate, Outdated, Misleading or Unverifiable — accompanied by the supporting regulatory evidence.

4.4 SME Compliance Risk Prediction

This module takes a description of a small business, estimates the probability that it will experience a compliance failure, identifies the factors responsible for that estimate, and expresses the result as guidance the owner can act on. The processing chain is deliberately compact and fully interpretable, because the output is intended to change an owner's behaviour rather than to rank cases for an administrator.

4.4.1 Input

The primary input is a structured business profile obtained through a purpose-built survey instrument. Each response describes a business along two groups of variables. The first group is demographic and records the sector, the number of years in operation, the number of employees, the location, whether an accountant or bookkeeper is used, and the education level of the owner. The second group describes the information environment and records when the owner last learned of a regulatory change relative to its deadline, whether official guidance is normally received in the language the owner works in, how promptly the owner acts on notices, and whether the owner relies on formal or informal information channels.

The outcome variable is derived from three further questions covering the preceding twelve months: whether a filing deadline was missed, whether a penalty or warning notice was received, and whether legal or defaulter action was faced. A business is recorded as having experienced a compliance failure if any of the three applies. These three questions are excluded from the predictor set to prevent label leakage.

The dataset used comprises 300 responses collected from businesses in the grocery retail, food service and general retail sectors, of which 138, or 46 per cent, reported a compliance failure within the preceding twelve months. During operation of the deployed platform the same profile is supplied directly by the owner through a web form, and a second input stream is introduced in the form of regulatory change events published by the Regulatory Change Awareness module.

4.4.2 Process

Raw survey responses are first recoded into a canonical schema. Verbose answer text is mapped to short analytical codes, ordered variables such as awareness lag and language match are encoded as ordinal integers rather than as sets of indicator columns in order to conserve degrees of freedom, and the outcome label is constructed from the three disclosure questions. Unrecognised answer values are reported rather than silently converted to missing values, since an unnoticed missing value in the outcome variable is the most damaging failure mode in the pipeline.

Modelling proceeds as a ladder of three specifications fitted on identical respondents so that their differences are attributable to the features alone. The first, denoted M0, is a naive cross-tabulation rule over sector, business age and accountant use, included as a baseline so that the performance of the learned models can be interpreted against a trivial reference. The second, M1, uses the demographic block. The third, M2, adds the information-barrier block and is the specification of interest. The estimator throughout is regularised logistic regression, selected for coefficient stability at this sample size rather than for maximum accuracy, and performance is measured on cross-validated out-of-fold predictions because the sample does not support an independent hold-out partition.

The hypothesis is evaluated by comparing M2 with M1 using DeLong's test for correlated receiver operating characteristic curves. Feature attribution is computed using SHAP values, which for the linear specification are obtained in closed form and aggregated from encoded columns back to their source variables so that each variable appears once in the ranking. Two further analyses guard the conclusion: a positive-unlabelled learning correction that re-estimates the result without assuming that non-disclosure implies compliance, and a set of descriptive association tests using permutation-based significance and Cramér's V effect sizes so that the central relationships can be verified without reference to any model.

Within the deployed platform the same fitted model is applied to an individual profile. The risk probability, the signed factor contributions and the resulting guidance are generated for that business, and the profile is retained so that it can be matched against subsequent regulatory change events. Matching combines the sector of the business with the obligations implied by its sector and size, so that an alert is raised only where the change genuinely applies.

4.4.3 Output

For the research component the module produces a quantified answer to its hypothesis: a comparison of the three model specifications, a significance test for the incremental contribution of the information block, a ranked attribution of variables with each classified as demographic or informational, a robustness result under the positive-unlabelled correction, and an assessment of how far the sample resembles the national SME population.

For the deployed platform the module produces, for each registered business, a compliance risk probability and an accompanying risk band, an ordered set of factors raising and lowering that risk, and a plain-language explanation with recommended actions in English, Sinhala or Tamil. It further produces targeted regulatory alerts: when a change is published, every affected business receives a notification in its owner's language stating what has changed and what action is required. These outputs are what convert a statistical finding into an intervention against the barrier the research identified.

4.5 Regulatory Misinformation Spread

The module implemented covers all the steps from data collection to final classification. First, social media posts on Facebook, Reddit, Twitter/X, and fact-checking websites that mentioned the regulations were manually gathered. The corpus was centered on Sri Lankan regulatory information that made the corpus suitable for the desired application domain, SMEs.

The gathered posts were then pre-processed manually and structurally cleaned for the purposes of annotation and modelling. Duplicates were eliminated, language detection was carried out, information that is not relevant for the topic was removed, personally identifiable information was removed, and regulation domains were assigned. The data was cleaned and later structured into a master dataset and then a classifier dataset and finally a train/test split for the development of the model.

Support in the form of annotation was provided by means of translation only. The original posts were kept in the dataset but Sinhala and Tamil posts were translated into English with the help of NLLB-200 so that the annotators can label the posts uniformly. This allowed for the final corpus to be multilingual but maintain reliable decisions for annotation.

Annotation was first performed on 200 posts, which were independently annotated by two annotators in Label Studio. The Cohen's Kappa was then computed, which showed a very high level of agreement with overall veracity Kappa of 0.934. The agreement was great—one annotator finished the last 800 posts and created a final annotated dataset consisting 1,000 posts.

Then, the annotated corpus was loaded in Google Colab to delete unnecessary metadata columns, to create the dataset for the classifier and to divide it into a training set of 800 posts and a testing set of 200 posts. Three methods were then applied and tested: fine-tuned XLM-RoBERTa, RAG with ChromaDB and Module 2, and simple prompting of Gemini. The output of the module is a veracity prediction, which assists SME users in determining the veracity of a regulatory claim, either completely accurate or partially accurate, or harmful.

4.5.1 Input

The major input to the module is a social media post with a regulatory claim. Such posts may come from Facebook, Reddit, Twitter/X, or a fact-checking site and were pulled because they included information about the regulations in Sri Lanka which are applicable to SMEs.

English, Sinhala, Tamil and Code mixed texts are supported. Sinhala and Tamil content can be translated during the process of annotating content, however, the original language forms are preserved in the dataset for modelling and analysis.

The module also leverages on regulation categories that are used to structure the data, such as tax rate change, import and export regulation, regulation by sector, EPF and ETF change, labour law, product standards, business registration and enforcement of penalties. Besides, the training and testing data sets are also included in the input of the module in the modelling step, where 800 posts are used for training and 200 for testing.

4.5.2 Process

The internal workflow starts with a manual collection of posts from the platforms selected and a fact checking from a number of sources. Duplicates are removed during delete and posts that are not relevant to the regulatory misinformation task are removed. Then, language detection is performed for appropriately processing multilingual posts, and personally identifiable information is deleted for privacy protection.

Translation support is the next step. Sinhala and Tamil posts are translated into English during the annotation process, using NLLB-200 only for the annotation process, to reduce the impact on the consistency of the task as well as to retain posts in the dataset in their native language. After the translation, the posts are annotated by Label Studio. 200 posts are labelled independently by two annotators and Cohen's Kappa is used to assess the degree of agreement.

The agreement is tight and the completion of the rest of the 800 posts is done by one annotator. The annotated dataset is then cleaned to prepare it for the modelling: The unneeded metadata is removed and the classifier dataset is created. The labels are also reorganized during model development: the first four labels were simply merged into the 3-way classification space of the three approaches. This is because that this step should ensure that misleading and false are represented in the model comparison as harmful.

The training process is then dependent on the approach. The training set is used to fine-tune XLM-RoBERTa; RAG retrieves evidence from module 2 before making a prediction; the baseline directly classifies using Gemini. After making predictions for the held-out test set, the final classification is made by comparing predictions to the actual data and performance of each approach is evaluated in the same way.

4.5.3 Output

The main output of the module is a veracity class. In the implemented classifier workflow, the prediction is accurate, partly accurate or harmful.

In the retrieval-based approach, the output may also feature an explanation based on retrieved regulatory evidence. This will make the outcome useful, since it will provide some context to the user for why a claim was determined a certain way. The output will benefit SMEs in practice because it enables them to evaluate a claim before reacting to it and lessen their reliance on informal advice.

4.6 Summary

The four modules share one contract: Module 1 guarantees a verified, classified, sector-tagged and language-complete regulation record, and the other three modules build decision support on top of it without re-parsing source documents. This is what allows the platform to be evaluated module by module and still be defensible end to end — each module's output quality is bounded by, and measured against, the quality of the record it received.

Chapter 5 - Analysis and Design

5.1 Introduction

This chapter presents the design of the platform. Section 5.2 gives the overall architecture together with the data-flow, class, sequence and database designs. Section 5.3 gives the module-level designs, with Module 1 in full detail and structured placeholders for the other modules.

5.2 High-Level Architecture of the Overall System

The platform is designed as a four-layer architecture, with the four research modules occupying the application layer and communicating through a shared event bus and a shared verified knowledge base.

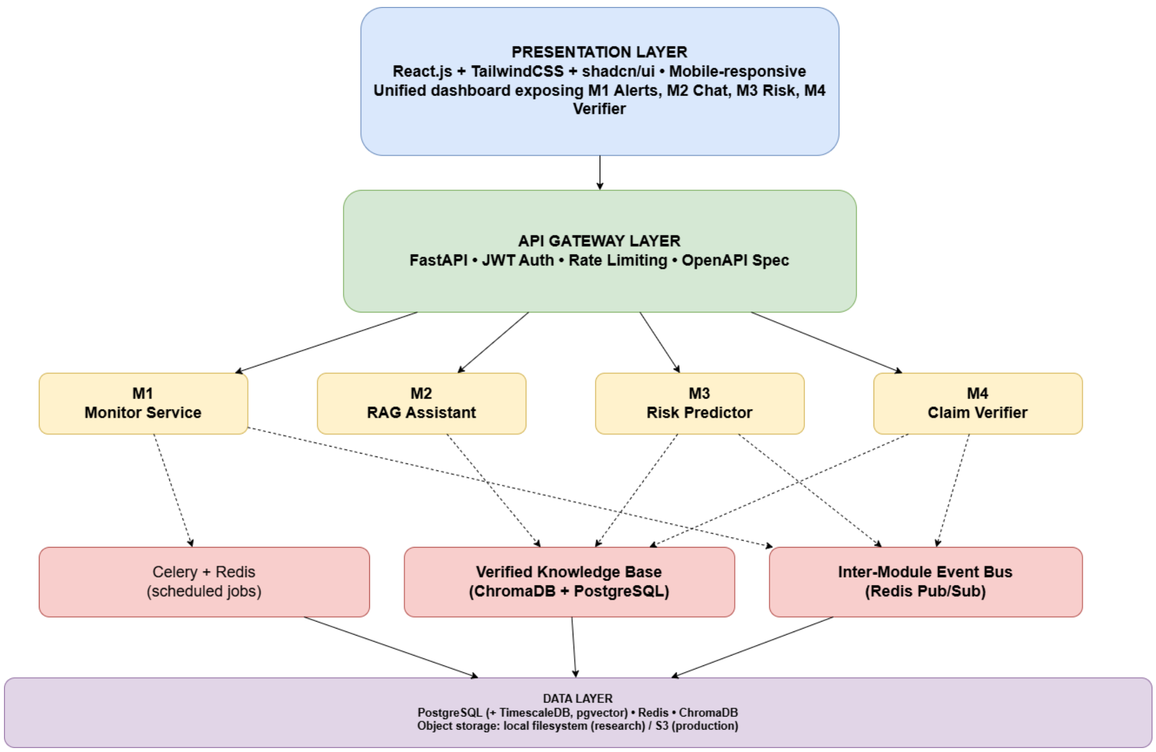


Figure 3 Four-layer high-level architecture of the Enigmatrix platform

The presentation layer is a single application rendering all four modules through a unified dashboard, so that the SME sees one product rather than four tools. The API gateway layer terminates authentication, role-based authorisation and rate limiting, and routes to module services. Each module is a separately deployable service; in the research prototype they share a process, but the boundaries are drawn so that they could be scaled independently. Three shared infrastructure components bind the modules: Celery for scheduled jobs, the verified knowledge base for ground truth, and the event bus for cross-module triggers. The data layer consolidates relational, vector and cache storage.

The layered view in Figure 5.1 is realised concretely as follows.

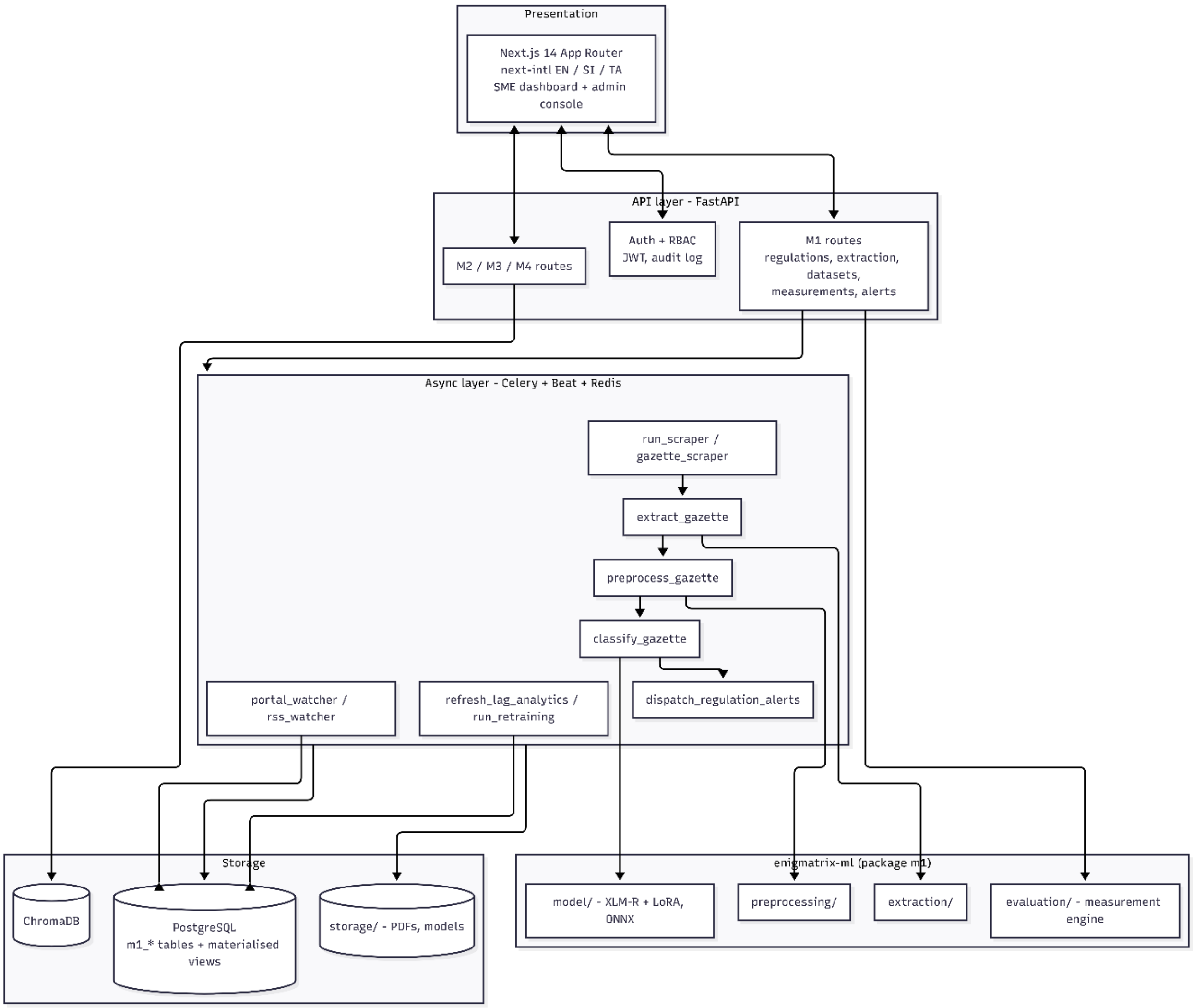


Figure 4 Deployment-level component view of the implemented platform

5.2.1 Data flow design

The platform's data flow is presented at two levels. The Level 0 context diagram shows the platform as a single process exchanging data with external actors; the Level 1 diagram decomposes it into the four modules with the verified knowledge base and the event bus as shared stores.

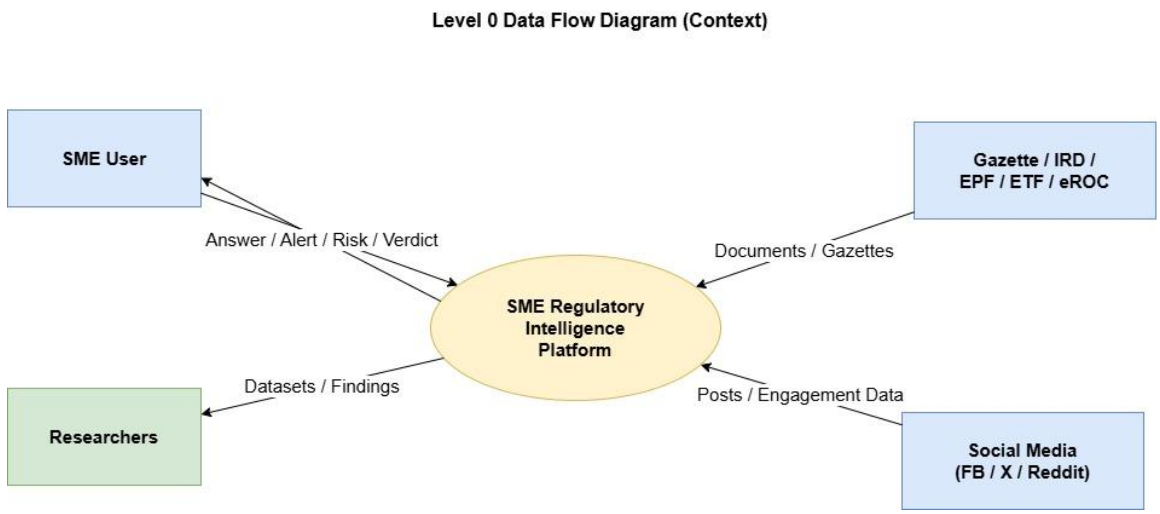


Figure 5 Level 0 (context) data flow diagram

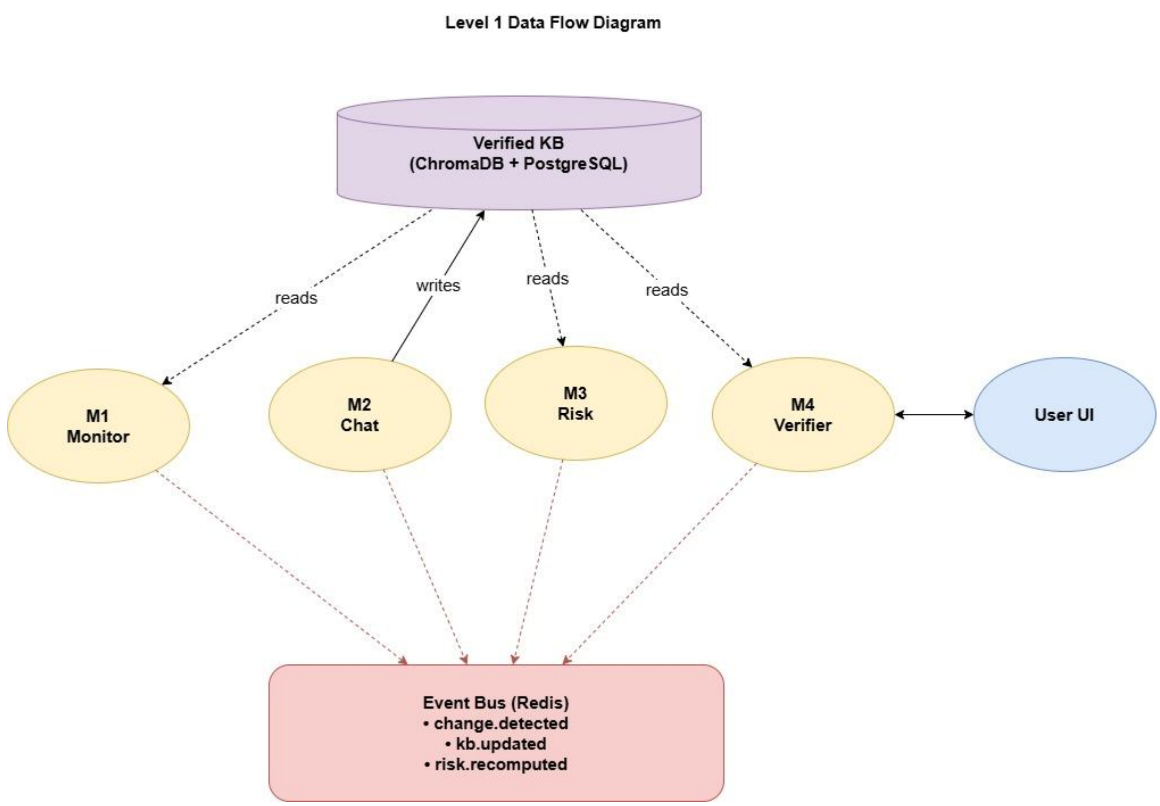


Figure 6 Level 1 data flow diagram

5.2.2 Domain model

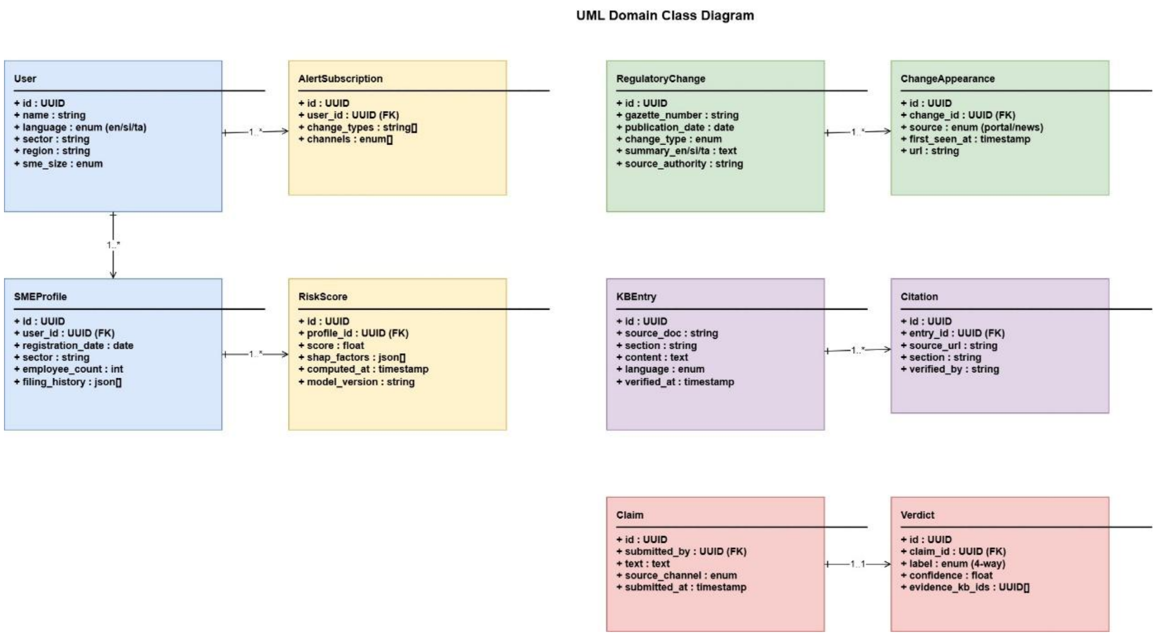


Figure 7 Domain class diagram

5.2.3 Representative interaction

The most representative cross-module interaction is a user submitting a compliance question, with a parallel risk recomputation triggered by an inbound regulatory change.

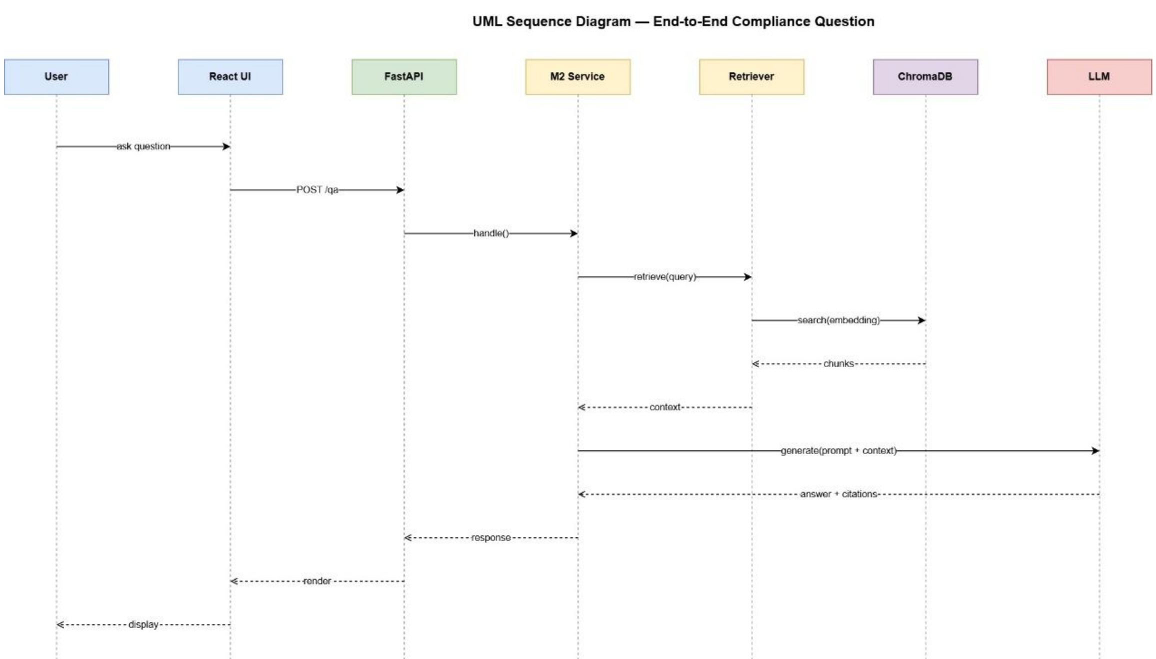


Figure 8 Sequence diagram for an end-to-end compliance question

5.2.4 Database design

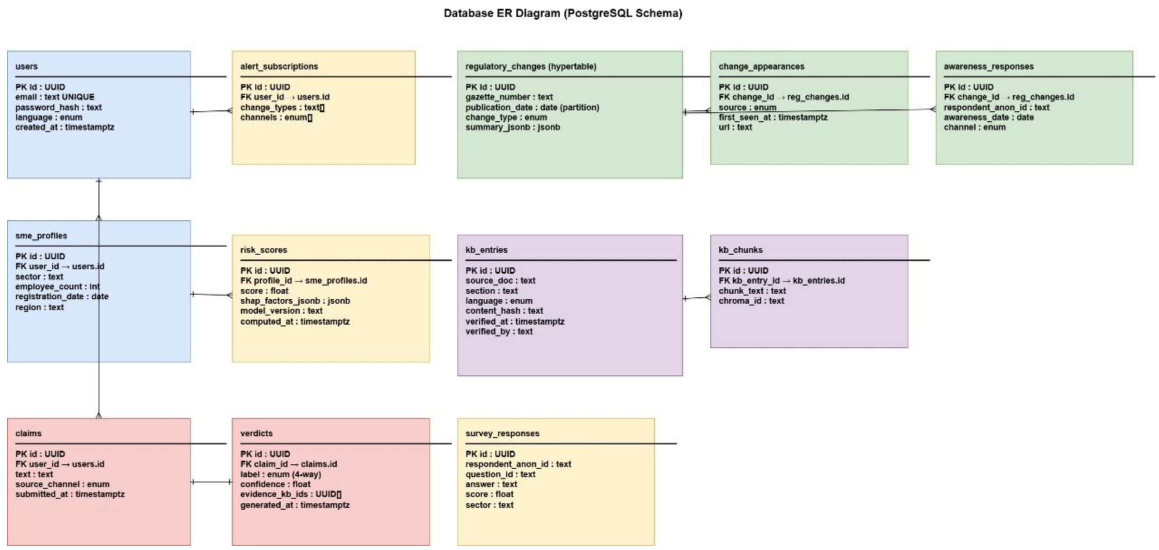


Figure 9 Entity relationship design of the shared database

The implemented Module 1 schema is summarised below. Platform tables shared with the other modules include users, sme_profiles, audit_log, the survey_* family, sectors and regulatory_domains.

Table 5.1 — Principal Module 1 database objects

Object	Purpose
m1_regulations	Core record with the status machine, extraction and classifier columns
m1_regulation_sectors	Many-to-many regulation ↔ sector relevance
m1_regulation_penalties	Extracted penalty structures, including the multi-penalty case
m1_sub_documents	Independent instruments split from a single gazette issue
m1_gazette_items	Item-level index of a gazette issue
m1_extraction_profiles / m1_extraction_runs	Versioned extractor configurations and their executions
m1_datasets / m1_dataset_versions / m1_dataset_rows	Dataset registry with immutable sealed versions and SHA-256 content hashes
m1_measurement_runs / m1_measurement_scores	Measurement executions and per-field scores
m1_propagation_events	Diffusion observations, unique on (regulation, source), with first_seen_at
m1_alerts	Dispatched alerts; sme_id IS NULL denotes a public broadcast

Object	Purpose
m1_retraining_runs	Retraining executions and promotion decisions
v_m1_regulation_lag_summary, v_m1_channel_effectiveness	Materialised views refreshed nightly for lag analytics

5.3 Module-wise Design

5.3.1 Regulatory Change Awareness Gap

Module 1 is structured as a five-stage Celery pipeline with a parallel measurement loop, as designed at interim stage and implemented since.

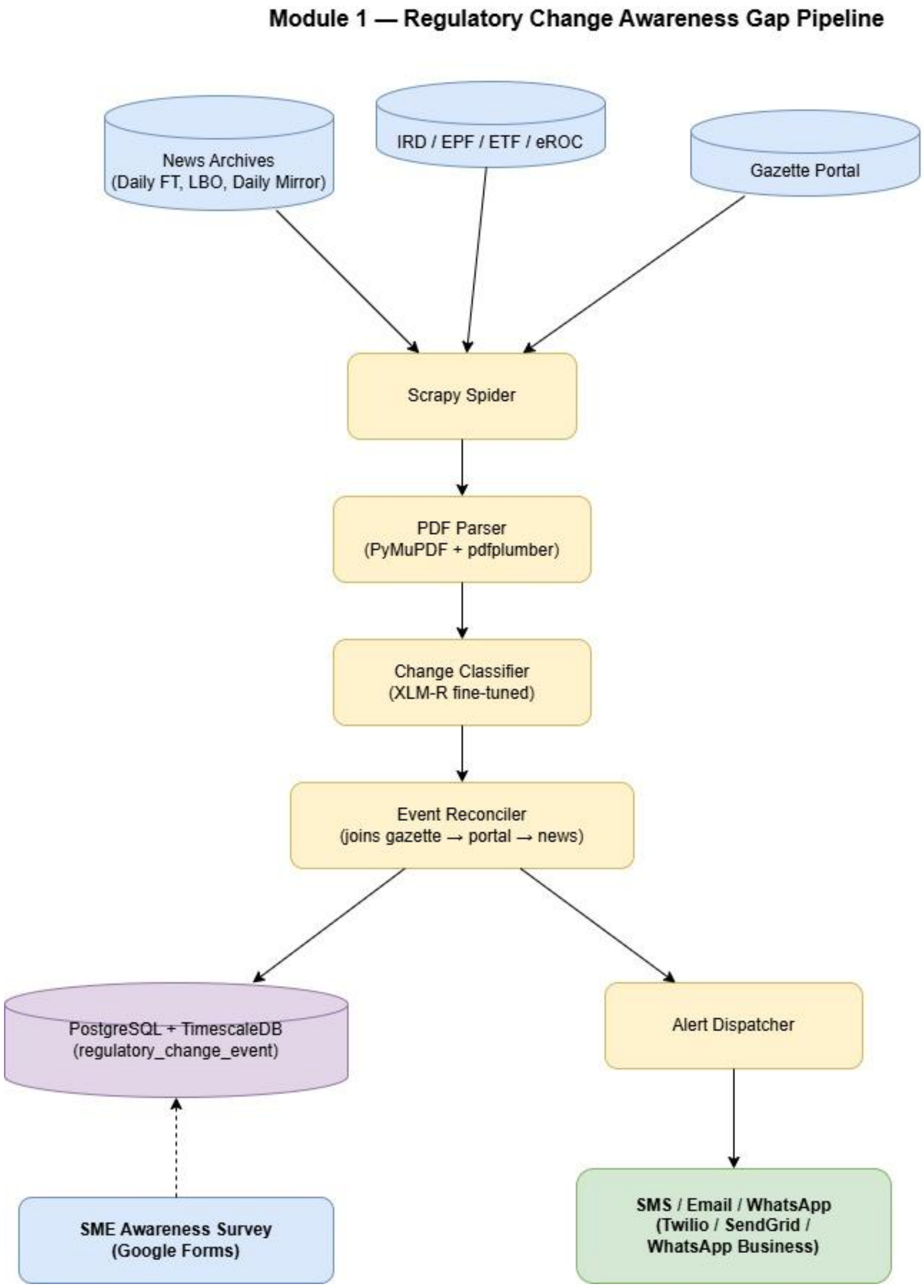


Figure 10 Module 1 pipeline design

The regulation status machine. Every record advances through a strict status machine. Status is a first-class evaluated field: a record that should have reached classified but stopped at preprocessed is a stage-progression failure, tracked separately from field accuracy.

Table 5.2 — Regulation status machine and the fields introduced at each stage

Status	Entered by	Fields introduced
ingested	Scrapy spider	m1_gazette_items, raw_pdf_path, gazette_number,

Status	Entered by	Fields introduced
		document_number, source_url
extracted	extract_gazette	raw_text, extraction_method, extracted_at, file_size_bytes, sha256, pdf_pages, language
preprocessed	preprocess_gazette	cleaned_text, classification_chunk, amendment_type, metadata_confidence, m1_sub_documents, m1_regulation_penalties
classified	classify_gazette	domain_code, change_category, severity_level, is_sme_relevant, classifier_confidence, classified_at

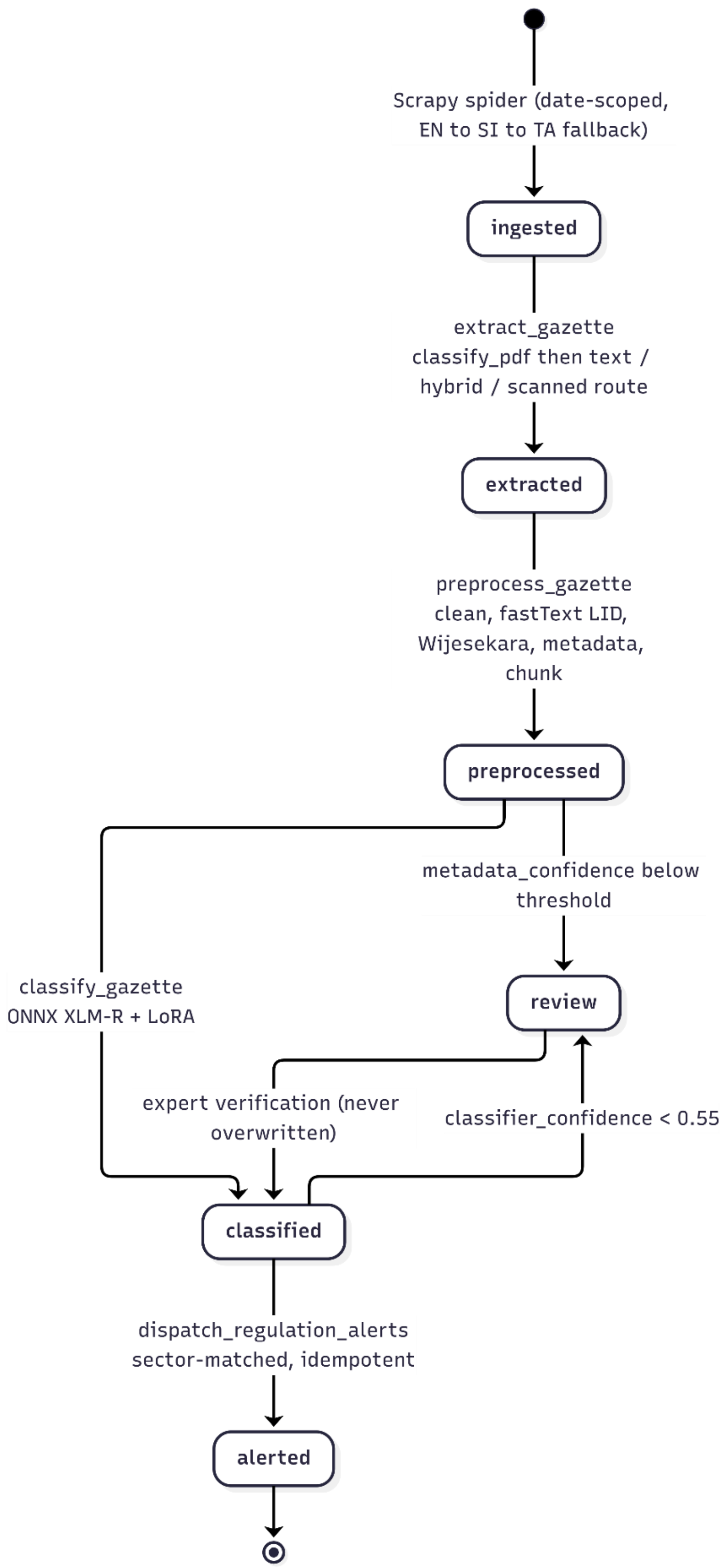


Figure 11 Regulation status machine with review routing

Extraction design. Engine choice must be made per page, not per document. classify_pdf inspects text yield and image coverage and assigns one of three routes; within the hybrid route each page is dispatched individually. Extraction profiles are registered entities, so a measurement result is always attributable to a named, versioned configuration.

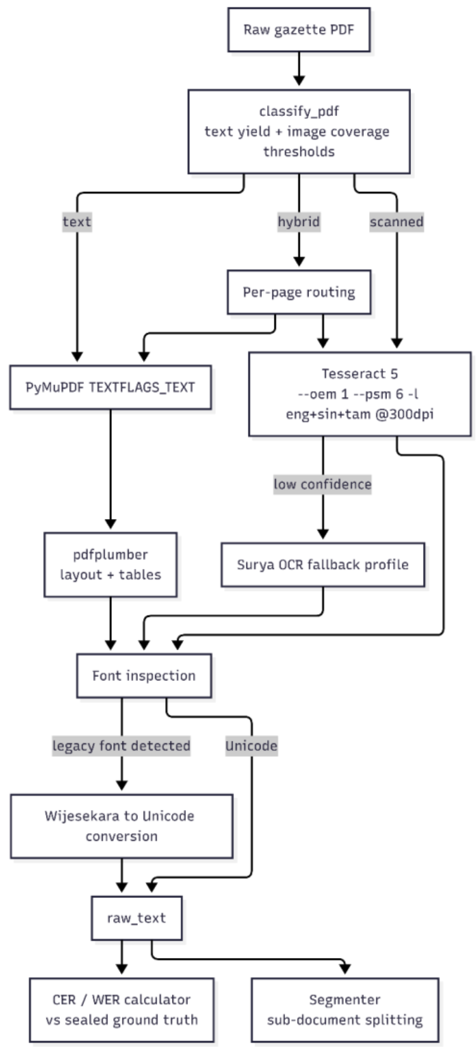


Figure 12 Extraction and OCR routing chain

Classification design. The classifier is a single XLM-RoBERTa encoder with LoRA adapters and two output heads.

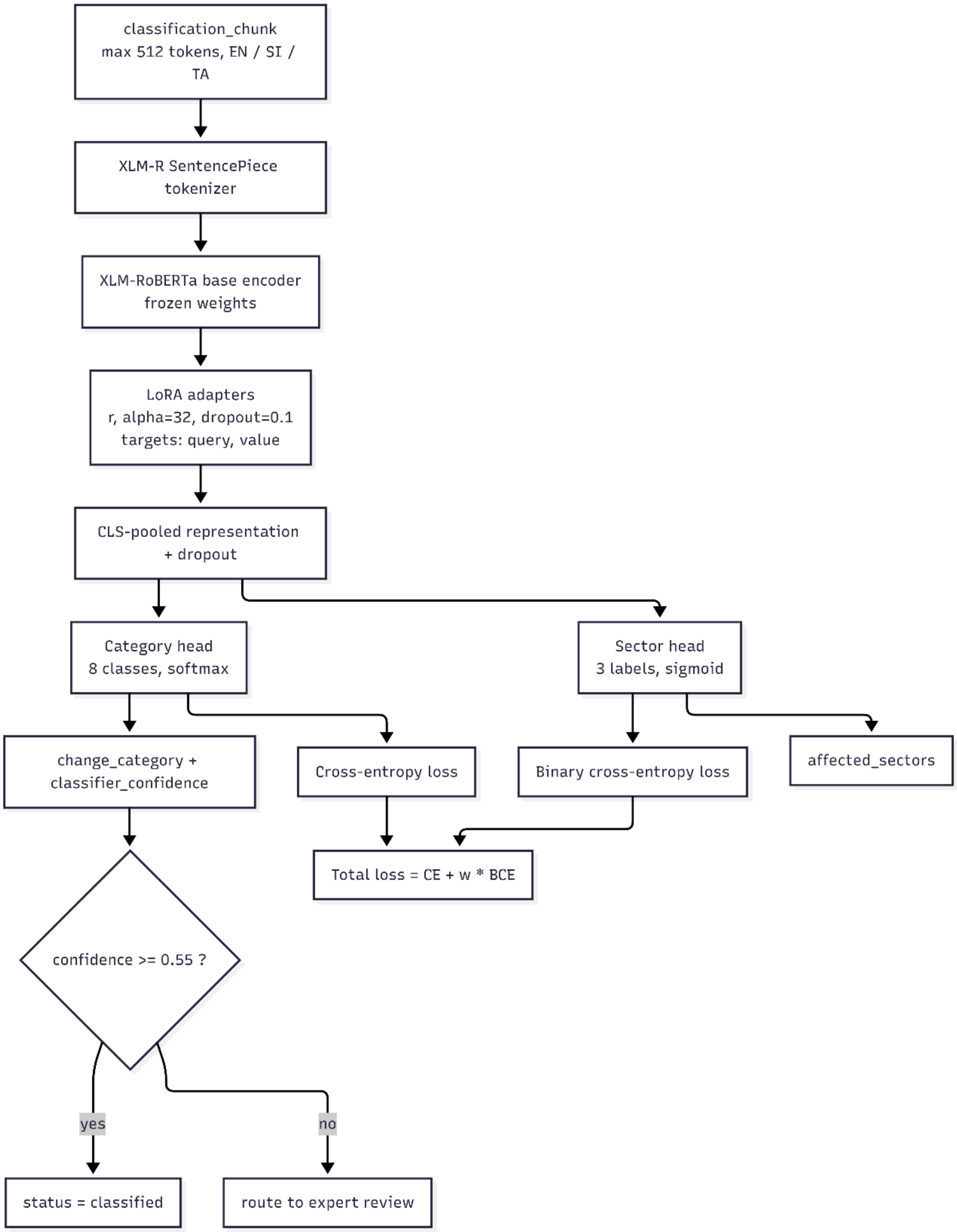


Figure 13 XLM-RoBERTa with LoRA adapters and a dual classification head

Table 5.3 — Classification design decisions and their rationale

Decision	Rationale
Single multilingual encoder	Labelled data per language is scarce; cross-lingual transfer is worth more than per-language specialisation at this scale
LoRA rather than full fine-tuning	Reduces trainable parameters so a single free-tier GPU session suffices; base weights stay frozen and reusable
Dual head, not two models	Category and sector share the same textual evidence; joint training regularises both and halves the deployment surface
Softmax for category, sigmoid for sectors	Category is mutually exclusive by construction; a regulation may affect several sectors simultaneously
Confidence threshold 0.55 for review	Routes the uncertain tail to human review without overwhelming reviewers; calibrated in Chapter 7
Expert-verified rows never overwritten	Human verification is authoritative; a later model run must not silently regress a checked record
ONNX with optional INT8	Production hosting has no GPU; INT8 reduces latency and memory at a measured accuracy cost

Propagation, alerting and analytics design. The measurement instrument is a two-step matcher that links content observed on secondary channels back to the primary gazette record.

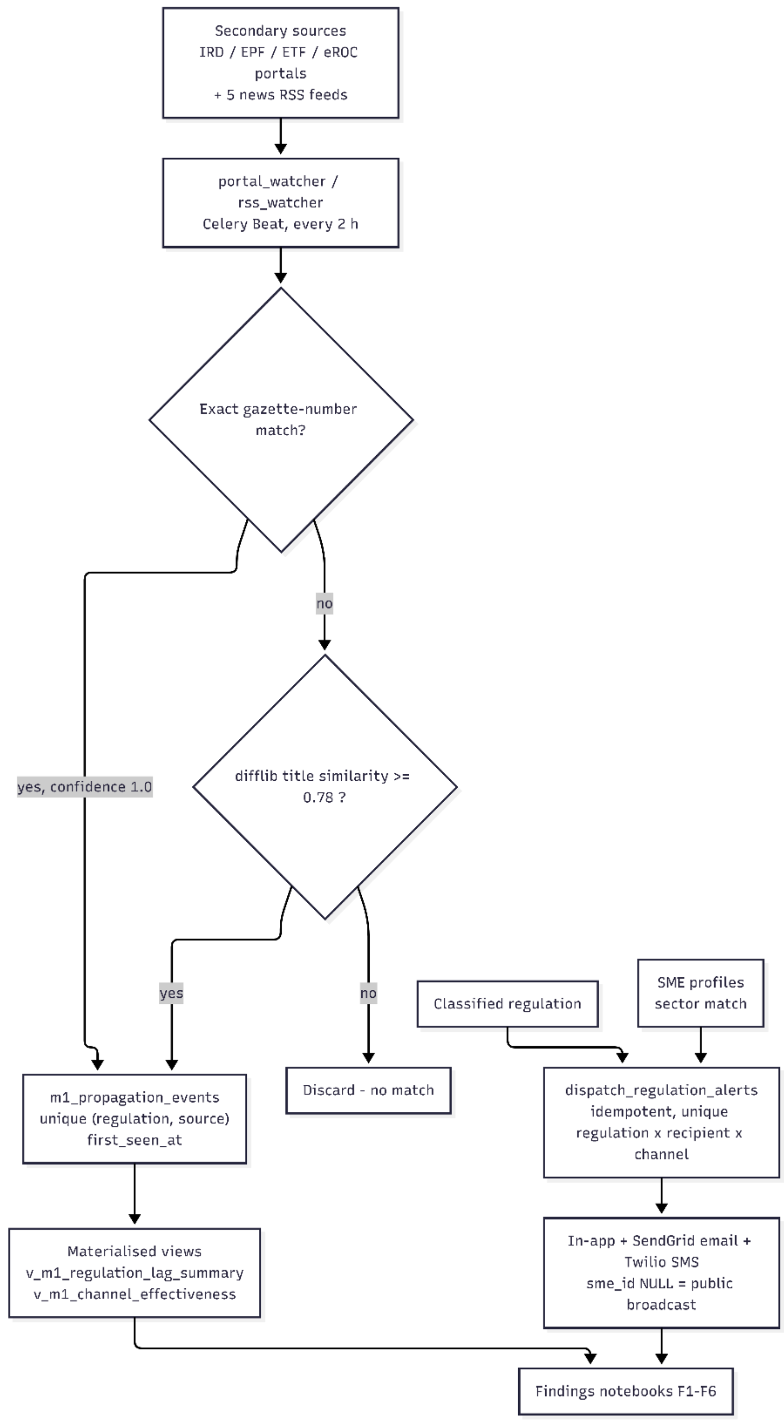


Figure 14 Propagation measurement and alert dispatch design

Alert dispatch is idempotent and keyed uniquely on (regulation, recipient, channel). A re-run must never double-send, both because duplicate notifications destroy user trust and because a duplicated alert would corrupt the difference-in-differences analysis of alert effectiveness.

5.3.2 Compliance Guidance Platform

The architecture of the module is organised as a layered pipeline. At the top is a Next.js frontend exposing a chat interface, a survey portal and a research dashboard. Requests pass to a FastAPI backend whose routing layer is driven entirely by a single domain registry that defines three sectors, eight categories and twenty regulatory domains. This registry is the only place in which sector membership is defined, so the retrieval classifier and the survey gate cannot drift apart.

Beneath the routing layer sits the retrieval layer. Each domain has its own text corpus, embedded with a multilingual sentence-transformer and stored as a dedicated ChromaDB collection. A query is routed to the relevant collection or collections, and the most similar passages are returned with their relevance scores.

The generation layer is a Hugging Face Space serving the fine-tuned Llama-3.1 model on ZeroGPU, reached from the backend through the Gradio client. A parallel translation service in the same Space provides the Sinhala and Tamil layer using NLLB-200. Because the language model is separated from the backend as a network service, it can be upgraded or replaced without redeploying the API, and a stable production Space can serve Module 4 while experimental variants are evaluated in isolation.

Knowledge enters the system through an offline pipeline: official procedures are scraped from government sources, compiled into the domain text files with explicit verification markers, embedded into the vector store, and shipped inside the deployment image. Expert-verification state is held in a separate ledger and written back into the corpus, so that re-scraping a source never destroys a chartered accountant's sign-off.

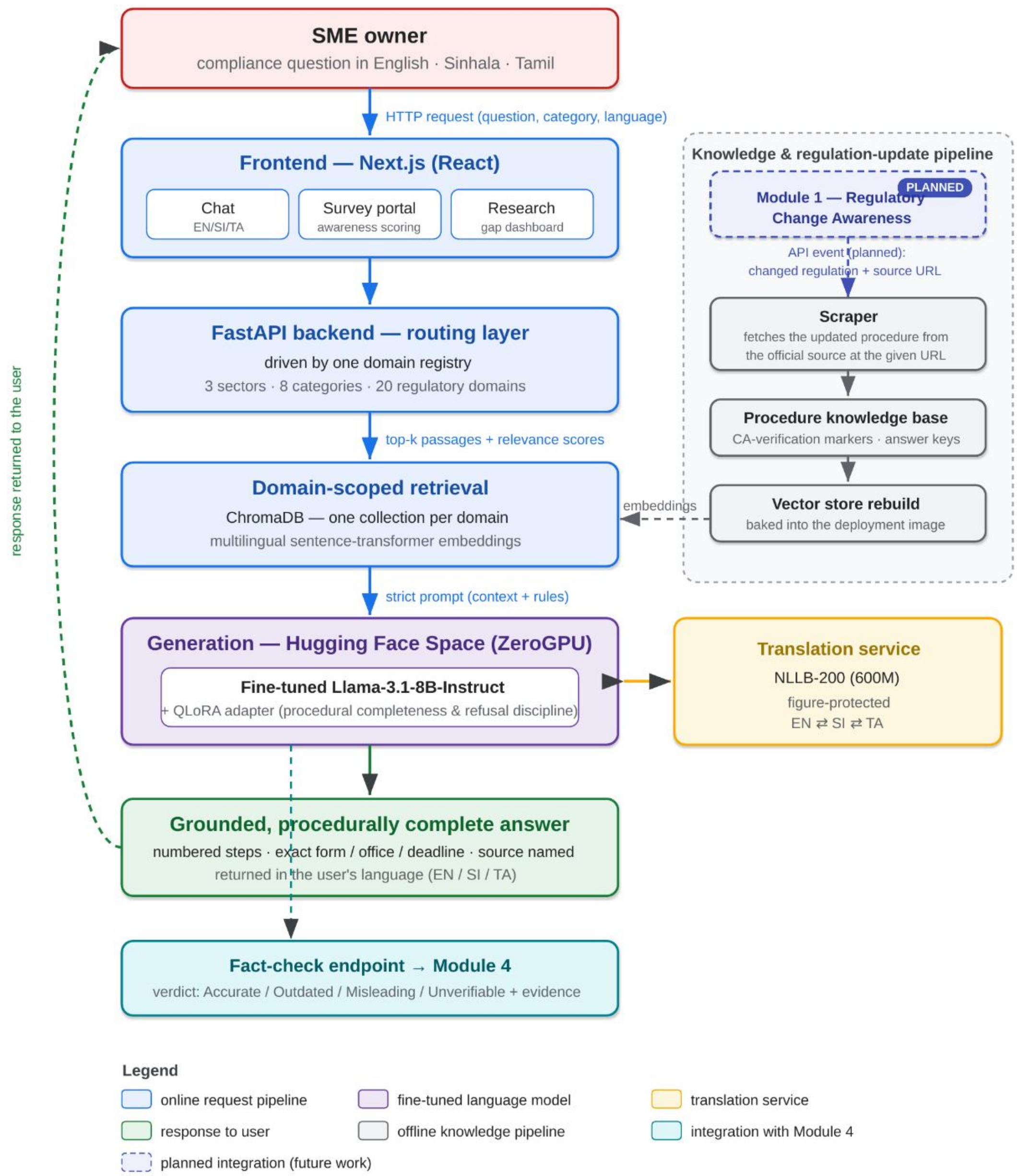


Figure 15 Module 2 pipeline design

5.3.3 SME Compliance Risk Prediction

The architecture of the SME Compliance Risk Prediction module is best read from left to right as two connected paths that share a single model: an offline research path that establishes and validates the risk model, and an online service path that applies it to individual businesses and keeps them informed over time.

The leftmost element of the research path is the survey instrument, which is the module's primary data source. Responses enter a recoding layer that maps answer text to analytical codes, encodes ordered variables as ordinals, constructs the outcome label from the three disclosure questions and reports any data-quality problems rather than absorbing them silently. This layer emits the canonical per-business dataset and should be drawn immediately after the survey source, since every downstream component depends on its schema.

The dataset then feeds a modelling layer containing the three specifications. The naive cross-tabulation baseline, the demographic model and the information-augmented model are drawn as three parallel blocks fed from the same dataset, which makes visually explicit that they are fitted on identical respondents. Their outputs converge on an evaluation block containing the DeLong comparison, from which the hypothesis result is obtained. A separate attribution block computes SHAP values from the information-augmented model and groups them into demographic and informational contributions; this block should be connected to that model rather than to the evaluation block, because attribution and discrimination are distinct questions. Two further validation blocks, the positive-unlabelled correction and the representativeness assessment, are drawn as checks attached to the evaluation stage.

The service path begins where the research path ends. The fitted information-augmented model is serialised into a deployable artefact together with the feature ordering and the baseline values required for attribution, and this artefact is the single point of connection between the two paths. It should be drawn as a bridge element, since it guarantees that the model serving live users is the same model the evaluation validated.

On the service side, a business owner supplies a profile through the web interface. The prediction service loads the artefact, produces a risk probability and computes the signed factor contributions for that individual business. These structured outputs pass to an explanation layer that renders them as plain language in English, Sinhala or Tamil. The explanation layer is deterministic and template-based rather than generative, which should be indicated in the diagram, because guidance concerning statutory obligations must not be subject to fabrication.

The final element is the alerting path, which is what allows the module to act over time rather than only at the moment of assessment. Registered profiles are held in a persistence layer alongside published regulatory changes and the alerts generated from them. Regulatory change events arrive from the Regulatory Change Awareness module and enter a matching engine, which determines the affected businesses by combining sector with the obligations implied by sector and size. Matched businesses receive an alert in their own language on their dashboard. This external input should be drawn entering the matching engine from outside the module boundary and labelled as originating from Module 1, in the same way that the misinformation module draws its retrieval input from Module 2.

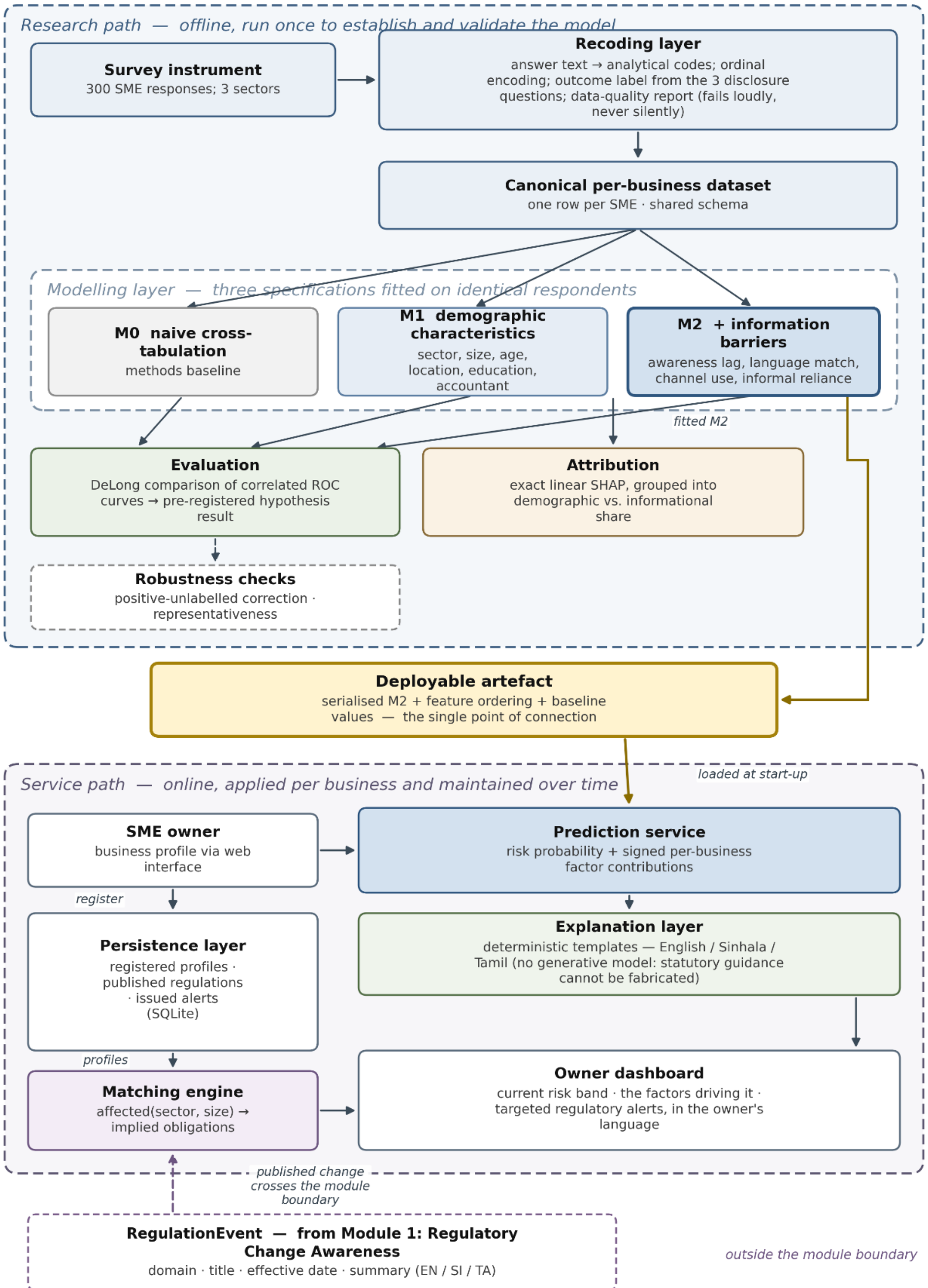


Figure 16 Module 3 pipeline design

5.3.4 Regulatory Misinformation Spread

The Regulatory Misinformation Spread module's high-level architecture starts with the external data sources that are fed into the module to form the corpus. These sources are Facebook, Reddit, Twitter/X, and fact-checking websites that furnish raw social media and claim-based content pertinent to regulatory issues in Sri Lanka which impact SMEs. The leftmost elements in the architecture should be these sources as they are the point of entry into the whole pipeline.

Then the posts are collected and sent to the data preprocessing layer. This layer is meant to convert the noisy raw social media data into a structured data which can be annotation and modelling. These preprocessing activities were done to ensure that the corpus is relevant to the classification goal of the module: Removal of irrelevant posts, duplicate removal, language detection, personally identifiable information removal, regulation domain assignment, and dataset cleaning. This component should be positioned immediately following the external data sources and linked directly to the sources.

After preprocessing, there is an architecture layer that includes a translation layer. Part of the collected data was written in Sinhala and Tamil; therefore, NLLB-200 was used to translate these texts into English for annotation purposes only, which was then added to the dataset without losing the original Sinhala and Tamil text. This should hence be presented as an additional stage in between the preprocessing and the annotation, and should be noted as such: "Translation used for the purpose of consistent annotation instead of the original multilingual text."

The following layer is the annotation layer. 200 posts were double-annotated independently in Label Studio in this module to evaluate the consistency of the annotations, and the rest of the posts were then annotated after this stage, using Google Sheets. This layer should be connected to an agreement calculation component for reasons of measuring the inter-annotator reliability on the double-annotated subset before the completion of the rest of the dataset, where Cohen's Kappa was computed. The annotation layer and the agreement calculation block should be closely related in the architecture because the measurement of agreement came directly from the double-annotation block.

The architecture then moves on to Dataset preparation. Further data cleaning of the annotated data set is performed at this stage for the purpose of modelling: irrelevant metadata columns are removed from the data set, the classifier data set is prepared, and the final train-test split for the experimental comparison is generated. This component should be depicted as a preparation block in the center receiving the annotated dataset and generating the dataset(s) needed for the development and evaluation of the model.

The splitting sets are designed to feed the three types of models evaluated in the module, but not in exactly the same manner. Approach 1 is the fine-tuned model XLM-RoBERTa which is trained on the training dataset and then tested on the testing dataset. Approaches 2 and 3 are not evaluated in the same way in this module, as they do not need the same training process, but rather are evaluated on the prepared test set using prompt-based or retrieval-grounded inference. This is why the architecture should be such that the training data is only connected to Approach 1, and all three approaches are connected to the testing data.

Approach 2 is the RAG based pipeline and should be depicted as having an additional input from Module 2 in the larger platform. This is because the RAG approach relies on regulatory evidence that is retrieved from the verified knowledge base that exists outside this module and not from a local vector database within the Regulatory Misinformation Spread module. Thus, rather than a separate ChromaDB or vector storage block drawn in as Module 2 or regulatory evidence retrieved into Approach 2, the architecture should have an external evidence or retrieval input that is separate from Approach 2 and labeled as Module 2 or retrieved regulatory evidence.

The direct Gemini prompting baseline is Approach 3. This is in contrast to the RAG-based approach, which does not include retrieved regulatory evidence, but rather a classification process based on the post prompt provided. It should thus be depicted, in the architecture diagram, as another path in parallel with Approach 1 and Approach 2, which takes the testing dataset as input and provides an output classification prediction.

All three approaches are then fed into the prediction stage and the system returns the veracity decision for each input post. The final output layer should deliver the predicted misinformation classification and enable downstream SME decision making by allowing users to make their decisions on if a circulating claim from a regulation is trustworthy or not before they act on it. The overall left-to-right flow should thus be in the following order: external data sources, preprocessing layer, translation layer, annotation layer, agreement calculation, dataset preparation, training and testing datasets, the three approaches evaluated, prediction, and final output with an additional external line from Module 2 to the RAG-based approach.

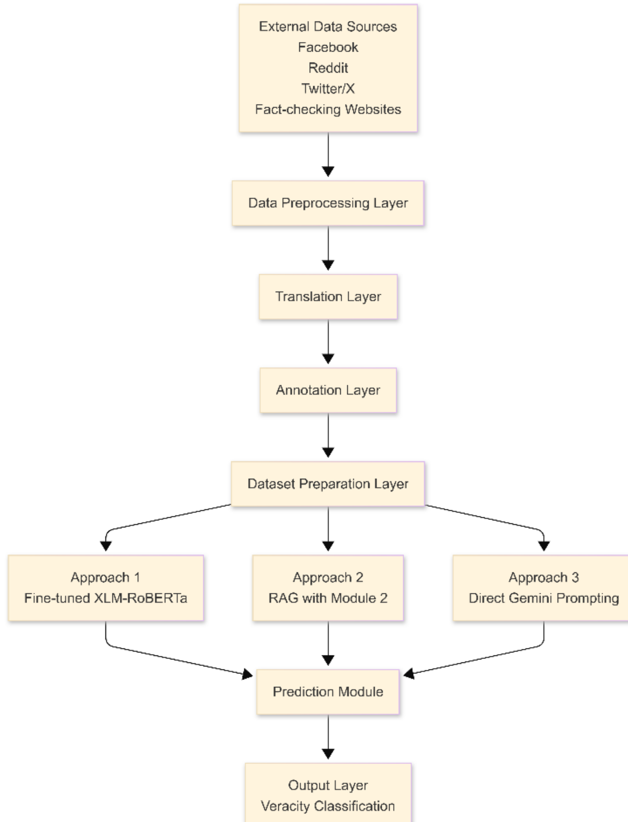


Figure 17 Module 4 pipeline design

5.4 Summary

The design separates three concerns that are frequently conflated: the operational pipeline that produces regulatory intelligence, the measurement framework that quantifies how well it does so, and the research instrument that observes how that intelligence propagates. Each has its own tables, its own metrics and its own failure modes. That separation is what allows Chapter 7 to report extraction accuracy, model accuracy and diffusion lag as independent, individually defensible results.

Chapter 6 - Implementation

6.1 Introduction

This chapter documents how the design of Chapter 5 was realised. Section 6.2 describes data collection, including the construction of the private labelled dataset that is the empirical foundation of Module 1, together with the three further datasets on which Modules 2, 3 and 4 rest — a verified compliance knowledge base, two SME survey instruments, and an annotated corpus of regulatory misinformation. None of the four existed before this project, and each module's central claim is answerable only because its dataset was built first. Collection is therefore documented per module, and the provenance constraints that bound what each dataset may be used to claim are stated in section 6.2.6 rather than deferred. Section 6.3 documents the implementation of each module, with the principal source files named so that every claim can be verified against the repository, and with representative code listings and interface captures.

6.2 Data Collection

Three distinct datasets were collected for Module 1: a raw document corpus, a private labelled gold dataset, and a sealed evaluation baseline. Modules 2, 3 and 4 each contribute a further dataset, listed alongside them in Table 6.1. The seven entries divide into three kinds: source corpora that the system reads, labelled datasets that models are trained and tested on, and sealed baselines that exist solely so a result can be checked against something that cannot move.

Table 6.1 — Datasets used in this project

Dataset	Size	Provenance	Role
Raw gazette corpus	800 PDFs across 11 extraction batches	gazette.lk, documents.gov.lk	Source material for extraction and preprocessing
Gold labelled dataset v1	800 reconciled records	Dual annotation in Label Studio, adjudicated	Training, validation and test data for classification
Sealed evaluation baseline	~800 curated field-level rows (Jan–Apr 2026)	Manually curated workbook, SHA-256 checksummed	Ground truth for extraction and stage accuracy
Database regression snapshot	~204 rows, most at preprocessed	Production database export, February 2026	Stage-progression and regression tracking
Calibration set	20 trilingual documents with expert labels and rationales	Domain expert	Annotator alignment before production labelling
Verified compliance	20 regulatory domains under 8 categories; rules	Official IRD, EPF, ETF, Customs, CAA and SLSI	Retrieval corpus and ground truth for Modules 2 and 4

Dataset	Size	Provenance	Role
knowledge base	and procedure files per domain	publications; expert-verified	
Compliance knowledge survey	76 responses; 60 items, 78 rubric points	SME owners and finance staff, administered through the platform	Declarative-versus-procedural knowledge benchmark (Module 2)
SME vulnerability survey	300 responses; 138 reported failures (46%)	SME owners in retail and food service, two provinces	Training and evaluation data for the risk model (Module 3)
Regulatory misinformation corpus	1,000 posts; 800 development / 200 test	Facebook, Reddit and X, plus fact-checking platforms	Training and evaluation data for claim classification (Module 4)

6.2.1 Document collection

Four Scrapy spiders - gazette_spider, weekly_gazette_spider, acts_spider and bills_spider - collect from the two official sources. Each supports date scoping and closes on scope exhaustion, falls back English→Sinhala→Tamil when an edition is missing, and exposes completeness verification and re-fetch endpoints so gaps are detected rather than silently tolerated. Collection is scheduled every six hours through Celery Beat and can also be triggered for a specific date range from the admin console.

Module 2 operates a second and narrower collection surface, because it reads a different object. Module 1 collects regulatory change events; Module 2 collects standing procedure — the form number, the issuing office, the deadline, the fee and the sequence of steps a business must actually complete. Per-authority scrapers write one structured file per regulatory domain, and the two differ in cadence as well as in content: a change event is time-critical and arrives unpredictably, whereas a procedure changes rarely and is re-scraped on a slow schedule.

One collection rule governs both surfaces and is enforced rather than encouraged: only official government sources may supply a rule. A news report may be used to locate a gazette number or a publication date, never as the source of the rule itself. Any regulatory content entering the knowledge base before expert sign-off is marked as pending verification, and that marking travels with the text into every answer generated from it, so an unverified figure carries its own caveat to the reader.

```
Principal files:
enigmatrix-backend/scraper/spiders/*.py, app/tasks/m1/run_scraper.py, app/tasks/m1/gazette_scraper.py, app/api/v1/m1_completeness.py
```

6.2.2 Annotation and gold dataset construction

No labelled dataset of Sri Lankan regulatory changes exists, so one was created. The protocol was designed so that the result would be defensible as ground truth.

1. **Calibration.** A 20-document trilingual calibration set with expert labels and written rationales was completed by every annotator before production labelling. Calibration disagreements were discussed and the decision hints in the guideline were refined.
2. **Sampling.** Documents were drawn using stratified and k-means samplers with explicit minority-domain targeting, plus a hybrid active-learning mode, so annotation effort concentrated where it changes the model rather than on whatever arrived first.
3. **Dual annotation.** Batches 02–05 were labelled independently by two annotators. Each task captured change category, affected sectors, SME relevance, an annotator confidence rating and free-text notes.
4. **Automated reduction.** scripts/resolve_iaa.py paired annotations, computed agreement statistics and emitted agreed rows directly to gold.
5. **Manual adjudication.** The 40 disagreement rows were adjudicated by a resolver and recorded in manual_resolutions.csv with a resolution method and resolver identifier. The final gold file contains **zero lead-annotator fallback rows** — every disagreement was decided explicitly, not defaulted.
6. **Freezing.** The result was frozen as gold_standard_v1_800.csv with an accompanying iaa_report_v1_800.json.

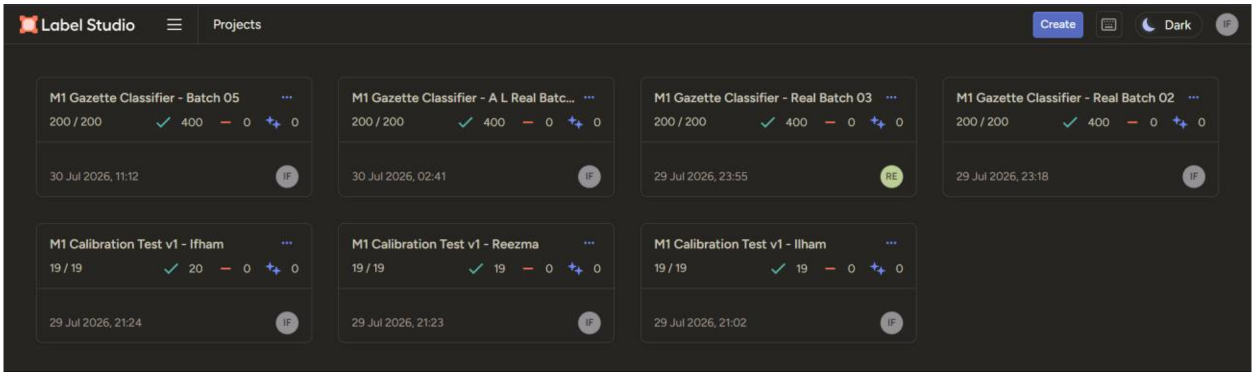


Figure 18 Label Studio annotation interface for the Module 1 labelling schema

The label schema is defined once in code and mirrored into the Label Studio configuration and the database enumerations.

Listing 6.1 — Canonical label schema (`enigmatrix-ml/m1/model/labels.py`)

```python
CATEGORIES: list[str] = [
    "TAX_RATE_CHANGE", "IMPORT_EXPORT", "SECTOR_SPECIFIC", "EPF_ETF_CHANGE",
    "LABOUR_LAW", "PRODUCT_STANDARD", "BUSINESS_REGISTRATION",
    "PENALTY_ENFORCEMENT",
]
SECTORS: list[str] = [
    "grocery_retail", "food_service", "general_retail",
]

CAT_TO_ID: dict[str, int] = {c: i for i, c in enumerate(CATEGORIES)}
ID_TO_CAT: dict[int, str] = {i: c for c, i in CAT_TO_ID.items()}
SECTOR_TO_ID: dict[str, int] = {s: i for i, s in enumerate(SECTORS)}


def encode_sectors(value) -> list[int]:
    """Multi-hot vector of length len(SECTORS)."""
    vec = [0] * len(SECTORS)
    for s in parse_sectors(value):
        vec[SECTOR_TO_ID[s]] = 1
    return vec

```

The three study sectors were selected because they are numerous among Sri Lankan SMEs, are affected by all eight categories, and are distinguishable by an annotator without specialist domain training. Restricting scope to three sectors is a deliberate trade-off: it keeps annotation feasible at the required agreement level, at the cost of limiting immediate generalisation. This is stated as a limitation in Chapter 8 rather than concealed.

Table 6.2 — Columns of the frozen gold dataset

Group	Columns
Identity	batch_id, regulation_id, regulation_key, gazette_number, year, language
Model input	classification_chunk
Labels	change_category, affected_sectors, is_sme_relevant
Annotator signal	confidence, confidence_mean, annotator_ids, annotator_notes
Adjudication provenance	resolution_method, resolver_id, disagreement_fields, resolver_notes, source_exports

6.2.3 Compliance knowledge base and expert verification

Module 2 requires a corpus that is not merely retrieved from but can be cited. It was assembled from the official publications of the Inland Revenue Department, the Employees' Provident Fund and Employees' Trust Fund, Sri Lanka Customs, the Consumer Affairs Authority, the Sri Lanka Standards Institution and the Registrar of Companies, covering 20 regulatory domains grouped under the same eight categories Module 1 classifies into and scoped to the three study sectors.

Each domain carries two artefacts. A rules file holds the human-readable statement of the obligation and is what is segmented and embedded for retrieval. A procedures file holds the scraped operational detail — form, office, deadline, fee and steps — and is injected into the rules text by an idempotent build step, so that a re-scrape updates the knowledge base without a manual merge. The separation matters because the two have different lifetimes and different authorities behind them.

Content is verified by a chartered accountant across two desks: one for the survey question bank and its marking rubrics, and one for the scraped procedures and their checkpoints. Sign-off is recorded in ledger tables and is never written back into the scraped files, because the scrapers overwrite those files on every run and a verification flag stored inside a scraped document would be destroyed by the next scrape. Display status is therefore *computed* rather than stored, with the precedence rejected, then stale, then verified, then partial, then pending. A verified item becomes stale automatically when the hash of its content no longer matches the hash that was signed, so an edit invalidates the sign-off instead of silently inheriting it. This is the single design decision that allows expert verification to remain trustworthy while the underlying corpus continues to be re-scraped.

6.2.4 Survey instruments

Both surveys were delivered through the platform's own trilingual survey module rather than an off-the-shelf form tool. Responses therefore land directly in the platform database, which permits richer response logging, conditional branching driven by the respondent's registered profile, and a single consent record — none of which a generic form service supports.

The Module 2 compliance knowledge survey is deliberately two-layered, because the module's hypothesis is about a distinction that a single-layer instrument cannot detect. A multiple-choice layer measures *declarative* knowledge — whether the respondent knows that an obligation exists — while an open-response layer is graded against rubrics to measure *procedural* knowledge — whether they know the form, the office and the deadline that discharge it. Sixty items across the 20 domains carry 78 rubric points in total. Items are gated twice before being shown: a sector gate requires the domain to apply to the respondent's sector, and a profile gate requires the answer key's applicability rule to match the registered profile. Both gates fail closed, so missing information means "not applicable" rather than "ask anyway". The report-card denominator applies the sector gate only, and this asymmetry is deliberate: it keeps scores comparable across respondents whose individual profiles differ.

The Module 3 SME vulnerability survey obtained 300 usable responses across the three sectors. It records the demographic block — sector, sub-sector, location, business age, headcount, owner education and whether an accountant is used — and the information-barrier block, comprising awareness lag, whether guidance is normally received in the owner's working language, responsiveness to notices, and reliance on informal channels. Self-reported compliance history over the preceding twelve months supplies the outcome. Responses are recoded from the survey export into a fixed model schema by prefix matching on question identifiers rather than on question wording, so that later rewording of an item does not silently break the pipeline. The outcome label is constructed by a rule fixed before analysis: a failure is recorded where the respondent missed a filing deadline, received a penalty or warning notice, or faced enforcement action within the period. That rule, and its asymmetry, are carried into the evaluation in section7.1.15.

6.2.5 Misinformation corpus collection

Module 4's corpus was assembled manually rather than through platform APIs. Research access tiers for the major social platforms are no longer dependable, and a methodology contingent on them would have been a single point of failure for the module. Publicly visible posts making a specific claim about Sri Lankan regulation — value-added tax, income tax, employer contributions, filing deadlines and gazette notices — were identified by keyword search across Facebook, Reddit and X, and supplemented from fact-checking platforms, which supply a seed of claims whose accuracy has already been assessed independently.

Only posts that were publicly visible and carried an identifiable regulatory claim were retained; general commentary, questions and posts referencing no regulatory content were excluded, and no private group or account was accessed. Post text was captured together with the platform, the date and whatever engagement figures were publicly displayed, the latter supporting the spread analysis rather than the classification task. Manual collection bounds the corpus at 1,000 posts, which is modest; it is, however, a corpus whose every item can be accounted for, which an API dump of comparable size would not be.

6.2.6 Provenance, consent and data-governance constraints Three constraints bound what these datasets may be used to claim, and are recorded here rather than in the limitations chapter alone, because each one governs how the data may be handled during the work and not merely how the results should be read. First, the Module 2 live survey corpus is split-source, and the two layers do not carry equal evidential weight. The multiple-choice layer is field data and is treated as evidence. The written short-answer layer originates from a separate collection submission and is flagged in the record as not verbatim; it is retained as a grading fixture only, and no sentence from it is quoted anywhere as a respondent's words. The flag follows from a check rather than a suspicion: paired same-respondent prose similarity was 0.3539 against a random-pair similarity of 0.3513, $t = 1.15$, $p = 0.25$ — that is, prose attributed to one respondent is not measurably more similar to itself than to another respondent's. That finding is carried into the limitations as a methodological result about data quality, not concealed as an inconvenience. Second, identifiable information is confined by design. Business names and owner names exist only in the two source workbooks and are never carried into any processed record; the conversion script reads them solely to set a boolean indicating that identity is on file, and never copies them forward. No private enterprise data of any kind — financial statements, filing records or account data — was collected for any module. Every measurement in this dissertation rests on public records, ethically administered survey responses, or publicly visible posts. Third, consent and ethics were handled at programme level. A single ethics application covering all four modules was prepared, together with a participant information sheet and a consent form, and every survey instrument presents an informed-consent statement before its first question. Free-text survey responses and raw social-media content are retained only for the period the analysis requires.

6.3 Implementation of Individual Modules

6.3.1 Regulatory Change Awareness Gap

Module 1 was implemented in five phases. All five are complete in code; the human and data gates that remain are stated explicitly in Chapter 7.

6.3.1.1 Phase 1 - Platform foundation

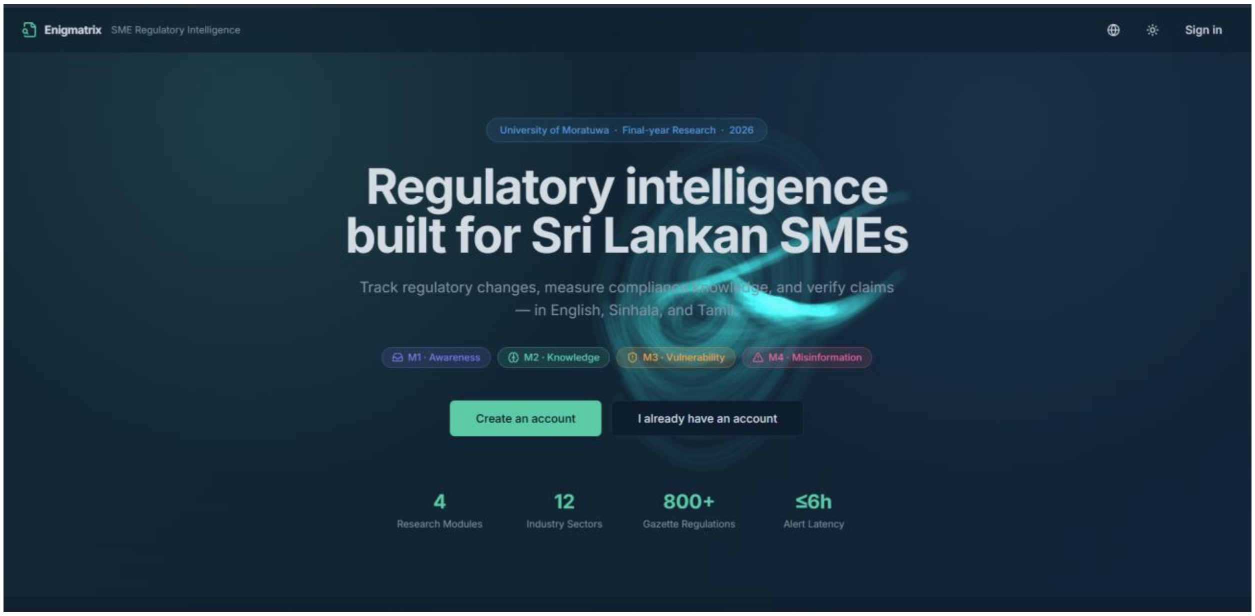
JWT authentication (bcrypt hashing, HS256 access and refresh tokens) with role-based access control across sme, admin and annotator roles; slowapi rate limiting; audit logging on every authentication event and administrative mutation; regulation CRUD at /admin/regulations with soft deletion via is_active, an expert-verification gate, bulk verification and restore; sector mapping and seed data.

Principal files: app/api/v1/m1_regulations.py, app/services/m1_regulation_service.py, app/models/regulation.py.

6.3.1.2 Phase 2 - Ingestion, extraction, preprocessing and measurement

This is the largest completed block. The canonical extraction implementation lives in enigmatrix-ml/m1/extraction/; the backend app/extraction/ is a thin re-export adapter that preserved existing imports and tests while removing roughly ninety lines of duplicated logic. Components: the classify_pdf router; per-page engines (PyMuPDF, pdfplumber, pypdfium2, Tesseract, Surya); registered profiles legacy_v1, page_routing_v1, surya_fallback_v1 and wijesekara_routing_v1; a font-aware Wijesekara-to-Unicode converter; a CER calculator; and a segmenter. Preprocessing (enigmatrix-ml/m1/preprocessing/) performs cleaning, fastText language identification, metadata and penalty extraction, chunking and sub-document splitting.

The operational console at /admin/m1/pipeline streams live progress over a WebSocket at /ws/extraction/{task_id} and provides a date-range picker, run history, cancel and rollback, a PDF Records page and a per-regulation pipeline trace.



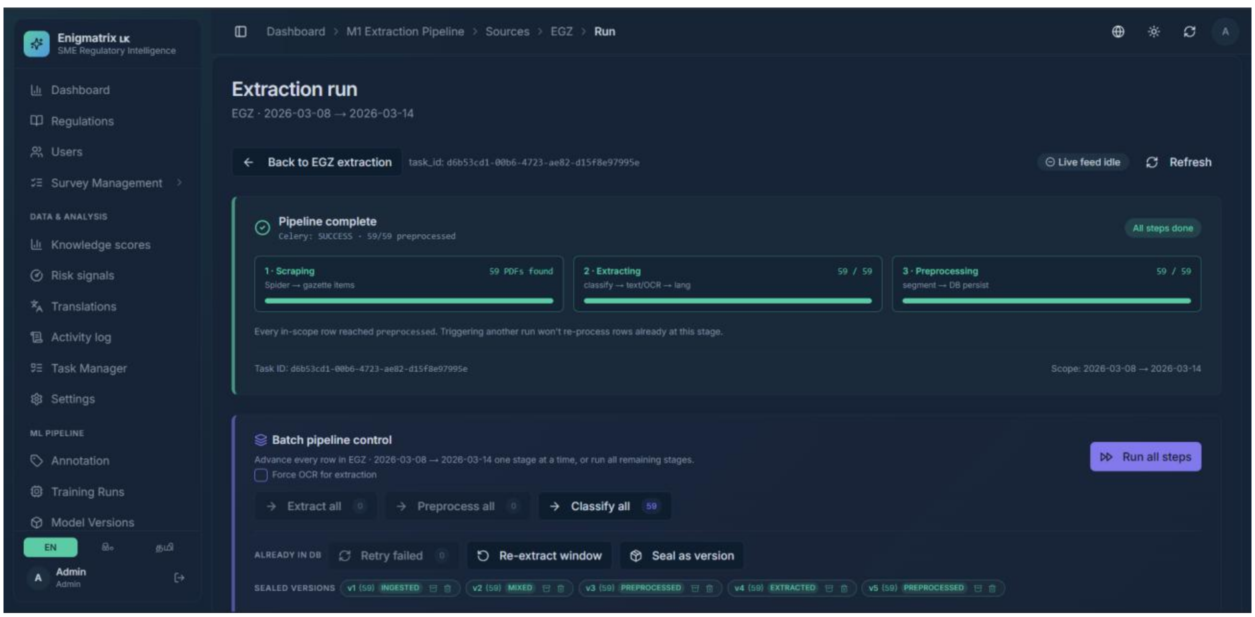


Figure 19 Administrative extraction pipeline console

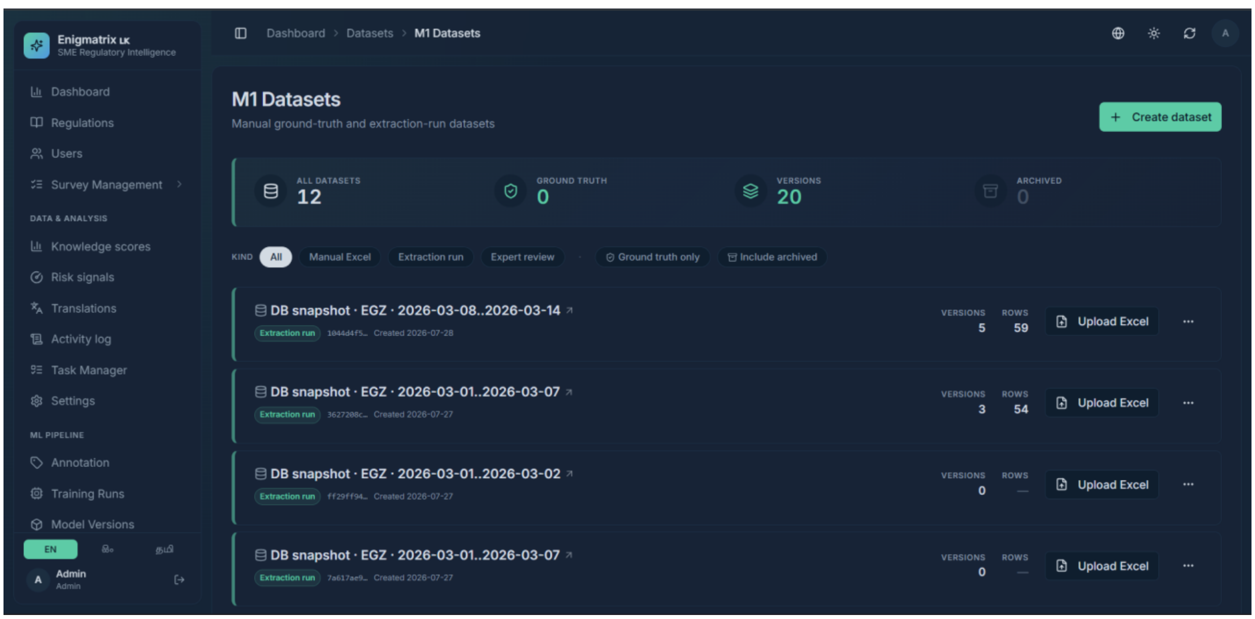
Principal files: app/api/v1/m1_gazette_extraction.py (17 routes), app/api/v1/m1_extraction_ws.py.

6.3.1.3 Phase 2 — Extraction accuracy measurement subsystem

This subsystem answers the question "how good is our extraction?" quantitatively and is a core artefact of the dissertation.

- **Dataset registry** — m1_datasets, m1_dataset_versions, m1_dataset_rows: named datasets with immutable sealed versions carrying SHA-256 content hashes, Excel ground-truth upload, retire and restore, and a nightly retention policy that keeps the current version, the previous version and anything flagged keep.

- **Extraction profile registry and run dispatcher** — runs any profile against a defined scope, with overlap detection and automatic v1→v2 versioning when a new run's date range overlaps an existing version.
- **Measurement engine** (enigmatrix-ml/m1/evaluation/) — per-field comparators for categorical, date, numeric, string, semantic and text-summary types; aggregates; strata; raw-text scoring; completeness; and a date-scope filter so that a date-scoped run is not penalised against the full ground truth.
- **Measurement UI** at /admin/datasets/m1/measurements* — run form with optional date-range and source scoping, dashboard, per-run detail, per-regulation drill-down, worst-N view, calibration view, sortable columns, sparklines and keyboard shortcuts.
- **Accuracy report export** — GET /api/v1/m1/measurements/{run_id}/report.md returns a downloadable Markdown report produced by a pure function in m1_measurement_report.py.
- **Data-quality suites** — Great-Expectations-style JSON expectations in data_quality/expectations/ validated automatically after sealing by the validate_dataset_version task.
- **Thesis artefact generator** — scripts/regenerate_thesis_tables.py, invoked by make thesis-artifacts.



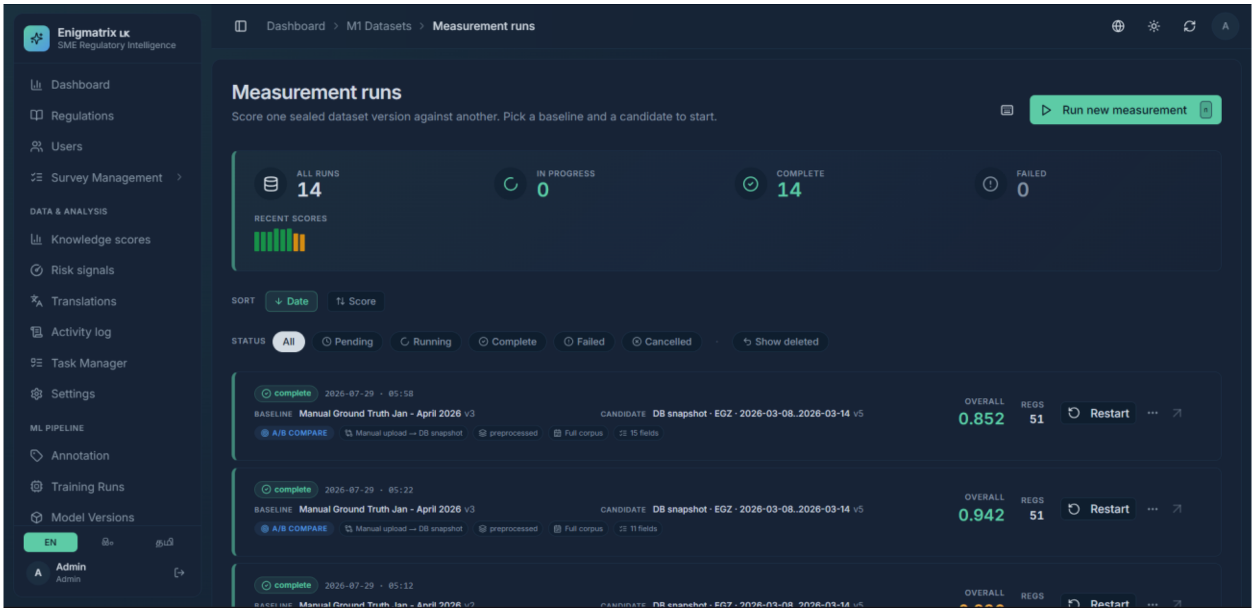


Figure 20 Extraction accuracy measurement dashboard

6.3.1.4 Phase 3 — Annotation, dataset preparation and classification

Dataset preparation is implemented in m1.model.data, which produces train/validation/test Parquet splits either deterministically on regulation_key or temporally on gazette_published_date. TF-IDF baselines are implemented in m1.model.baselines.

The classifier itself is a single encoder with two heads.

Listing 6.2 — Model architecture (`enigmatrix-ml/m1/model/architecture.py`)

```python
class GazetteClassifier(nn.Module):
    """CLS-pooled XLM-R (LoRA-adapted) -> single-label category head +
    multi-label sector head. Head widths come from ModelConfig."""

    def __init__(self, cfg: ModelConfig | None = None):
        super().__init__()
        self.cfg = cfg or ModelConfig()
        encoder = AutoModel.from_pretrained(self.cfg.base_model)
        self.encoder = get_peft_model(encoder, LoraConfig(
            r=self.cfg.lora_r, lora_alpha=self.cfg.lora_alpha,
            lora_dropout=self.cfg.lora_dropout,
            target_modules=list(self.cfg.lora_targets), bias="none",
            task_type="FEATURE_EXTRACTION",
        ))
        hidden = encoder.config.hidden_size
        self.dropout = nn.Dropout(0.1)
        self.category_head = nn.Linear(hidden, self.cfg.num_categories)
        self.sector_head = nn.Linear(hidden, self.cfg.num_sectors)

    def forward(self, input_ids, attention_mask):
        out = self.encoder(input_ids=input_ids, attention_mask=attention_mask)
        pooled = self.dropout(out.last_hidden_state[:, 0])   # [CLS]
        return self.category_head(pooled), self.sector_head(pooled)

```

```python

def compute_loss(category_logits, sector_logits, category_target,
                 sector_target=None, sector_weight: float = 1.0,
                 class_weights=None):
    """CrossEntropy on the category head + BCE on the multi-label sector head."""
    ce = nn.functional.cross_entropy(category_logits, category_target,
                                     weight=class_weights)
    if sector_target is None:
        return ce
    bce = nn.functional.binary_cross_entropy_with_logits(
        sector_logits, sector_target.float())
    return ce + sector_weight * bce

```

Training hyper-parameters are held in a single dataclass so that every run is fully described by one serialisable object, which is written into the model registry alongside the metrics.

Listing 6.3 — Training configuration (`enigmatrix-ml/m1/model/config.py`)

```python
@dataclass
class ModelConfig:
    base_model: str = "xlm-roberta-base"
    num_categories: int = len(CATEGORIES)   # 8
    num_sectors: int = len(SECTORS)         # 3
    max_length: int = 512

    # LoRA
    lora_r: int = 16
    lora_alpha: int = 32
    lora_dropout: float = 0.1
    lora_targets: tuple[str, ...] = ("query", "value")

    # optimisation
    lr_head: float = 2e-5
    lr_lora: float = 1e-4
    weight_decay: float = 0.01
    warmup_ratio: float = 0.1
    epochs: int = 8
    batch_size: int = 16
    early_stop_patience: int = 3
    fp16: bool = True
    sector_loss_weight: float = 1.0
    seeds: tuple[int, ...] = (42, 1, 2)

```

Two learning rates are used deliberately: the randomly initialised classification heads take the higher rate (1e-4 is applied to the LoRA parameters, 2e-5 to the head), warmup is applied over the first 10 % of steps, and early stopping with patience 3 guards against overfitting on a small dataset.

Evaluation and export. m1/model/eval.py computes per-slice macro-F1 by language, quarter and text length, applies the 8-percentage-point slice-cliff check and writes an error-analysis CSV. m1/model/export_onnx.py merges the LoRA weights into the base model and exports to ONNX with optional INT8 quantisation.

Production inference. GazetteInference wraps ONNX Runtime; app/m1/services/classifier_service.py loads the artefact from M1_MODEL_ONNX_DIR (default storage/models/m1/onnx/v1) and applies the 0.55 minimum-confidence review threshold. When the directory is absent or empty, classifier_status() returns no_model and classify_gazette_task leaves the record at preprocessed - the correct and safe behaviour before a model is promoted. Migration 202606300001 adds the classifier confidence columns.

6.3.1.5 Model training on free GPU platforms

Training is executed on Google Colab and Kaggle Notebooks. The command is identical on both; only the environment preparation differs.

```
# Colab / Kaggle setup cell
pip install uv
cd /content/xyz/enigmatrix-ml          # Kaggle: /kaggle/working/xyz/enigmatrix-ml
uv sync --extra training --extra research

# three-seed training run
uv run python -m m1.model.train_xlmr \
  --data datasets/m1_regulations \
  --seeds 42 1 2 \
  --base xlm-roberta-base \
  --lora-r 16 --epochs 8 --fp16 \
  --out ../storage/models/m1/xlmr_lora_v1

# per-slice evaluation and ONNX export
uv run python -m m1.model.eval \
  --model ../storage/models/m1/xlmr_lora_v1 \
  --test datasets/m1_regulations/test.parquet \
  --report ../storage/models/m1/eval_v1
uv run python -m m1.model.export_onnx \
  --model ../storage/models/m1/xlmr_lora_v1 \
  --out ../storage/models/m1/onnx/v1 --int8
```

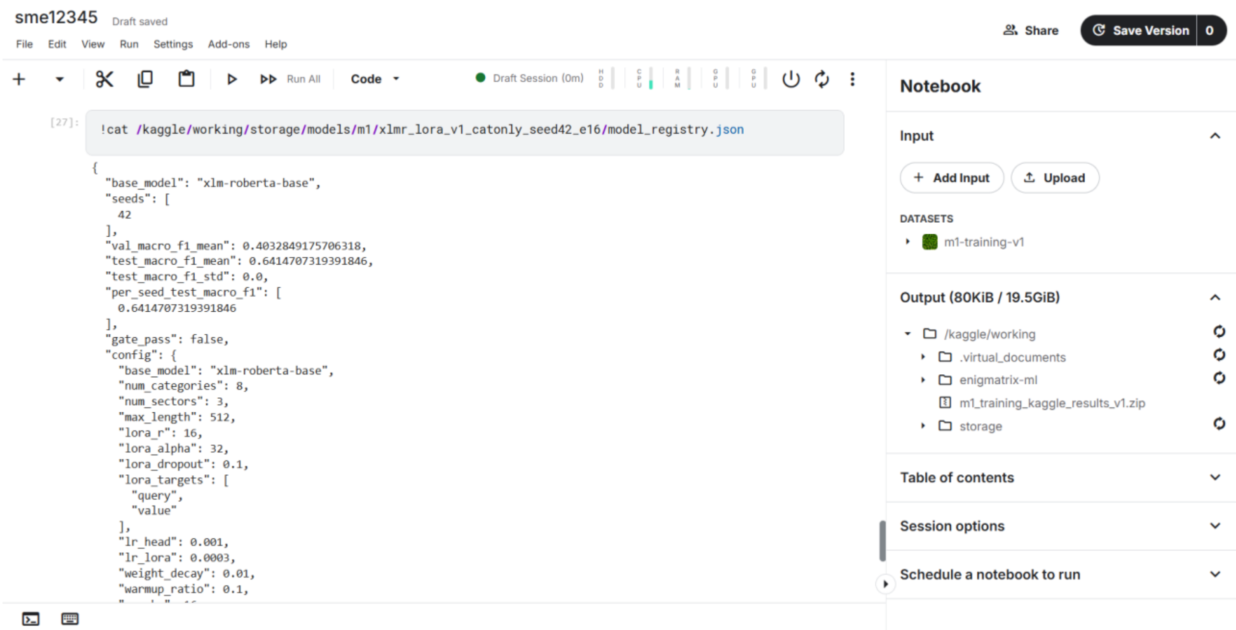


Figure 21 GPU training session on the free notebook platform

On Kaggle the frozen gold dataset and the generated Parquet splits are attached as a versioned Kaggle Dataset, which pins the exact data a run consumed; on Colab the repository is cloned or Google Drive is mounted. Both paths write the same model_registry.json, so the two platforms produce directly comparable artefacts.

6.3.1.6 Phase 4 — Watchers, alerts and analytics

- **Propagation.** m1_propagation_events with a uniqueness constraint on (regulation, source) and a first_seen_at timestamp; a secondary-source registry covering the IRD, EPF, ETF and eROC portals plus five news RSS feeds; a two-step matcher (exact gazette-number match at confidence 1.0, then difflib similarity ≥ 0.78), unit-tested; portal_watcher and rss_watcher on a two-hourly offset Beat schedule.
- **Alerts.** m1_alerts with a uniqueness constraint on (regulation, recipient, channel) and sme_id IS NULL denoting a public broadcast; a pure alert-content builder; SendGrid and Twilio providers that degrade gracefully to a skipped status without API keys; an idempotent alert service; a batched dispatch task; API routes /m1/alerts/public, /m1/alerts and mark-read; and the frontend /alerts page.
- **Analytics.** Materialised views v_m1_regulation_lag_summary and v_m1_channel_effectiveness; a Kullback–Leibler confidence-drift helper alerting above 0.15; nightly refresh_lag_analytics.

6.3.1.7 Phase 5 — Findings and retraining

The diffusion analysis is preregistered in enigmatrix-ml/research/preregistration.md at $\alpha = 0.05$ with bootstrap confidence intervals, fixing hypotheses F1–F6 before the data is unblinded. findings_common.py provides tested loaders and bootstrap CI computation with a database-or-synthetic-demo mode, and four notebooks cover lag analysis, secondary diffusion, alert effectiveness by difference-in-differences, and classifier evaluation.

Retraining is recorded in m1_retraining_runs and gated by a pure, unit-tested decision function.

Listing 6.4 — Canary promotion decision (`enigmatrix-ml/m1/model/promotion.py`)

```python
def decide(prod_f1, candidate_f1, gate: float = 0.92, regression_tol: float = 0.01):
    """Return (action, reason) where action in {promote, rollback, hold}."""
    if candidate_f1 is None:
        return ("hold", "no candidate metric")
    if candidate_f1 < gate:
        return ("rollback", f"candidate {candidate_f1:.3f} below gate {gate:.2f}")
    if prod_f1 is not None and candidate_f1 < prod_f1 - regression_tol:
        return ("rollback",
                f"candidate {candidate_f1:.3f} regresses vs prod {prod_f1:.3f}")
    return ("promote", f"candidate {candidate_f1:.3f} clears gate {gate:.2f}")

```

Keeping this function free of framework and database dependencies is what makes the promotion policy exhaustively testable — the decision that governs whether a model reaches users is the one component that must not be able to fail silently.

6.3.1.8 Summarisation and translation

Implemented components: database, model and schema fields for title_en, title_si, title_ta, summary_en, summary_si and summary_ta; an administrative translation queue at app/api/v1/admin_translations.py; an NLLB-200 helper at scripts/lib/nllb_translate.py; a gazette title scraper at scripts/lib/title_scraper.py; and field-metric support for title and summary fields in the measurement tooling.

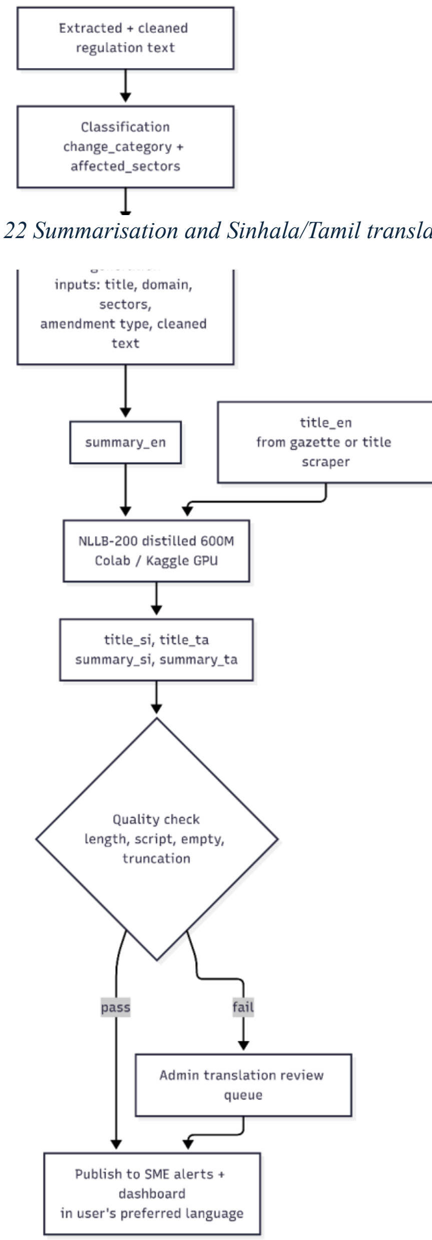


Figure 22 Summarisation and Sinhala/Tamil translation flow

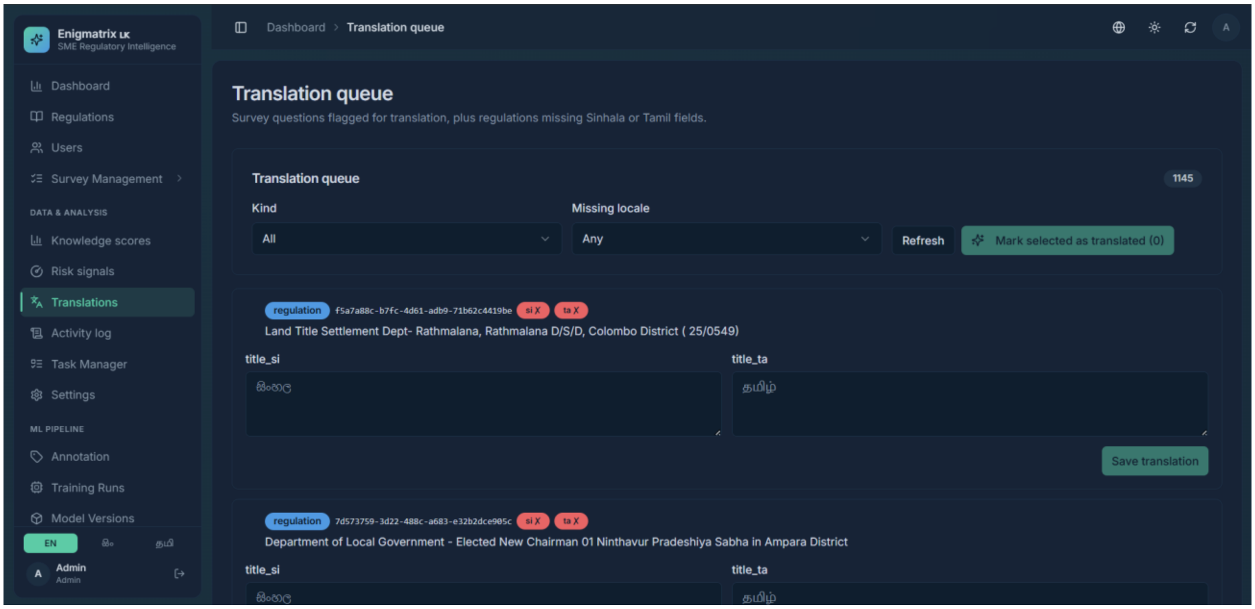


Figure 23 Administrative translation review queue

Status statement. The schema, the review queue, the NLLB helper and the title scraper are implemented. The production batch summary generator and the bulk translation backfill are the remaining pieces; Section 7.2.1 gives the commands to complete and evidence them, and this report does not claim large-scale automatic summarisation output that has not been produced.

6.3.1.9 Frontend

The SME-facing surface provides registration with a business profile, a trilingual dashboard, a regulation feed, the alerts page and the survey wizard. The administrative surface provides regulation CRUD and verification, the pipeline console, the dataset registry, the measurement dashboards and drill-downs, the translation queue, user management and the activity log. Localisation is handled by next-intl across English, Sinhala and Tamil with script-appropriate fonts.

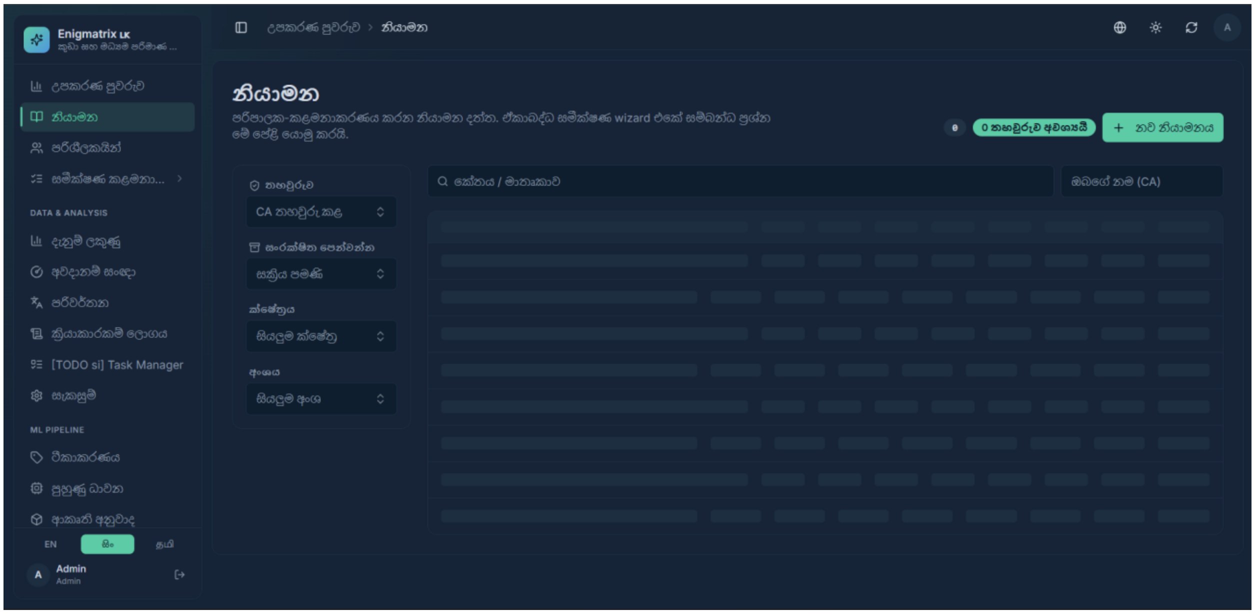


Figure 24 Trilingual SME dashboard

6.3.2 Compliance Guidance Platform

The module was implemented as a FastAPI service exposing endpoints for querying, the survey, fact-checking and internal review tooling. Retrieval was implemented with LangChain and ChromaDB using a multilingual MPNet embedding model, with regulatory text split into overlapping chunks tagged with domain, category and sector metadata. Twenty domain corpora were compiled exclusively from official government sources, each carrying explicit "pending / partially / fully verified" markers so that an unverified figure announces its status inside the retrieved text itself.

The prompt was engineered in several layers to enforce procedural completeness and to suppress the base model's tendency to refuse legitimate questions as "financial advice." It frames the task as reporting published regulation, permits exactly one refusal sentence for genuinely unanswerable questions, and requires a numbered procedure, a per-step form/office/deadline, an explanation of every named form, and a worked example.

The language model was specialised using QLoRA. Llama-3.1-8B-Instruct was loaded in 4-bit precision with its weights frozen, and low-rank adapters (rank 16) were trained on all seven projection matrices, amounting to roughly 42 million trainable parameters — about 0.52% of the model. Training used a purpose-built instruction corpus of grounded question–answer pairs generated from the domain text and answer keys, split by item group to prevent paraphrase leakage between the training and validation sets. Training ran on an NVIDIA A100 for three epochs (about sixteen minutes); the checkpoint with the lowest validation loss, reached at the second epoch, was retained.

8. Load the base model in 4-bit and attach LoRA

QLoRA: the 8B base is frozen at 4-bit NF4, and only small low-rank matrices train. On an 8B model that is roughly 42M trainable parameters — about 0.5% — which is why this fits on a Colab GPU at all, and why the resulting adapter is ~160 MB instead of a 16 GB merged checkpoint.

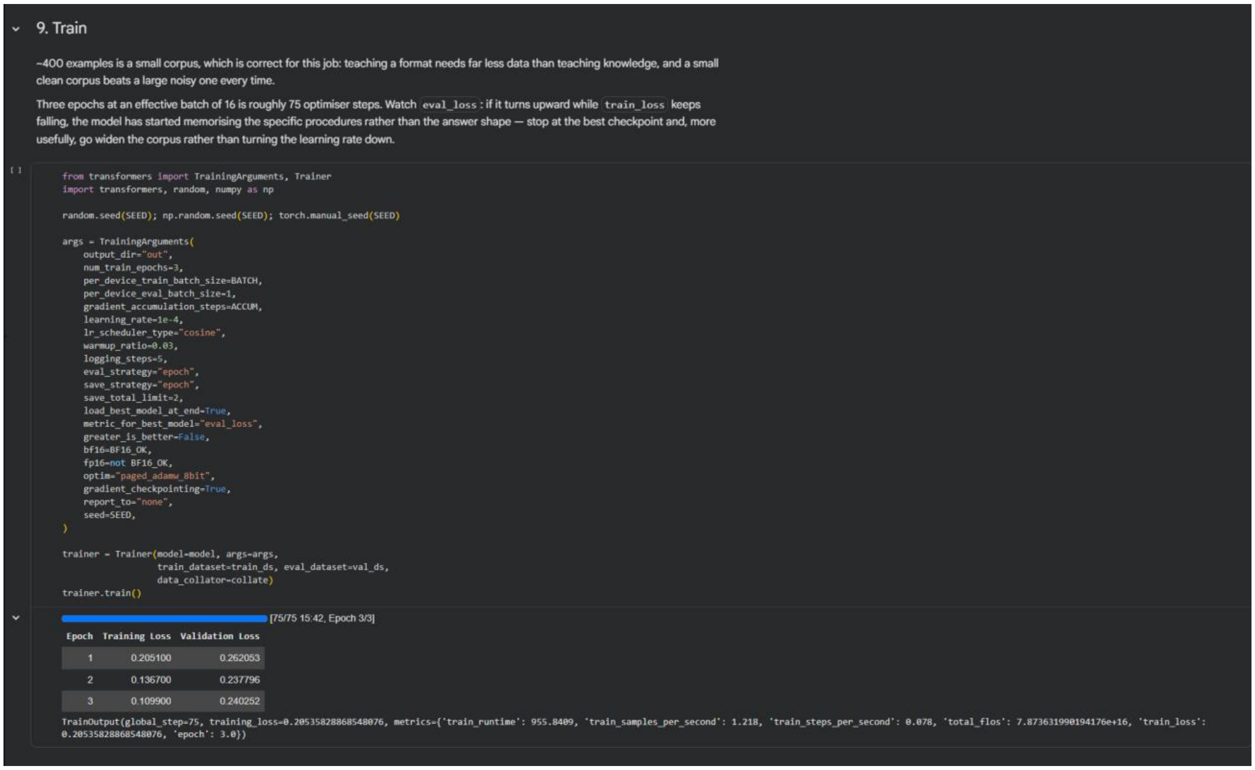
The ~160 MB matters practically: it makes retraining a 2-minute upload instead of an hour, so you can afford to iterate on the corpus after seeing the evaluation, which is where most of the real improvement will come from.

LoRA is attached to **all** attention and MLP projections. Attention-only (`q_proj` / `v_proj`) is the common shortcut, but the behaviour being taught here is output formatting, which lives largely in the MLP blocks.

```
# — Preflight: prove 4-bit actually works on the GPU before downloading 16 GB —
# The failure this guards against is not loud. bitsandbytes prints a WARNING when
# it cannot find a CUDA binary and then carries on with its CPU build, so the
# first sign of trouble used to arrive four safetensors shards and four minutes
# later, as a traceback from deep inside transformers. Sixty-four numbers through
# a real Linear4bit is a complete test of the path that matters, and it costs
# nothing.
import bitsandbytes as _bnb
print("bitsandbytes", _bnb.__version__)
try:
    _lin = _bnb.nn.Linear4bit(64, 64, bias=False, quant_type="nf4",
                              compute_dtype=torch.float16).cuda()
    _ = _lin(torch.randn(2, 64, device="cuda", dtype=torch.float16))
    del _lin, _
    print("4-bit NF4 on GPU: OK\n")
except Exception as _e:
    raise SystemExit(
        f"bitsandbytes cannot do 4-bit on this GPU: {type(_e).__name__}: {_e}\n"
        "Re-run the install cell, then Runtime -> Restart session, then re-run "
        "from cell 1. If it still fails, the runtime has no GPU attached."
    )

from transformers import AutoModelForCausalLM, BitsAndBytesConfig
from peft import LoraConfig, get_peft_model, prepare_model_for_kbit_training

COMPUTE_DTYPE = torch.bfloat16 if BF16_OK else torch.float16

bnb = BitsAndBytesConfig(
    load_in_4bit=True,
    bnb_4bit_quant_type="nf4",
    bnb_4bit_use_double_quant=True,
    bnb_4bit_compute_dtype=COMPUTE_DTYPE,
)

model = AutoModelForCausalLM.from_pretrained(
    BASE_MODEL, quantization_config=bnb, device_map={"": 0}, torch_dtype=COMPUTE_DTYPE,
)
model = prepare_model_for_kbit_training(model, use_gradient_checkpointing=True)
model.config.use_cache = False          # required with gradient checkpointing. Turn back ON to serve.

lora = LoraConfig(
    r=16, lora_alpha=32, lora_dropout=0.05, bias="none", task_type="CAUSAL_LM",
    target_modules=["q_proj", "k_proj", "v_proj", "o_proj",
                    "gate_proj", "up_proj", "down_proj"],
)
model = get_peft_model(model, lora)
model.print_trainable_parameters()
```

```
bitsandbytes 0.49.2
/usr/local/lib/python3.12/dist-packages/bitsandbytes/backends/cuda/ops.py:213: FutureWarning: _check_is_size will be removed in a future PyTorch release along with guard_size_oblivious.    Use _check(i >= 0) instead.
  torch._check_is_size(blocksize)
4-bit NF4 on GPU: OK

config.json: 100% 855/855 [00:00<00:00, 106kB/s]
`torch_dtype` is deprecated! Use `dtype` instead!
model.safetensors.index.json: 23.9k/? [00:00<00:00, 2.75MB/s]
Fetching 4 files: 100% 4/4 [00:40<00:00, 17.51s/it]
model-00001-of-00004.safetensors: 100% 4.98G/4.98G [00:36<00:00, 114MB/s]
model-00004-of-00004.safetensors: 100% 1.17G/1.17G [00:14<00:00, 79.7MB/s]
model-00003-of-00004.safetensors: 100% 4.92G/4.92G [00:36<00:00, 99.8MB/s]
model-00002-of-00004.safetensors: 100% 5.00G/5.00G [00:40<00:00, 145MB/s]
Loading checkpoint shards: 100% 4/4 [00:17<00:00, 3.73s/it]
/usr/local/lib/python3.12/dist-packages/bitsandbytes/backends/cuda/ops.py:213: FutureWarning: _check_is_size will be removed in a future PyTorch release along with guard_size_oblivious.    Use _check(i >= 0) instead.
  torch._check_is_size(blocksize)
generation_config.json: 100% 184/184 [00:00<00:00, 25.1kB/s]
trainable params: 41,943,040 || all params: 8,072,204,288 || trainable%: 0.5196
```

Figure 25 QLoRA fine-tuning of Llama-3.1-8B-Instruct in Google Colab on an A100 GPU. -1



9. Train

~400 examples is a small corpus, which is correct for this job: teaching a format needs far less data than teaching knowledge, and a small clean corpus beats a large noisy one every time.

Three epochs at an effective batch of 16 is roughly 75 optimiser steps. Watch `eval_loss` : if it turns upward while `train_loss` keeps falling, the model has started memorising the specific procedures rather than the answer shape — stop at the best checkpoint and, more usefully, go widen the corpus rather than turning the learning rate down.

```python
from transformers import TrainingArguments, Trainer
import transformers, random, numpy as np

random.seed(SEED); np.random.seed(SEED); torch.manual_seed(SEED)

args = TrainingArguments(
    output_dir="out",
    num_train_epochs=3,
    per_device_train_batch_size=BATCH,
    per_device_eval_batch_size=1,
    gradient_accumulation_steps=ACCUM,
    learning_rate=1e-4,
    lr_scheduler_type="cosine",
    warmup_ratio=0.03,
    logging_steps=5,
    eval_strategy="epoch",
    save_strategy="epoch",
    save_total_limit=2,
    load_best_model_at_end=True,
    metric_for_best_model="eval_loss",
    greater_is_better=False,
    bf16=BF16_OK,
    fp16=not BF16_OK,
    optim="paged_adamw_8bit",
    gradient_checkpointing=True,
    report_to="none",
    seed=SEED,
)

trainer = Trainer(model=model, args=args,
                  train_dataset=train_ds, eval_dataset=val_ds,
                  data_collator=collate)
trainer.train()
```

[75/75 15:42, Epoch 3/3]

Epoch	Training Loss	Validation Loss
1	0.205100	0.262053
2	0.136700	0.237796
3	0.109900	0.240252

```
TrainOutput(global_step=75, training_loss=0.20535828868548076, metrics={'train_runtime': 955.8409, 'train_samples_per_second': 1.218, 'train_steps_per_second': 0.078, 'total_flos': 7.873631990194176e+16, 'train_loss': 0.20535828868548076, 'epoch': 3.0})
```

Figure 26 QLoRA fine-tuning of Llama-3.1-8B-Instruct in Google Colab on an A100 GPU. - 2

The Sinhala and Tamil layer was implemented as a mask–split–translate–restore–verify pipeline over NLLB-200. Numbers, percentages, monetary amounts, form codes, URLs and e-mail addresses are replaced with sentinel tokens before translation and restored afterwards; any sentence in which a critical token fails to survive is reverted to English, so the system never ships a translated answer containing a corrupted figure.

The frontend was implemented in Next.js with a chat view, a sector-scoped survey applying client-side double gating, and a research dashboard. The backend was containerised with Docker — the vector store being built into the image — and deployed on Railway behind a health-checked root endpoint.

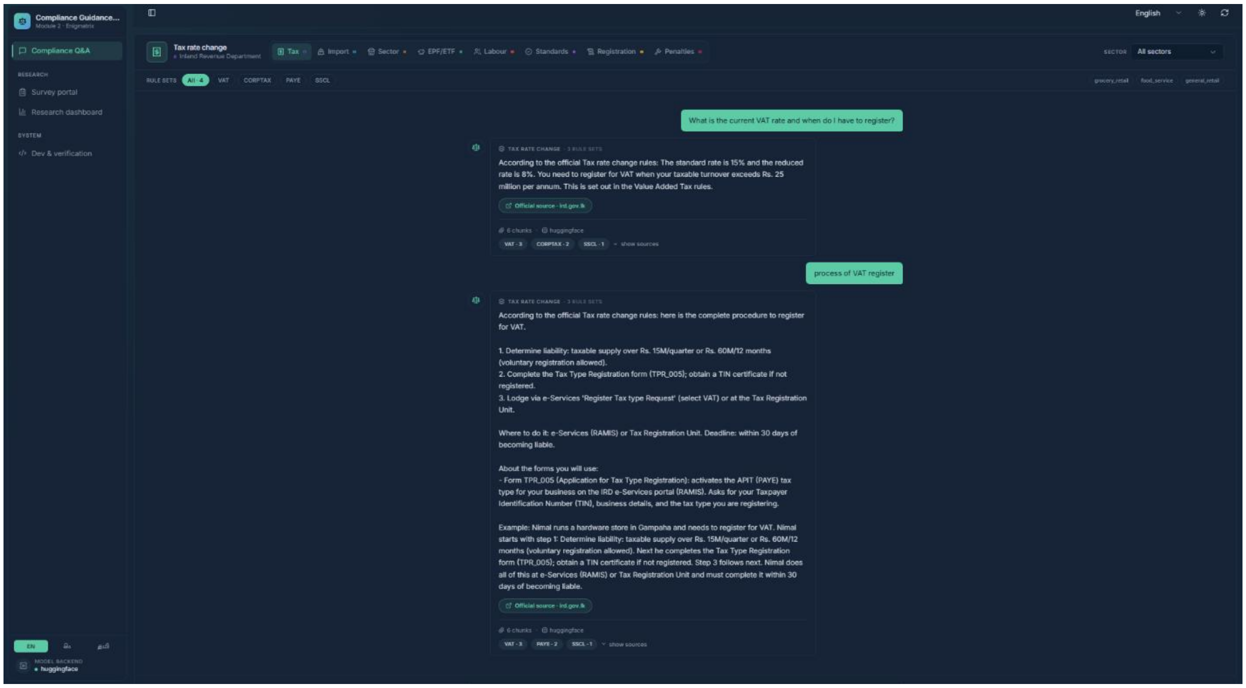


Figure 27 Chat interface returning a grounded, procedurally complete answer with the source authority named.

6.3.3 SME Compliance Risk Prediction

The module was implemented in Python. Data handling uses pandas, modelling uses scikit-learn, figures are produced with matplotlib, and the web service is built on FastAPI with a lightweight SQLite store. Development followed a test-driven approach, and the module is covered by sixty-one automated tests spanning recoding, modelling, statistical procedures, explanation generation and the end-to-end platform workflow.

The recoding component is implemented as a declarative mapping from survey answer text to analytical codes, together with the ordinal encodings and the outcome-label rule. Two implementation details proved consequential. Answer strings are normalised for case, whitespace and Unicode punctuation before lookup, because exports differ in their use of typographic quotation marks and dashes. Separately, the literal answer "None" to the missed-deadline question had to be protected explicitly, since it is treated as a missing-value token by default in pandas; without this protection a valid and meaningful answer would have been silently discarded. Values that fail to map are reported and cause the build to fail rather than being converted to missing data.

The statistical procedures were implemented directly rather than taken from a package, both to avoid unnecessary dependencies and to permit verification. DeLong's test was implemented following the efficient formulation of Sun and Xu [19] and cross-checked against an independently written implementation, with agreement to within numerical tolerance. SHAP values are computed in closed form for the linear model, and the implementation was verified against the additivity property that defines the method, holding to machine precision. Significance for the descriptive associations is obtained by permutation rather than by the asymptotic chi-squared distribution, which remains valid with the small contingency cells present at this sample size, and every reported significance value is accompanied by a Cramér's V effect size so that a statistically detectable but negligible association cannot be mistaken for a substantive one.

The positive-unlabelled correction implements the Elkan and Noto estimator [16]. A portion of the labelled positives is withheld, a classifier is trained against the unlabelled pool, and the label frequency is estimated as the mean score assigned to the withheld positives. The implementation was validated by recovering a known label frequency from data constructed with separable classes. Because the correction is a monotonic rescaling, discrimination is preserved by construction, which is precisely why it constitutes a robustness argument.

The analytical pipeline is exposed as a sequence of command-line entry points that build the dataset, fit and compare the model ladder, compute attributions, run the robustness analysis, produce the descriptive results and assess representativeness. Each writes both a console report and a persistent artefact, so that every figure and table in Chapter 7 can be regenerated from the raw survey export.

The deployed platform is organised so that all substantive logic resides in framework-independent modules, with the web layer acting only as a thin transport. The prediction service loads the serialised model and returns a risk estimate with its factor attribution; the explanation layer converts that structure into text in three languages; the matching engine determines which businesses a regulatory change affects; and the persistence layer records registered profiles, published changes and the resulting alerts. This separation allows the same scoring entry point to be reused by an alternative delivery channel, such as a messaging interface, without modification. The interface accepting regulatory change events is defined as an explicit event structure, so that events published by the Regulatory Change Awareness module can be consumed without further adaptation.

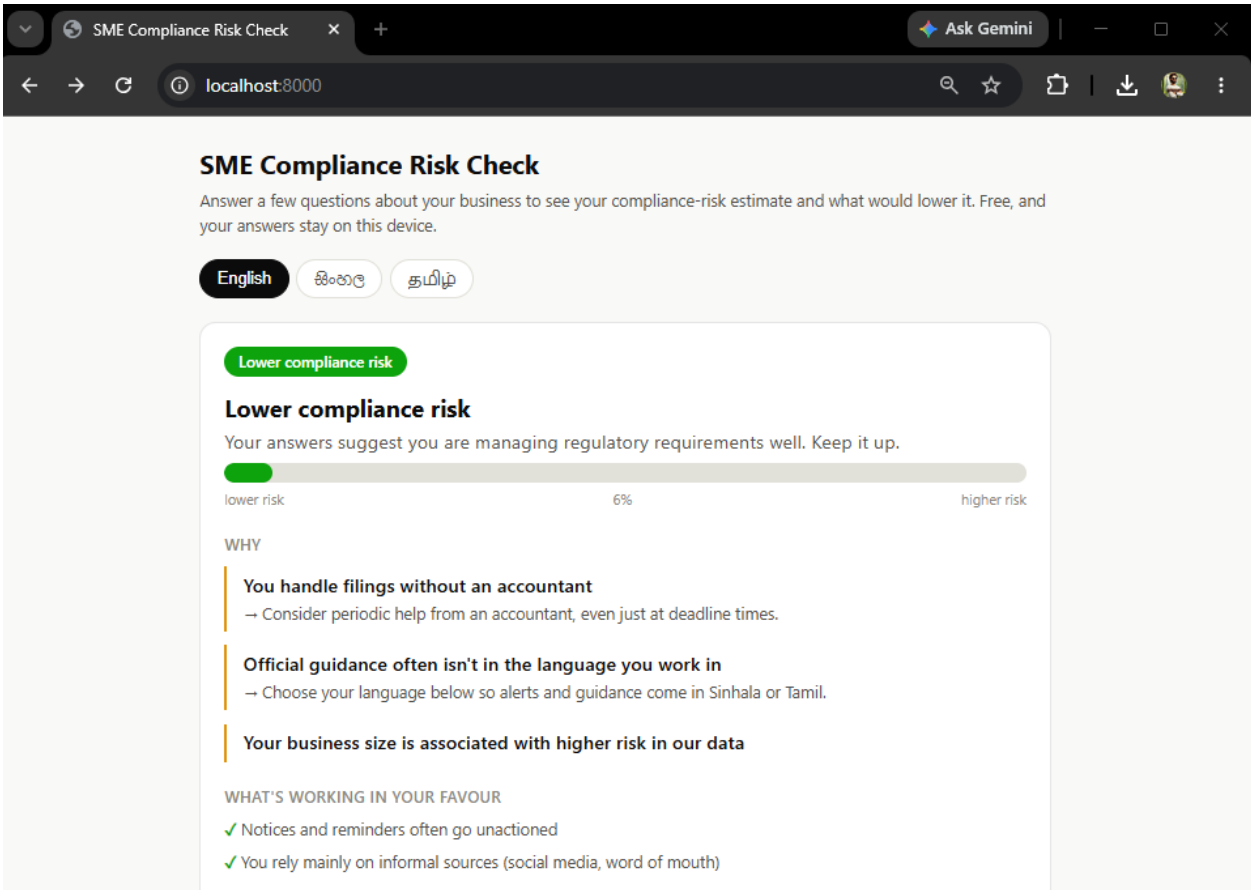


Figure 28 Dashboard of scoring the SME

6.3.4 Regulatory Misinformation Spread

The module was written mostly in Python, the development environment was Google Colab. The dataset was preprocessed, prepared, and the model was developed, evaluated, and orchestrated using Python, while the manual experimentation and usage of GPUs were carried out using the Google Colab platform notebook environment.

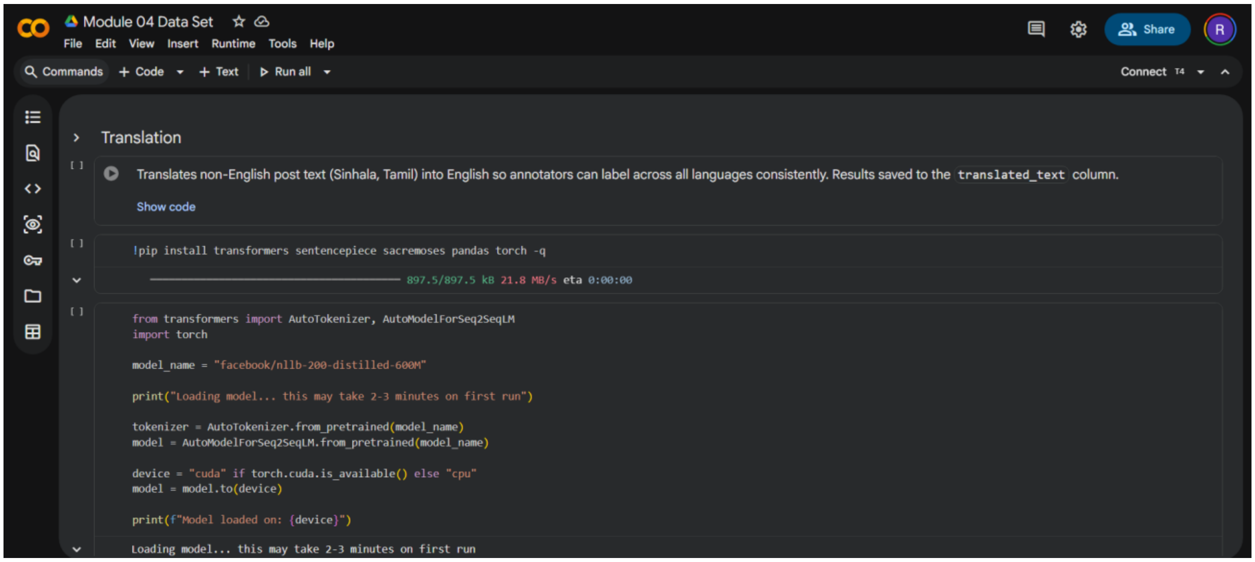


Figure 29 Google Colab notebook used for model development, configured with T4 GPU.

The first 200 posts were independently annotated by two annotators using Label Studio and the same examples were then annotated again by both annotators—after which the agreement was computed. For the other 800 posts, Google Sheets was used due to the fact that the annotation schema was already stabilised and the larger portion of the data could be annotated in a lightweight spreadsheet-based workflow.

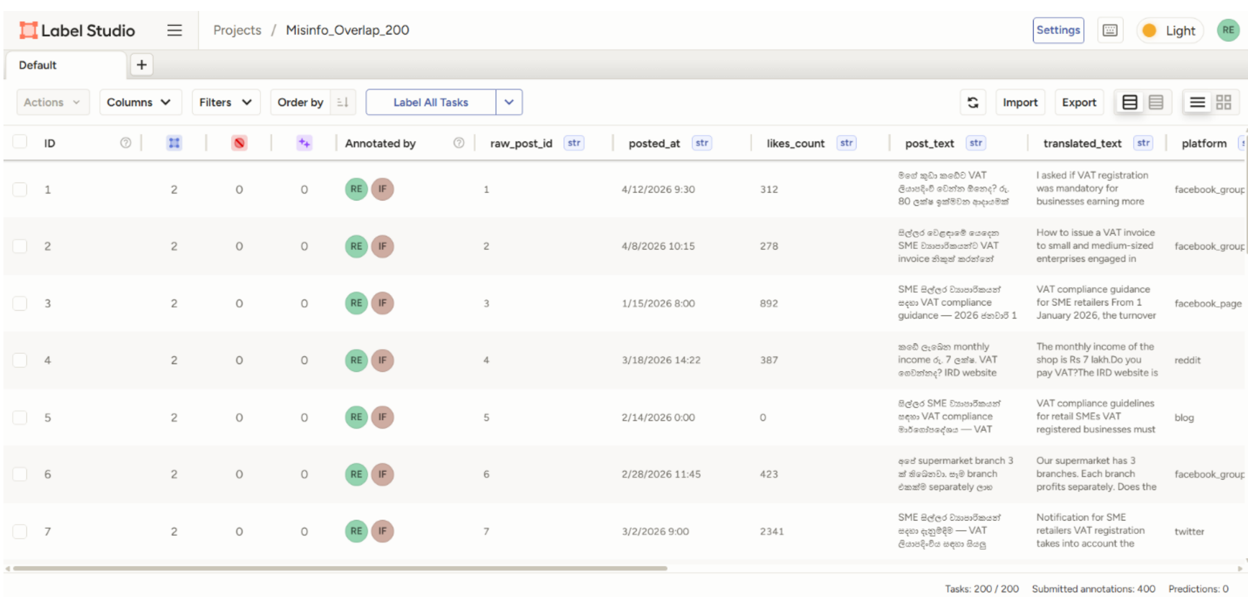


Figure 30 Annotation interface – Label Studio

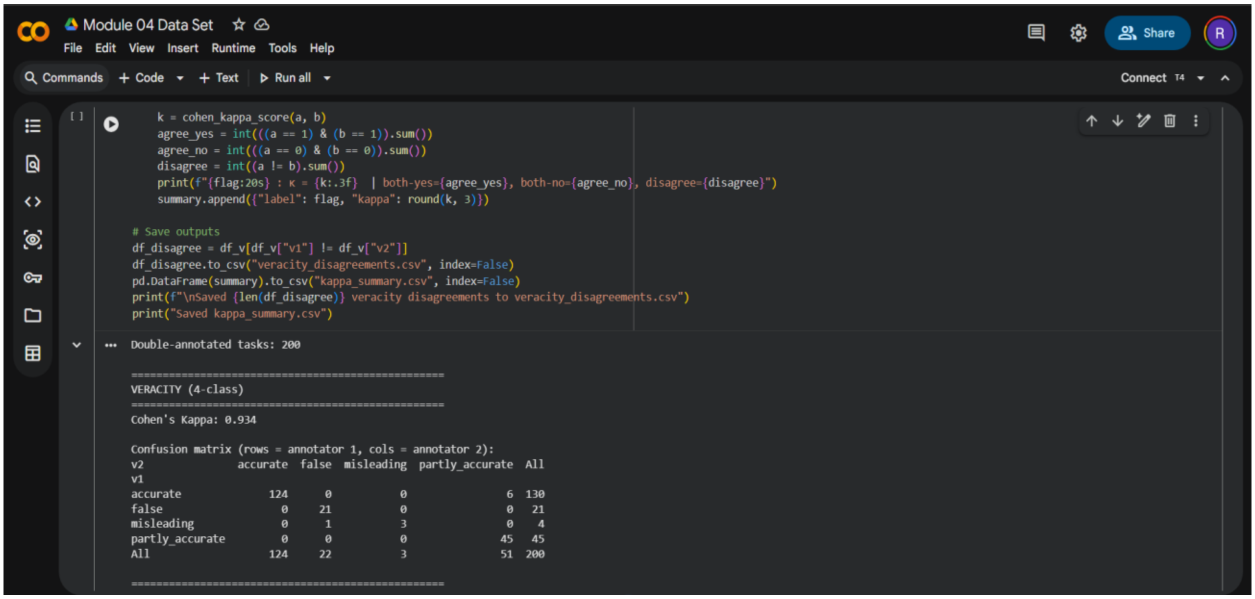


Figure 31 Inter-annotator agreement on the 200-post double-annotated subset

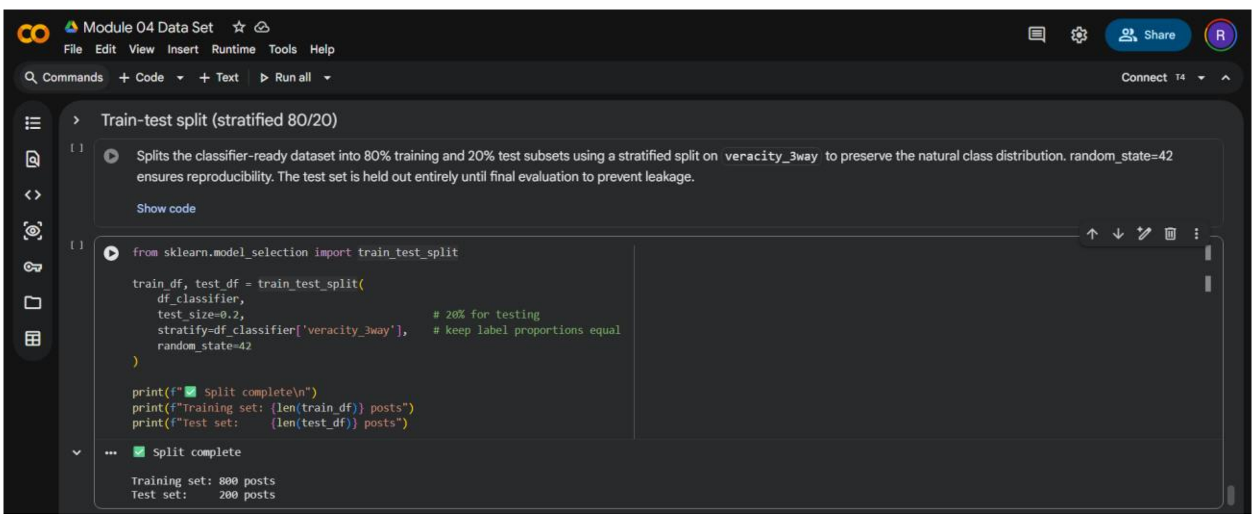


Figure 32 Train–test split (800/200) with stratified label distribution

XLM-RoBERTa was used for the multilingual transformer approach. From the uploaded module reference, the fine tuning was done on 800 training posts, and evaluated on 200 test posts using class weighting to balance out class imbalance and main comparison metric being macro-F1. This was a sensible implementation decision as the corpus contained English, Sinhala, Tamil, and code-mixed text, which are all covered in the multilingual scope of XLM-R.

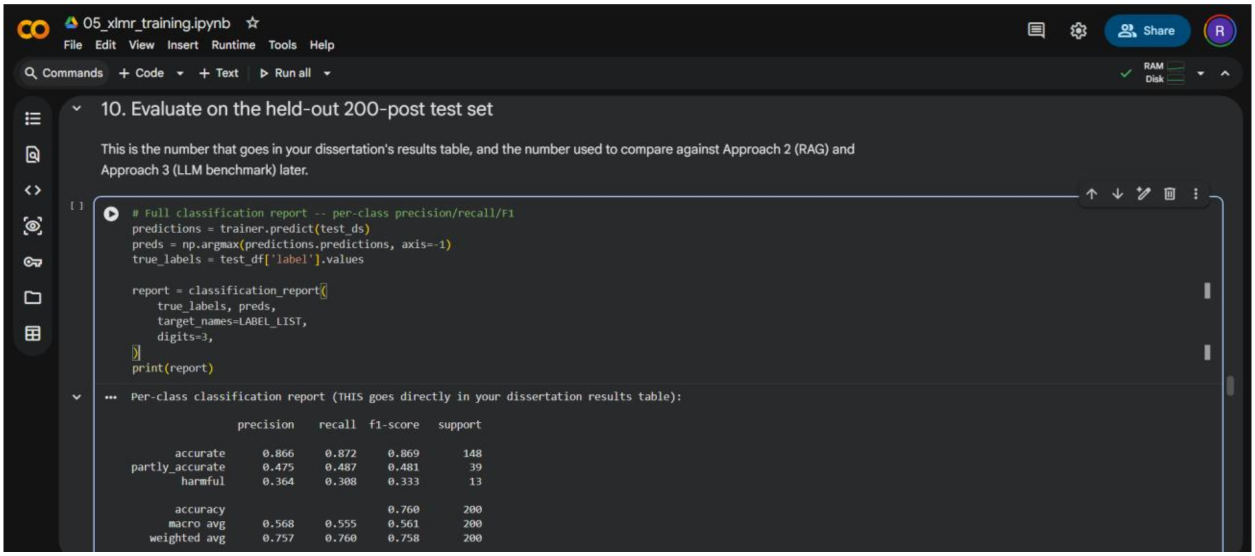


Figure 33 XLM-RoBERTa fine-tuning in Colab – class-weighted training, macro-F1 evaluation, and confusion matrix on the 200-post test set.

Retrieval-based implementation involved using Sentence Transformers for embeddings, ChromaDB for vector storage in the final prediction stage. The RAG design was chosen due to its superior performance over other designs when tested on the held-out test set, and because it was more appropriate for this domain to access retrieved regulatory information rather than to directly prompt the relevant facts.

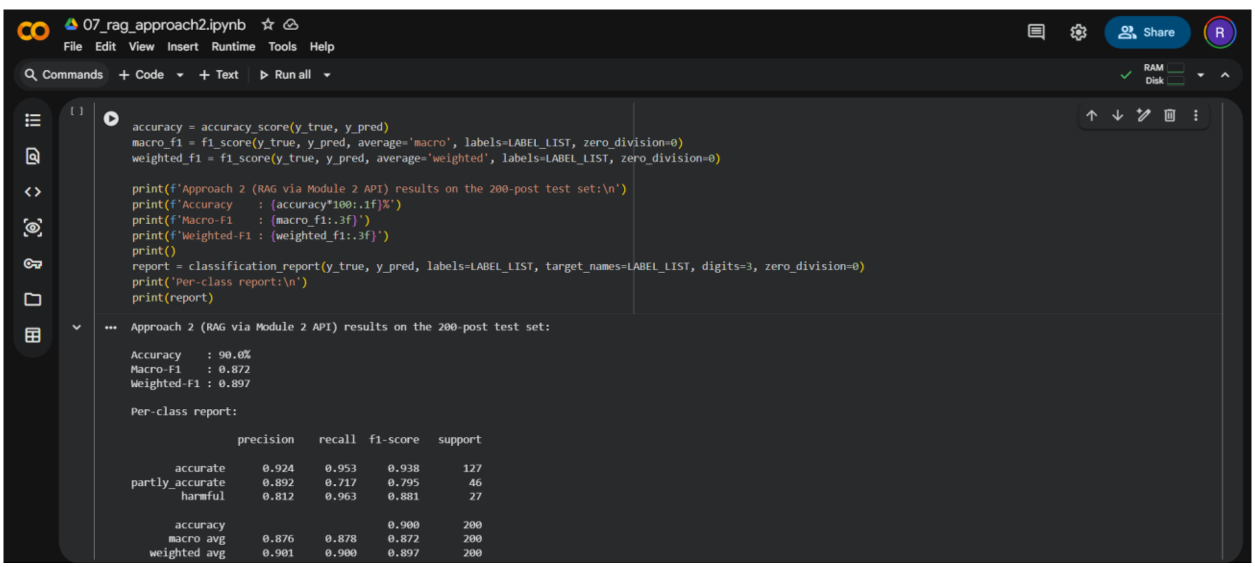


Figure 34 RAG verifier in Colab – class-weighted training, macro-F1 evaluation, and confusion matrix on the 200-post test set.

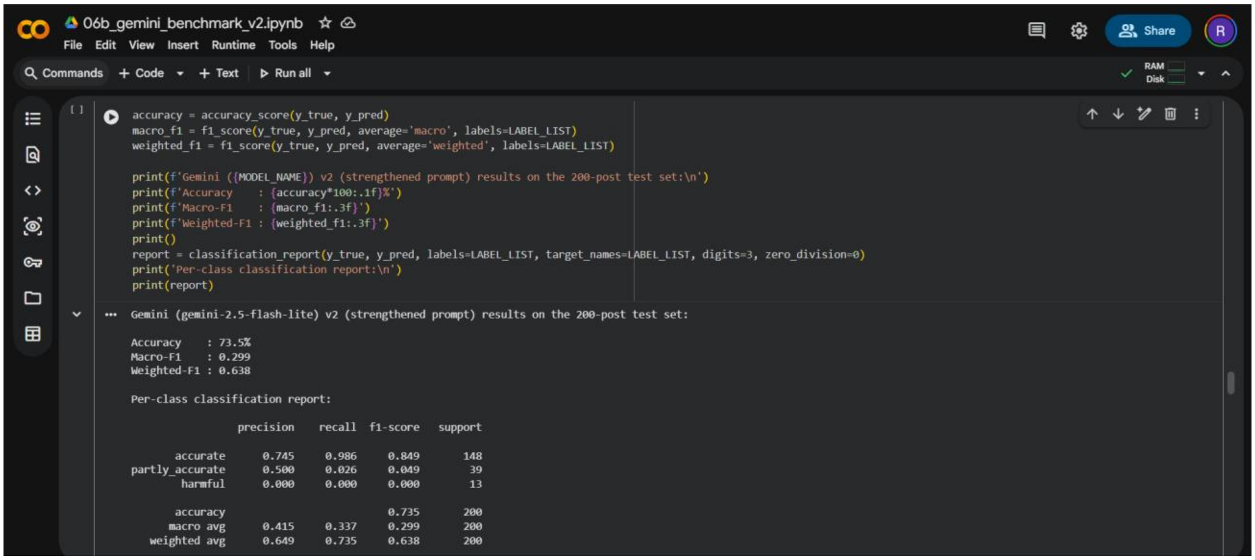


Figure 35 Gemini Benchmark verifier in Colab – class-weighted training, macro-F1 evaluation, and confusion matrix on the 200-post test set.

Care has been taken in handling the data sets to avoid leakage. All the 1,000-post corpus underwent cleaning and annotation, followed by its division (without overlap) into 800 training posts and 200 testing posts. All three approaches were made with the same split to make for a fair comparison. The accuracy, macro-F1, weighted-F1, and classification on per-class reports were used for the evaluation, and the RAG approach was the best performing model overall.

Figure 36 Comparative evaluation of the three classifier approaches on the same 200-post test set.

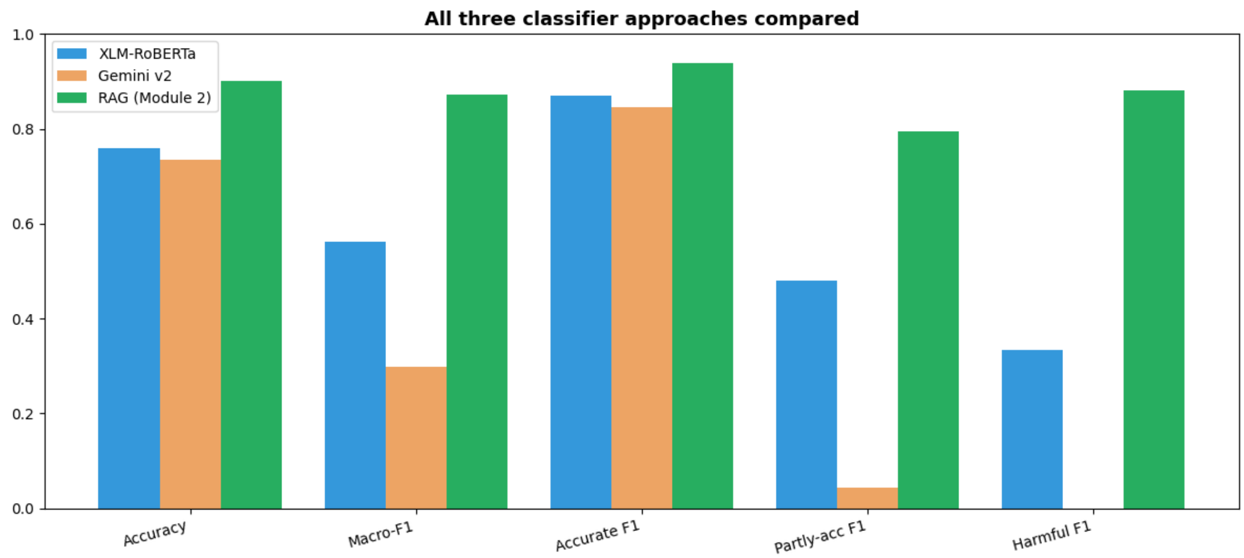


Figure 37 Bar chart comparing all the three approaches

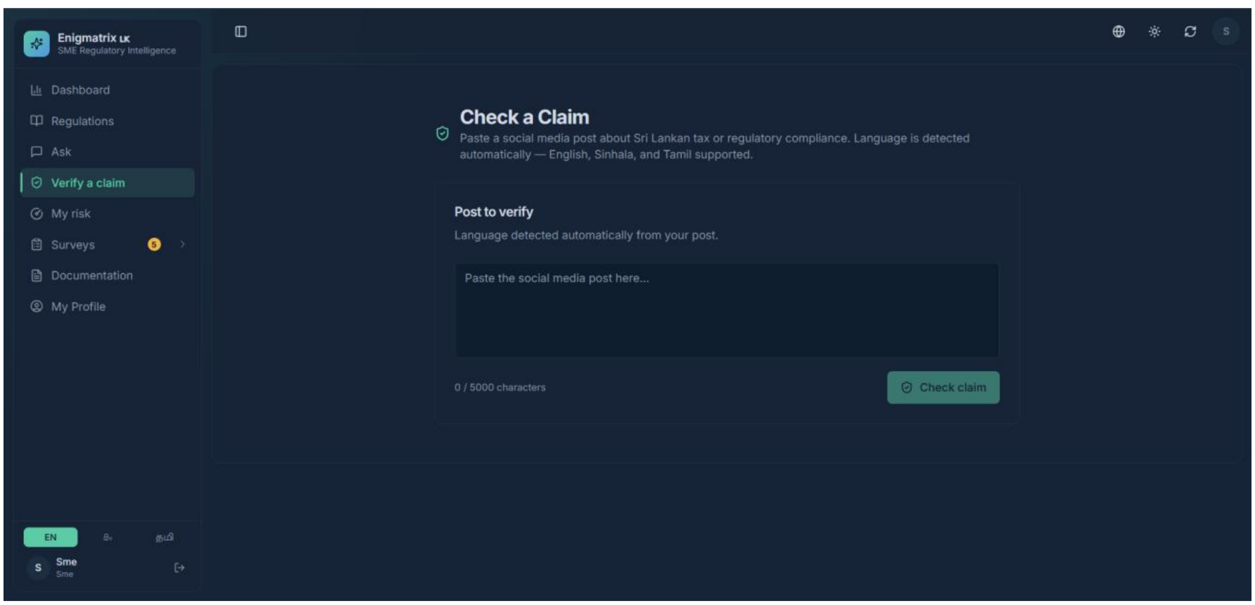


Figure 38 Frontend claim-verification interface

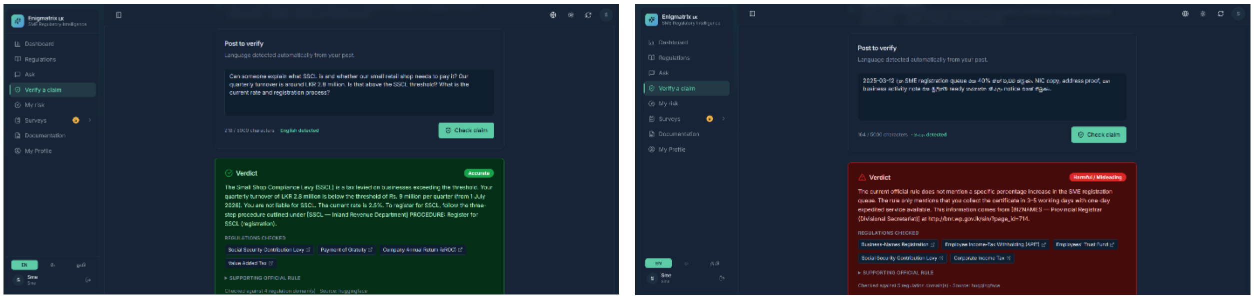


Figure 39 Verdict screen showing veracity classification, explanation, and cited regulatory evidence from the knowledge base.

Chapter 7 - Evaluation

7.1 Metrics Used

This section defines every metric used in the dissertation. Module 1 evaluates three different kinds of object - a classifier, an extraction pipeline and a diffusion process - and each requires its own measure. The remaining modules add two more. Module 2 evaluates generated guidance, where the unit of assessment is not a class label but an answer that either possesses or lacks a set of required properties. Module 3 evaluates a ranked risk estimate together with a hypothesis about which variables carry the signal, which calls for threshold-free discrimination, a paired significance test and an attribution measure. Module 4 evaluates a claim classifier and, separately, the reliability of the corpus it was trained on.

7.1.1 Accuracy

Accuracy is the proportion of predictions that are correct.

```
Accuracy = number_of_correct_predictions / total_predictions
```

Accuracy is reported for completeness but is **not** the primary metric for classification in this project. The gold dataset is severely imbalanced: SECTOR_SPECIFIC accounts for 83.9 % of records, so a degenerate model that always predicts that class would score approximately 0.87 accuracy while being useless. Accuracy is meaningful here only for balanced binary fields such as is_sme_relevant, and for stage-progression checks.

It is not used at all for the other two modelling modules, for different reasons. Module 3's outcome is close to balanced at 46 per cent, so imbalance is not the objection; the objection is that a single accuracy figure fixes an operating threshold that the deployment does not fix, and discrimination is therefore reported by the threshold-free measures of section7.1.11. Module 2's output is not a class at all, so accuracy is undefined for it; its answers are scored on the property set of section7.1.16.

7.1.2 Precision

Precision is the proportion of predicted positives that are correct. For category *i*:

```
Precision_i = TP_i / (TP_i + FP_i)
```

Precision matters most for alerting. A false positive is an alert sent to an SME for a regulation that does not affect them; a system with low precision trains its users to ignore it.

7.1.3 Recall

Recall is the proportion of actual positives that are found.

```
Recall_i = TP_i / (TP_i + FN_i)
```

Recall matters most for SME relevance. A false negative is a binding regulation that never reaches the business it binds — the exact failure this project exists to prevent. For that reason is_sme_relevant recall below 0.85 is treated as a blocking result regardless of the accuracy figure.

7.1.4 F1-Score

F1 is the harmonic mean of precision and recall, and is the appropriate single summary when both error types carry cost.

```
F1_i = 2 * Precision_i * Recall_i / (Precision_i + Recall_i)
```

Macro-F1 averages F1 over classes with equal weight and is the **primary metric** for category classification in this project:

```
Macro-F1 = (1 / K) * sum_i F1_i
```

Macro-F1 is chosen precisely because it refuses to be flattered by the majority class: a model that ignores the seven minority categories cannot achieve a high macro-F1 no matter how well it predicts SECTOR_SPECIFIC. The project gate for RQ1 is macro-F1 ≥ 0.92 with no language slice more than 8 percentage points below the overall score.

For multi-label sector relevance, each sector is treated as an independent binary problem:

```
Micro-F1 = 2 * sum_s TP_s / (2 * sum_s TP_s + sum_s FP_s + sum_s FN_s)
Macro-F1 = mean_s F1_s
```

Module 4 reports per-class precision, recall and F1 on its four-way schema for the same reason: the four classes are unevenly represented, and a single aggregate would hide whichever class the classifier handles worst. Because the classes are ordered in severity, per-class recall on the misleading and false categories is the operationally decisive figure.

7.1.5 Cohen's Kappa

Cohen's kappa measures agreement between two annotators after discounting agreement expected by chance:

```
kappa = (p_o - p_e) / (1 - p_e)
```

where p_o is observed agreement and p_e is the agreement expected from the marginal distributions. Kappa rather than raw agreement is the headline reliability figure because raw agreement is inflated when one class dominates — which is exactly the case here. Sector labels are multi-label, so kappa is computed per sector and averaged, with a set-level exact agreement reported alongside. Interpretation follows the conventional bands: above 0.80 near-perfect, 0.61–0.80 substantial.

Cohen's kappa is defined for exactly two annotators and requires complete overlap, neither of which holds for the Module 4 corpus, where three annotators contributed and only a subset was double-annotated. Krippendorff's alpha generalises the same idea and is reported alongside it:

alpha = 1 - (D_o / D_e)

where D_o is observed disagreement and D_e the disagreement expected by chance. Alpha admits any number of annotators, tolerates missing values, and is defined for nominal, ordinal and interval data, so a single coefficient covers both corpora. Both statistics are computed on the double-annotated subset, and in both modules full-scale annotation proceeded only after the coefficient cleared the threshold fixed in advance — the reliability of the dataset is therefore a gate on the work, not a description of it after the fact.

7.1.6 Field, record and stage accuracy

Extraction quality is scored against a sealed baseline at three granularities:

```
field_accuracy(f)  = mean( s(r,f) for r where f in V(r) )
record_score(r)    = sum_{f in V(r)} s(r,f) / |V(r)|
record_score_w(r)  = sum_{f in V(r)} w(f) * s(r,f) / sum_{f in V(r)} w(f)

stage_only(S)      = mean( s(r,f) for r with gold_status >= S, f in stage_fields(S) )
cumulative(S)      = mean( s(r,f) for r with gold_status >= S, f in cumulative_fields(S) )
progression(S)     = |{r : pred_status >= S and gold_status >= S}| / |{r : gold_status >= S}|

overall_micro      = sum_r sum_f s(r,f) / sum_r |V(r)|
overall_macro      = mean_r record_score(r)
```

where s(r,f) is the comparator score in [0,1] for field *f* of record *r*, w(f) is the field weight and V(r) is the set of fields with a valid gold value. Weights are 3.0 for hard identifiers, 2.5 for core NLP outputs, 2.0 for structural and list fields, 1.0 for bulk text, 0.5 for timestamps and 0.0 for confidence fields. Releases are gated on the weighted cumulative score, with unweighted micro and macro published alongside as an honesty check.

7.1.7 Character and Word Error Rate

OCR-sensitive text fields are scored by edit distance rather than exact match:

```
CER = edit_distance(chars_pred, chars_gold) / len(chars_gold)
WER = edit_distance(words_pred, words_gold) / len(words_gold)
raw_text_score = clamp(1 - CER, 0, 1)
```

CER and WER are computed **per language**. A single averaged text score would allow a weak script to hide inside a strong one, which is precisely the failure mode RQ2 exists to detect.

7.1.8 Calibration: ECE and Brier score

Confidence is never mixed into accuracy. It is evaluated separately:

```
Brier = mean( (predicted_probability - actual_outcome)^2 )
ECE   = sum_b (|B_b| / N) * | accuracy(B_b) - confidence(B_b) |
```

where B_b is confidence bin *b*, using bins of width 0.1. Calibration matters operationally because the 0.55 threshold decides what fraction of the corpus reaches a human reviewer.

Calibration is reported for Module 3's risk model for a related but distinct reason. The positive-unlabelled correction of section7.1.15 shifts mean predicted risk materially while leaving the hypothesis test unchanged, so a model may be simultaneously well ordered and badly calibrated. Since the ordering is what the hypothesis concerns and the calibration is what any deployed decision threshold depends on, the two are reported separately and neither is allowed to stand in for the other.

7.1.9 Timeliness and diffusion metrics

```
extraction_latency     = extracted_at      - source_discovered_at
classification_latency = classified_at     - preprocessed_at
alert_latency          = alert_created_at  - gazette_published_at
diffusion_lag(channel) = first_seen_at(channel) - gazette_published_at
awareness_lag          = sme_first_awareness_at - gazette_published_at

alert_precision = relevant_alerts_sent / total_alerts_sent
alert_recall    = relevant_alerts_sent / total_relevant_regulations_for_sme
```

Lag distributions are reported as median and interquartile range rather than mean and standard deviation, because diffusion lag is strongly right-skewed.

7.1.10 Model drift

The confidence distribution of production predictions is compared against the reference distribution using Kullback–Leibler divergence; a value above 0.15 triggers retraining.

7.1.11 Ranked discrimination: ROC-AUC and PR-AUC

The area under the receiver operating characteristic curve is the probability that a randomly chosen positive case is ranked above a randomly chosen negative one. It summarises a model's ordering across every possible threshold rather than at one chosen threshold, which is the property Module

3 needs: the deployed system presents a risk band and a factor list, and does not commit to a single cut-off.

$$AUC = P(\ score(x\_positive) > score(x\_negative)\ )$$
$$PR\text{-}AUC = \text{area under the precision-recall curve}$$

An AUC of 0.5 is chance and 1.0 is perfect separation. PR-AUC is reported alongside because it is sensitive to performance on the positive class in a way ROC-AUC is not, and because their chance levels differ: ROC-AUC is 0.5 under chance regardless of the base rate, whereas PR-AUC under chance equals the base rate itself — 0.46 for this sample. Quoting a PR-AUC without that reference point would overstate it.

Both are computed on cross-validated out-of-fold predictions, so no observation contributes to fitting the model that scores it. With a sample of 300 this is not a refinement but a requirement; an in-sample AUC on a dataset this size would be optimistic by an unknown margin.

7.1.12 Comparing correlated ROC curves: the DeLong test

The central comparison in Module 3 is between two models scored on the same businesses. Their ROC curves are therefore correlated, and an unpaired comparison of two AUCs would be invalid — it would treat shared variance as independent evidence and overstate significance. DeLong's test estimates the covariance of the two AUC statistics from the same sample and forms:

$$z = (AUC\_2 - AUC\_1) / sqrt(\ Var(AUC\_1) + Var(AUC\_2) - 2 * Cov(AUC\_1, AUC\_2)\ )$$

The statistic is referred to the standard normal distribution. Reported for each comparison are the difference in AUC, its 95 per cent confidence interval, the z statistic and the two-sided p value. The pre-registered decision rule for the module's hypothesis is a gain of at least 0.03 with p below 0.05, both fixed before the analysis was run. The implementation uses the fast Sun and Xu formulation of the same estimator, which is algorithmically more efficient but statistically identical.

7.1.13 Association strength: Cramér's V

Before any model is fitted, the descriptive layer asks whether the claimed relationships are present in the raw cross-tabulations at all. Cramér's V measures the strength of association between two categorical variables, normalising the chi-square statistic so that values are comparable across tables of different size:

$$V = sqrt(\ chi2 / (\ n * min(rows - 1, cols - 1)\ )\ )$$

V ranges from 0 to 1. It is preferred to raw chi-square because chi-square grows with sample size and with table dimensions, so two chi-square values are not comparable across variables with different numbers of levels — which is exactly the comparison being made when ranking predictors by effect size. Each V is reported with its p value. This layer is computed independently of any model, so that the pattern is shown to exist in the data rather than only in a fitted object.

7.1.14 Attribution: Shapley values and block share

Feature attribution answers a different question from performance: not how well the model discriminates, but which variables carry the signal. For a linear model the Shapley value has a closed form, so the attribution is exact rather than sampled:

*phi_i(x) = beta_i * (x_i - E[x_i])   (in log-odds)*
block_share(B) = sum_{i in B} |phi_i| / sum_{all i} |phi_i|

where beta_i is the fitted coefficient and E[x_i] the training-set mean of feature i. Contributions on one-hot columns are summed back to their source feature, so an attribution is reported per variable rather than per encoded column. Because the quantity is analytic, it is reproducible to the last decimal and costs nothing to compute at request time — the same numbers shown to an SME in the deployed explanation are the numbers reported here.

Block share aggregates attribution over a group of features and is what makes the module's hypothesis quantitative rather than rhetorical: it states what proportion of total attributed importance the information-barrier block carries against the demographic block. The pre-registered condition accompanying it requires at least two information features among the five highest-ranked.

7.1.15 Positive-unlabelled correction and label frequency

Module 3's outcome label is asymmetric by construction, and every result depends on how that asymmetry is treated. A positive is a confirmed self-reported compliance failure. A negative is only the absence of a reported failure — the respondent is *unlabelled*, not confirmed compliant. Treating unlabelled cases as true negatives, which any standard classifier does implicitly, biases the estimate by an unknown amount.

The robustness analysis therefore re-estimates the comparison under a positive-unlabelled correction, which models the observed labelling as a corruption of the true outcome through a label frequency:

c = P(labelled = 1 | y = 1)

P(y = 1 | x) = P(labelled = 1 | x) / c

The reported quantity is whether the improvement attributed to the information block survives the correction, not the corrected AUC itself. The estimated label frequency c is reported as directional evidence only: the estimator is known to be unreliable at moderate levels of discrimination, and the figure must not be read as a population prevalence of undisclosed non-compliance. Stating that bound explicitly is part of the metric's definition, not a caveat attached to it afterwards.

7.1.16 Generated-answer quality

A generated compliance answer cannot be scored as correct or incorrect, because it is not a class label. It is scored instead on whether it possesses the properties that make procedural guidance usable and safe. An automated harness evaluates each answer on a held-out split against the following columns.

Table 7.1 — Answer-quality columns for generated guidance

Column	Definition	Why it is scored
Cites the source	Answer names the issuing authority or official document	An uncited compliance figure cannot be checked by the reader
Numbered procedure	Steps presented in explicit order	Procedural knowledge is the gap the module addresses
Worked example	At least one concrete instantiation is given	Abstract rules are the form SMEs already fail to act on
Explains named forms	Every form code mentioned is explained in plain language	A form code alone transfers no procedural knowledge
Grounded figures only	Every rate, threshold and deadline appears in the retrieved context	The principal hallucination risk in a compliance setting
Refuses the unanswerable	Declines when retrieval returns insufficient context	Silence is safer than invention
Refuses the answerable (lower is better)	Wrongly declines a question the context supports	Guards against improvement obtained by refusing more often
Content recall	Proportion of reference content reproduced	Completeness, independent of the properties above
Median answer length	Characters per answer	Diagnostic, not a score — see below

Two of these columns exist specifically to close loopholes in the others. Refusing the answerable is reported because a model can raise its correct-refusal rate simply by refusing more often, and the two figures must move independently for the gain to be real. Median answer length is reported as a diagnostic rather than scored, because length falling while content recall rises indicates a denser answer, whereas both falling together would indicate truncation — the same numbers admit opposite readings without it.

Attribution follows a three-configuration design rather than a before-and-after comparison. The stock base model is evaluated under identical prompt and identical retrieval as a single-variable control, so that the difference between it and the adapted model is attributable to the adaptation alone. A direct comparison against the previously deployed model would move two variables at once and support no such attribution.

The bound on these measures is stated with them: the graders are automated and pattern-based, so they detect the presence of a required element rather than adjudicating its correctness. They measure faithfulness to the verified corpus, not independent real-world truth.

7.1.17 Figure survival in translation

Translation of compliance text has a failure mode that no general translation-quality score captures. A fluent translation that alters a rate, a threshold, a deadline or a form code produces confidently wrong guidance, and is worse than no translation at all. Figure survival measures it directly:

$$survival(L) = preserved\_protected\_spans(L) / total\_protected\_spans(L)$$

A protected span is any currency amount, percentage, form code, bare number, date, URL or email address identified in the source. Spans are classed critical or cosmetic, and survival is reported per language, because a single averaged figure would let a strong script conceal a weak one — the same reasoning that governs per-language CER in section7.1.7.

Survival is never reported alone. A system can reach a survival of 1.0 trivially by translating nothing, so it is published jointly with translation coverage — the proportion of sentences actually delivered in the target language. The pair is the measure; either number in isolation is uninformative.

7.2 Module-wise Evaluations

7.2.1 Regulatory Change Awareness Gap

7.2.1.1 Experimental setup

Two environments were used, and the distinction is material to how the results should be read. The local workstation has no CUDA device and was used only for pipeline execution, extraction and measurement runs, baseline training and a single-epoch smoke test. **No model-quality claim rests on the local machine.** All transformer training and evaluation is executed on free GPU notebook platforms.

Table 7.2 — Experimental environments

Item	Local workstation	Colab / Kaggle GPU
CPU	11th Gen Intel(R) Core(TM) i5-1135G7 @ 2.40GHz	Provided by platform
Logical cores	8 logical processors, 4 physical cores	Provided by platform
RAM	**19.70 GB**	~13 GB (Colab) / ~29 GB (Kaggle)
GPU	None — torch.cuda.is_available() is False	**2 x NVIDIA Tesla T4, 15,360 MiB each**
Operating system	**Microsoft Windows 11 Pro, version 10.0.26100**	Linux 6.12.90+ x86_64, glibc 2.35
Python / PyTorch / Transformers	**Python version not captured; PyTorch 2.12.0+cpu; Transformers 4.57.6**	**Python ;PyTorch 2.10.0+cu128; Transformers 5.0.0**
Role	Extraction, preprocessing, measurement, TF-IDF baselines, CPU smoke test	XLM-R + LoRA training, slice evaluation, ONNX export, NLLB backfill
Runtime per seed	not applicable	minutes for 560 training rows

7.2.1.2 Dataset composition

Table 7.3 — Change category distribution in the v1 gold dataset (n = 800)

Category	Count	Share
SECTOR_SPECIFIC	671	83.9 %
TAX_RATE_CHANGE	56	7.0 %
IMPORT_EXPORT	32	4.0 %
LABOUR_LAW	27	3.4 %
BUSINESS_REGISTRATION	5	0.6 %
PENALTY_ENFORCEMENT	5	0.6 %

Category	Count	Share
PRODUCT_STANDARD	4	0.5 %
EPF_ETF_CHANGE	0	0.0 %
Total	**800**	**100 %**

Table 7.4 — Train / validation / test split (deterministic key split, 70/15/15)

Split	Rows	of which SECTOR_SPECIFIC
Train	560	462
Validation	120	104
Test	120	105
Total	**800**	**671**

Two properties govern the interpretation of every classification result that follows. First, the class imbalance described in Section 7.1.1. Second, **the split is deterministic, not temporal**: the gold CSV carries no usable gazette_published_date, so the generated Parquet files contain a null date for every row. The split is reproducible and leak-free with respect to regulation_key, but it does not simulate deployment on future documents. Section 7.2.1.9 gives the procedure to backfill dates and regenerate a temporal split; until that is done, this report does not describe the split as temporal.

7.2.1.3 Annotation reliability results

Source artefact: research/data/labeling/iaa_report_v1_800.json.

Table 7.5 — Overall inter-annotator agreement

Measure	Value
Tasks	800
Annotations	1,600
Paired tasks	800
Gold rows	800
Disagreement rows	40
Category Cohen's kappa	0.8715
Category raw agreement	0.9600
Mean sector kappa	0.8638
Sector-set exact agreement	0.9525
SME-relevance Cohen's kappa	0.7235
SME-relevance raw agreement	0.9550

Table 7.6 — Per-sector agreement

Sector	Cohen's kappa
grocery_retail	0.7845
food_service	0.8549
general_retail	0.9519
Mean	**0.8638**

Table 7.7 — Disagreement counts by field

Field	Disagreements
affected_sectors	38
is_sme_relevant	36
change_category	32

Table 7.8 — Agreement by annotation batch

Batch	Category kappa	Mean sector kappa	SME-relevance kappa
Batch 02	0.7509	0.8721	0.7458
Batch 03	0.7079	0.8541	0.6602
Batch 04	0.9555	0.8459	0.6700
Batch 05	0.8837	0.9156	0.9386

Interpretation. Category agreement of 0.8715 and mean sector agreement of 0.8638 fall in the near-perfect band and establish that the taxonomy is applied consistently. The gap between raw agreement (0.9600) and kappa (0.8715) is exactly the chance correction the imbalanced distribution demands, and is why kappa is the headline figure.

SME relevance is the weakest field at 0.7235 — substantial rather than near-perfect. This is a real property of the task, not a defect in the process: the boundary between a notice that binds a small business and one that is purely administrative is genuinely contestable. It is also the highest-consequence label in the system, because it gates whether an SME receives an alert at all. The response was procedural rather than statistical: all 40 disagreement rows were adjudicated explicitly with zero rows defaulting to the lead annotator, and the resolver rules were tightened after Batch 03 — visible in the rise of SME-relevance agreement from 0.6602 in Batch 03 to 0.9386 in Batch 05. The trajectory across batches is itself evidence that calibration worked.

7.2.1.4 Baseline classification results

Source artefact: storage/models/m1/baselines_v1/baselines.json.

Table 7.9 — TF-IDF baseline results on the test split (n = 120)

Model	Features	Test macro-F1
Logistic regression	TF-IDF	0.4980
LinearSVC	TF-IDF	0.6167

LinearSVC is the stronger baseline at 0.6167 macro-F1, ahead of logistic regression at 0.4980. The margin is consistent with the known behaviour of a max-margin linear classifier on sparse, high-dimensional text with few examples per minority class: the probabilistic objective of logistic regression is pulled harder toward the dominant class. Both figures sit well below the RQ1 gate of 0.92, which is the useful result — the baselines quantify roughly 30 percentage points of macro-F1 headroom that the transformer must cover, and establish that the task is not solvable by surface lexical features alone.

7.2.1.5 Transformer results — CPU smoke test

Source artefact: storage/models/m1/xlmr_lora_smoke/model_registry.json.

Table 7.10 — CPU smoke-test configuration and outcome

Field	Value
Base model	xlm-roberta-base
Seeds	42 (single seed)
Epochs	1
Batch size / max length	16 / 512
LoRA rank *r* / alpha / dropout	8 / 32 / 0.1
LoRA target modules	query, value
Learning rate (head / LoRA)	2e-5 / 1e-4
Weight decay / warmup ratio	0.01 / 0.1
Mixed precision	Disabled (CPU)
Categories / sectors	8 / 3
Validation macro-F1 (mean)	0.1111
Test macro-F1 (mean)	0.0000
gate_pass	false
Created at	2026-07-30T12:57:25

The smoke test executed one epoch on a reduced dataset with a single seed on CPU. It validated dependency resolution, tokenizer initialisation, LoRA injection into the attention projections, the dual-head loss computation, the training loop, checkpoint serialisation and model-registry generation. A test macro-F1 of 0.0 after one CPU epoch on a severely imbalanced dataset is the expected outcome — the model has not yet left the majority-class regime — and gate_pass = false is the promotion gate of Listing 6.4 correctly refusing an unfit model. **These numbers are reported for completeness and are used nowhere as evidence of model quality.**

7.2.2 Compliance Guidance Platform

The fine-tuned model was evaluated with an automated harness that scores each answer on a held-out validation split of the instruction corpus. Rather than a single accuracy figure, the harness measures a set of properties aligned with the module's aim of procedural completeness: whether the answer cites its source, presents a numbered procedure, includes a worked example, explains every named form, uses only figures present in the retrieved context, refuses genuinely unanswerable questions, avoids refusing answerable ones, and how much of the reference content it recalls.

To attribute any improvement to the fine-tuning itself rather than to a change of base model, three configurations were compared on the same validation set: the previously deployed merged model (baseline_merged); the stock Llama-3.1-8B-Instruct with the same prompt and retrieval but no adapter (baseline_base, the single-variable control); and the base model with the trained adapter (finetuned). Comparing baseline_base with finetuned isolates the effect of the adapter.

Table 7.11 — Principal Metrics for the Three Model Configurations on the Validation Split

Metric	**Baseline (merged)**	**Base model**	**Fine-tuned**
Cites the source	56%	85%	100%
Presents a numbered procedure	100%	96%	100%
Includes a worked example	96%	89%	100%
Explains every named form	100%	100%	100%
Uses only grounded figures	89%	81%	95%
Refuses the unanswerable	75%	75%	100%
Refuses the answerable (lower is better)	2%	2%	2%
Content recall vs reference	57%	64%	86%

Median answer length (characters)	478	690	501

Comparing the single-variable control with the fine-tuned model, source citation rose from 85% to 100%, grounded-figure adherence from 81% to 95%, correct refusal of unanswerable questions from 75% to 100%, and content recall from 64% to 86%, while the rate of wrongly refusing answerable questions stayed flat at 2%. Median answer length fell from 690 to 501 characters even as recall rose, indicating denser rather than truncated answers. Because the two safety-related properties — grounding and correct refusal — improved together, the gains cannot be explained by the model simply learning to refuse more often.

The awareness survey was administered to a sample of SME owners, and the responses were scored to compare declarative recall against procedural knowledge across the applicable domains, providing the empirical basis for the module's central hypothesis. The multilingual layer was validated with an automated figure-survival test across Sinhala and Tamil answers, which confirmed that every rate, threshold, deadline and form code was preserved through translation.

The main limitations are the modest size of the validation split, which makes single-point differences unreliable; the use of automated, regex-based graders rather than human judgement; a single training run without seed variation; and the fact that the corpus is derived from the same regulatory text the system retrieves, so the evaluation measures faithfulness to that corpus rather than independent real-world correctness. Expert (chartered-accountant) verification of the scraped procedures is ongoing.

7.2.3 SME Compliance Risk Prediction

Evaluation of this module addresses three questions in sequence: whether the relationships claimed are present in the raw data at all, whether the information-barrier variables improve prediction beyond business characteristics, and whether that improvement survives the assumptions made in constructing the outcome label. All results are computed on the 300 real survey responses, of which 138, or 46 per cent, reported a compliance failure within the preceding twelve months. Performance figures are obtained from cross-validated out-of-fold predictions.

The descriptive analysis, conducted without reference to any model, establishes the central pattern. Compliance-failure rates rise monotonically across every information-barrier variable. Among businesses whose owners act promptly on notices the failure rate is 22 per cent, rising to 46 per cent among those who act occasionally and 75 per cent among those who ignore or delay them. Where owners learn of a change before its deadline the failure rate is 20 per cent, rising through 45 and 59 per cent to 68 per cent among those who do not learn of it until a problem arises. Where guidance is normally received in the owner's working language the rate is 24 per cent, rising to 65 per cent where it rarely or never is. Ranked by effect size, the three strongest associations with compliance failure are all informational, with Cramér's V values of 0.41, 0.36 and 0.34, each significant at p below 0.001. Business sector, by contrast, returns a Cramér's V of 0.11 at p equal to 0.169 and is therefore not statistically associated with failure.

The model ladder quantifies the same effect. The naive cross-tabulation baseline reaches an AUC of 0.592, barely above chance, confirming that a simple tabulation of business attributes carries little predictive content. The demographic model reaches 0.637. Adding the information-barrier variables raises this to 0.720.

Table 7.12 — Model ladder performance on cross-validated out-of-fold predictions (n = 300).

Model	Features	AUC	PR-AUC
M0 — naive cross-tabulation (baseline)	3	0.592	0.541
M1 — demographic characteristics	12	0.637	0.560
M2 — demographics + information barriers	18	0.720	0.665

The increase of 0.083 attributable to the information-barrier block has a 95 per cent confidence interval of 0.033 to 0.133 and returns a DeLong test statistic of 3.254, corresponding to p equal to 0.0011. Repeating the analysis across five cross-validation seeds produces gains between 0.071 and 0.085, a spread of 0.014, confirming that the result is not an artefact of a particular partition. Feature attribution places responsiveness to notices and awareness lag as the two most influential variables, with informal information reliance fifth; three of the five highest-ranked variables are informational, and the information block accounts for 56 per cent of total attributed importance against 44 per cent for business characteristics.

The hypothesis was evaluated against thresholds fixed before the analysis was conducted. All three conditions are satisfied.

Table 7.13 — Evaluation of the module hypothesis against pre-registered thresholds.

Condition	Threshold	Observed	Met
Increase in AUC from M1 to M2	≥ 0.03	+0.083	Yes
DeLong test significance	p < 0.05	0.0011	Yes
Information features in top five by SHAP	≥ 2	3	Yes

The robustness analysis addresses the principal threat to this conclusion, namely that businesses reporting no failure are unlabelled rather than confirmed compliant. Re-estimating the comparison under the positive-unlabelled correction leaves the improvement at 0.077, still clearly above the pre-registered threshold. The finding therefore does not depend on treating non-disclosure as evidence of compliance. The correction also alters calibration, raising mean predicted risk from 0.46 to 0.68, which is material for any deployed decision threshold but does not affect the hypothesis. The estimator additionally implies a label frequency of approximately 0.55, which would indicate substantial undisclosed non-compliance; this figure is reported as indicative of direction only, since the estimator is known to be unreliable at moderate levels of discrimination and should not be read as a population prevalence.

Assessment against the Economic Census of 2013/14 [21] shows the sample to be consistent with the national establishment population in enterprise size, being dominated by micro-enterprises. Three limitations bound the interpretation of these results. The study frame covers retail and food service only, so manufacturing, which accounts for roughly a quarter of national establishments, is excluded by design. Responses were obtained from two of the nine provinces. The sample was not drawn by probability methods. The findings are therefore indicative for retail and food-service enterprises in comparable settings rather than nationally representative. It is worth noting that the recruitment method is likely to have under-reached the least digitally connected owners, who are precisely the group most exposed to the barriers under study; their under-representation would attenuate rather than inflate the measured association, so the reported effect is more probably conservative than overstated.

Taken together, the evaluation supports the module's hypothesis at every level of analysis. The relationship is visible in the raw cross-tabulations, it is significant and stable in the models, it survives correction for the labelling assumption, and its generalisability is bounded explicitly rather than assumed.

7.2.4 Regulatory Misinformation Spread

The evaluation was done on a labeled corpus of 1000 posts which were split into 800 dev and 200 test sets. It was hand-crafted from social media platforms (Facebook, Reddit, Twitter/X) and fact checking platforms, thus highly topical to the misinformation in the regulatory sphere in Sri Lanka in the context of SMEs. The quality of the corpus is suitable for a small-scale NLP oriented undergraduate research module, since it has been created manually and carefully filtered.

The quality of the annotations was good. 20% of the 200 posts were independently annotated by two annotators using Label Studio, resulting in overall veracity Cohen's Kappa of 0.934, or an excellent agreement. This implies that the annotation scheme was well established and could be scaled up by a single annotator to cover the entire dataset.

The method for evaluation was to evaluate three approaches on the same test set. The fine-tuned XLM-RoBERTa model (Approach 1) achieved a multilingual baseline performance with an accuracy of 76.0%, a macro-F1 score of 0.561 and a weighted-F1 score of 0.758. It is good at making inferences quickly, and at handling native multilingual, but it fails at the minority harmful class, which is the most relevant class for misinformation detection. The main drawback was that a pattern-based model does not allow for explicit regulatory evidence to be easily incorporated.

The RAG pipeline with ChromaDB with Module 2 gave the best performance with an accuracy of 90.0%, a macro-F1 of 0.872, and a weighted-F1 of 0.897. It outperformed the other approaches on the harmful class, with a class-wise performance of 0.881 F1 vs 0.333 F1 for XLM-R and zero F1 for direct prompting with Gemini. This shows that the retrieval grounding aspect of this model was useful, particularly for regulatory claims that depend on the legal facts and dates more so than surface words.

The top Direct Gemini baseline (Approach 3) has 73.5% accuracy and macro-F1 of 0.299. It worked well on the most correct class, but wasn't reliable enough on the wrong class to be used directly for prompting harmful posts, so direct prompting without retrieval is not suitable in this domain. This finding reinforces the argument for making retrieval a requirement and not an option.

The benefit of all three approaches was assessed and approach 2 was chosen as the final solution as it performed best on the metrics that are most relevant in the context of misinformation detection. The advantage of retrieval was that the model could access regulatory evidence and that this could generate a veridical and defensible verdict. This is more suitable to the regulatory domain than general language capability or learned patterns.

The major advantages of the module that was implemented are its domain-specificity, high level of reliability on its annotations, multi-lingual support, and evidence-based prediction. There are a few limitations, such as the limited size of the data set, the disproportionality of the harmful class, and the assumption that the retrieval corpus is complete. Future work should include extending the dataset, expanding the knowledge base of the regulatory data and further increase of coverage of the annotations so that the harmful class can be modelled more robustly.

Table 7.14 — Comparative evaluation of three misinformation classification approaches

Metric/Property	Approach 1 – XLM-R	Approach 2 – RAG + Module 2	Approach 3 - Gemini
Performance Metrics (200 Post Test Set)			
Accuracy	76.0%	90.0%	73.5%
Macro-F1	0.561	0.872	0.299
Weighted-F1	0.758	0.897	0.629
Per-class F1 Scores			
Accurate(F1)	0.869	0.938	0.851
Partly accurate(F1)	0.481	0.795	0.043
Harmful(F1)	0.333	0.881	0.000

7.3 Overall System Evaluation

7.3.1 Results

Table 7.15 — Summary of results against the project objectives

Objective	Evidence	Status
Labelled dataset with established reliability	800 rows; κ = 0.8715 / 0.8638 / 0.7235; 40 adjudicated disagreements	Achieved
Non-transformer baseline established	macro-F1 0.4980 (LogReg), 0.6167 (LinearSVC)	Achieved
Transformer pipeline validated end to end	CPU smoke test; registry written; gate_pass correctly false	Achieved
Transformer meets the RQ1 gate		Pending GPU run
Extraction accuracy quantified per field, stage and language	Measurement engine, registry, report export	Framework achieved; figures pending
Trilingual explanation delivered	Schema, NLLB helper, review queue implemented	Partially achieved
Diffusion lag measured empirically	Watchers, matcher, lag views, preregistration	Instrument achieved; findings pending data
Platform integration across four modules	Shared verified store; contracts defined	Achieved for Module 1 side

7.3.2 Discussion

What the results establish. Three things are demonstrated with committed evidence. The annotation process is reliable at the level required for the dataset to serve as ground truth. The classification task is non-trivial — the strongest lexical baseline reaches 0.6167 macro-F1, roughly 30 points short of the project gate — so any transformer improvement is measured against a meaningful floor. And the measurement framework localises failure to a specific field, stage and language rather than producing a single opaque score, which is what makes the extraction claims auditable.

Strengths. The separation of pipeline, measurement and research instrument is the design decision that pays off most. Because confidence is scored only by calibration and never folded into accuracy, a confidently wrong model cannot inflate the headline number. Because the promotion decision is a pure function with an absolute gate and a regression tolerance, an unfit model cannot reach users by accident — as the smoke test demonstrates in practice. Because every dataset version is sealed and checksummed, a measurement result can always be attributed to an exact input.

Weaknesses. The dataset is small for an eight-class problem and severely imbalanced; three categories have fewer than six examples and one has none, so per-class results for the rare categories will carry wide confidence intervals and EPF_ETF_CHANGE cannot be evaluated at all. The split is deterministic rather than temporal, so reported performance may be optimistic relative to deployment on future gazettes. SME relevance — the label that gates alerting — has the lowest annotator agreement, meaning part of the residual error the model must fit is genuine human disagreement rather than signal.

Threats to validity. *Internal:* class imbalance and the non-temporal split, as above. *Construct:* the subjectivity of SME relevance. *External:* the study covers three retail and food-service sectors over a bounded date range, so generalisation to manufacturing, construction or professional services is not demonstrated. *Measurement:* diffusion lag is measured by first observation on a monitored channel, which is a lower bound that depends on the two-hour polling interval and the completeness of the source registry, and self-reported SME awareness dates are subject to recall bias. *Reproducibility:* model non-determinism across hardware is mitigated by reporting three seeds with a standard deviation rather than a single run.

7.4 Summary

The evaluation establishes reliable annotation, a meaningful baseline, a functioning training pipeline and an auditable measurement framework. What remains outstanding is stated plainly rather than obscured: the GPU training result, the extraction measurement tables, the empirical diffusion findings and the translation review. Each has a command in Section 7.2.1.9 that produces it, and each placeholder in this chapter names the artefact from which its value is read.

Chapter 8 - Conclusion

This project set out to reduce a specific, measurable failure: the delay between the moment a Sri Lankan regulatory change becomes legally binding and the moment the small business it binds becomes aware of it. The response was Enigmatrix, a modular trilingual platform in which regulatory publications are converted once into verified structured intelligence and then reused for awareness, knowledge, risk and verification.

Module 1, the subject of this dissertation, implements the upstream half of that system. Four Scrapy spiders ingest gazettes, acts and bills. A per-page routing extraction chain handles a corpus that is part born-digital English and part scanned legacy-encoded Sinhala and Tamil, including the font-aware Wijesekara conversion that general-purpose extraction tools fail silently without. Preprocessing cleans, identifies language, extracts metadata and penalties, and chunks text for classification. An 800-record gold dataset was constructed through dual annotation and explicit adjudication. TF-IDF baselines and an XLM-RoBERTa model with LoRA adapters and a dual head were implemented, with ONNX CPU serving for production. Downstream, propagation watchers, sector-matched idempotent alerting and nightly lag analytics turn the platform into a research instrument.

Alongside the pipeline, two frameworks were built that are contributions in their own right: a sealed-baseline evaluation specification that scores the pipeline at field, record and stage granularity against a checksummed ground truth, and a preregistered findings programme that makes regulatory information diffusion an empirical quantity.

Achievement against objectives. Objectives 1, 2, 3, 4, 7, 8, 9 and 10 are achieved. Objective 5 is achieved for the baselines and for the training pipeline, with the final transformer result reported from the GPU run in Section 7.2.1.7. Objective 6 is partially achieved: the schema, translation helper, title scraper and review queue are implemented, while production batch summarisation and bulk backfill remain outstanding.

Contributions. First, an operational trilingual regulatory awareness pipeline for Sri Lankan SMEs that handles scanned and legacy-font documents. Second, a reliability-established labelled dataset of 800 Sri Lankan regulatory records with published agreement statistics and full adjudication provenance. Third, a sealed-baseline evaluation framework with per-field comparators, a fixed error taxonomy, calibration held separate from correctness, and an atomic fact table from which every headline number is a query. Fourth, a measurement instrument for regulatory information diffusion. Fifth, an explicit account of what is and is not yet demonstrated.

Limitations. The dataset is modest and severely imbalanced, with one category unevaluable. The split is deterministic rather than temporal. The study scope is three retail and food-service sectors. SME relevance, the highest-consequence label, has the lowest annotator agreement. Residual (cid:…) glyph spans remain a risk to Sinhala and Tamil extraction quality. The diffusion findings require live propagation data and survey fieldwork. The backend runs as a single container hosting web, worker and scheduler together, so scheduled tasks contend with API traffic under load.

Future work. In the immediate term: complete the multi-seed GPU run and apply the promotion gate; backfill gazette_published_date and regenerate a temporal split; ingest or hand-target additional EPF/ETF, product-standard, business-registration, penalty and import/export notices to lift the rare categories; re-extract documents with (cid:…) spans using the wijesekara_routing_v1 profile; confirm the watcher source URLs; complete the production summariser and NLLB backfill with manual review; and apply the outstanding migrations in the deployed environment. In the medium term: conduct the SME survey fieldwork with at least 100 respondents and run the F1–F6 notebooks on real data; wire alert dispatch to a verify-and-publish action with production messaging credentials; split the worker into its own service; and complete the remaining Sinhala and Tamil interface strings. In the longer term: extend the sector taxonomy beyond the three study sectors; close the active-learning loop by routing low-confidence production predictions to annotation and into quarterly retraining; move from document-level classification to obligation-level extraction of deadlines, thresholds and duties as structured fields; and test whether the diffusion instrument generalises to other jurisdictions with comparable gazette-based publication regimes.

Concluding remark. The central claim of this project is modest and specific: the gap between regulatory publication and SME awareness is an engineering problem with a measurable size, and building the transfer mechanism is only half the work — instrumenting it is the other half. What has been produced is reported as produced; what has not, is not. That distinction is maintained deliberately throughout Chapter 7, because a compliance system that overstates its own reliability would reproduce, in software, precisely the information failure it was built to correct.

Chapter 9 - References

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Appendix A - Individual Contribution

A.1 215075J - Mohomed M.R.I

I owned **Module 1, the Regulatory Change Awareness Gap**, end to end: the automated gazette pipeline and the diffusion-measurement research programme, together with the administrative tooling that operates them.

Design and architecture. I designed the five-stage regulation status machine and its field sets, the per-page extraction routing chain, the dual-head classification architecture, the propagation-measurement design with its two-step matcher, and the sealed-baseline evaluation specification that scores the pipeline at field, record and stage granularity.

Data collection and annotation. I built the Label Studio project covering eight domains, three sectors, SME relevance, annotator confidence and free-text rationale; produced the 20-document trilingual calibration set; implemented the stratified, k-means and hybrid active-learning samplers with minority-domain targeting; ran the dual annotation of batches 02–05; implemented resolve_iaa.py; and adjudicated all 40 disagreement rows into the frozen 800-row gold dataset with zero lead-annotator fallbacks.

Implementation. I implemented the four Scrapy spiders with date scoping, language fallback and completeness verification; the extraction chain across PyMuPDF, pdfplumber, pypdfium2, Tesseract and Surya with font-aware Wijesekara-to-Unicode conversion and CER measurement; the preprocessing stage with fastText language identification, metadata and penalty extraction, sub-document splitting and chunking; the dataset registry with immutable sealed versions and Excel ground-truth upload; the extraction-profile registry and run dispatcher with overlap detection and auto-versioning; the measurement engine and its per-field comparators; the measurement dashboards and the downloadable accuracy report; the admin extraction portal with live WebSocket progress; the propagation watchers and matcher; the idempotent sector-matched alert service and dispatcher; the lag-analytics materialised views and drift monitor; and the retraining loop with its canary promotion decision.

Modelling. I implemented the label schema, the data-splitting module, the TF-IDF baselines, the XLM-R + LoRA dual-head model and its training loop, the per-slice evaluator with the slice-cliff check, the ONNX exporter with INT8 quantisation, and the backend ONNX inference service with confidence-based review routing. I ran the CPU smoke test and the GPU training and evaluation runs on the free notebook platforms.

Research programme. I wrote the preregistration for findings F1–F6, implemented the shared findings loaders with bootstrap confidence intervals, and built the four analysis notebooks.

Documentation. I authored the Module 1 chapters of this dissertation, the evaluation specification, the annotation guidelines, the reproducibility command set and the artefact inventory.

A.2 215007F - Ahamadh M.S.A

I designed and implemented the Compliance Guidance Platform module, covering the full path from the underlying research question through the knowledge base, the retrieval and language-model pipeline, the multilingual layer, and the awareness survey built on the same foundation.

I framed the module around a distinction drawn from the compliance literature between declarative knowledge — knowing that a regulation exists — and procedural knowledge — knowing how to satisfy it. Starting from the observation that Sri Lankan SMEs more often fail compliance on the procedural side than the declarative side, I made answer completeness the central design goal: every answer must give the numbered procedure, the exact form, office and deadline for each step, a plain-language explanation of each named form, and a worked example. I defined the scope of the module as twenty regulatory domains grouped into eight categories across three retail sectors, and encoded that scope in a single registry so that the retrieval classifier and the survey could not drift apart.

I compiled the knowledge base exclusively from official Sri Lankan government sources and built the retrieval layer in Python using LangChain and ChromaDB with a multilingual sentence-transformer, so that a question asked in English, Sinhala or Tamil retrieves the relevant regulatory text. I engineered the prompt in several layers to enforce procedural completeness, to keep every quoted figure grounded in the retrieved source, and to permit only a single fixed refusal sentence for genuinely unanswerable questions, adding a post-processing stage that repairs spurious refusals and removes internal routing codes from the output.

I fine-tuned Llama-3.1-8B-Instruct with QLoRA to teach answer structure, completeness and refusal discipline while leaving all factual content to the retrieval layer, and I evaluated it with a purpose-built harness that scores citation, procedural formatting, grounding, refusal behaviour and content recall on a held-out validation split. To attribute the gains to the adapter rather than to a change of base model, I designed a three-configuration comparison against a single-variable control. I also implemented the Sinhala and Tamil layer as a figure-protecting translation pipeline over NLLB-200 that masks and restores numbers, form codes and amounts and falls back to English for any sentence in which a critical figure would otherwise be lost.

I built the frontend as a Next.js application with a chat interface, a sector-scoped awareness survey that scores an owner's declarative and procedural knowledge separately, and a research dashboard, and I deployed the backend as a Docker container on Railway. I also exposed the fact-checking endpoint used by the Regulatory Misinformation Spread module, and documented the module's architecture, research design, results and integration requirements for this report.

A.3 215008J - Ahamed T.I

I designed and implemented the SME Compliance Risk Prediction module, covering the full path from formulating the research question through data collection instruments, statistical analysis, and the deployed platform built on the resulting findings.

I began by narrowing the module from a broad interest in compliance risk to a single testable question: whether information-access variables predict compliance failure beyond business characteristics. I designed the survey instrument used to collect the data, including the questions measuring awareness lag and language match that the analysis later identified as principal predictors, together with the participant information sheet and consent form, and I prepared the programme-level ethics documentation covering all four modules. Before any data was collected I conducted a simulation-based power analysis to establish the sample size the study would require and fixed the decision thresholds for the hypothesis in advance.

I implemented the analytical pipeline in Python: the recoding of survey responses into the modelling schema, the three model specifications including the naive baseline used as a reference, and the statistical procedures. I implemented DeLong's test and the SHAP attribution directly rather than relying on packaged versions, and verified each against an independent implementation and against the additivity property respectively. I implemented the positive-unlabelled correction to test whether the conclusion depended on treating non-disclosure as compliance, and the descriptive and representativeness analyses that bound the findings.

I applied data-quality checks to the survey data before analysis and documented them, including the identification and correction of a defect in my own processing code that had affected an earlier reading of one variable. I regard recording that correction as part of the module's contribution rather than an omission from it.

I then designed and built the platform arising from the findings: a web application that scores a business, explains the specific factors driving its risk, and delivers guidance in English, Sinhala and Tamil, together with the matching engine that determines which registered businesses a newly published regulation affects and issues targeted alerts in the owner's language. I defined the event interface through which the Regulatory Change Awareness module supplies those regulatory changes, and verified the complete workflow end to end. The module is covered by sixty-one automated tests. I documented the module's architecture, research design, results and integration requirements, and prepared the material for its inclusion in this report.

A.4 215019T - Cader Z.R

As part of the larger platform for detecting misinformation, I designed and developed the Regulatory Misinformation Spread module. My efforts were dedicated to the end-to-end process of this module, which involved data collection and preprocessing (manually), annotation, preparation of the dataset, development of the model, its evaluation, and documentation. My aim was to develop a trustworthy module that can recognize and identify regulatory misinformation in social media content targeting Sri Lankan SMEs.

I did a manual search of posts related to the regulation aspect of social media platforms and fact-check websites with content related to taxation, import and export regulations, labour regulations, product standards, and penalty enforcement. I gathered the raw posts and cleaned and preprocessed the data by excluding irrelevant information, merging duplications, adding regulation domains, identifying language and discarding personally identifiable information (PII), if applicable. I took the trouble to organize the data in such a way, that the raw data file was easily separated from the master data file, from the classifier data file, and from the training/testing data file, so that they could be used in downstream processing.

The data set contained Sinhala and Tamil posts so I translated these posts into English using NLLB-200 and kept their original text in the data set. This enabled a consistent annotation across languages, while ensuring that the corpus remained multilingual. Then I set up Label Studio for the first double annotation pass (200 posts were double annotated by two annotators). Before continuing with the remaining dataset I calculated the Cohen's Kappa of the second annotator to check the agreement between the two and to assure that the scheme was reliable.

I then downloaded the rest of the posts into Google Sheets and annotated them, after they came strong in the agreement results. Next, I built the classifier dataset by reordering the labels for model creation, and by dividing the training data into training and test sets, to ensure that the evaluation set was not used during training. I applied and tested the three modelling strategies introduced in the module: fine-tuned XLM-RoBERTa, retrieval-augmented generation and the baseline approach of the Module 2. I evaluated them on the same test set that I held out, and chose the final model from the results.

I also recorded the process of implementing the modelling and the outcomes of the evaluations in the report. In general, I worked on all aspects of the Regulatory Misinformation Spread module ranging from data preparation, data annotation, model comparison, to the final solution selection.